

PERCENTAGE

PERCENTAGE (प्रतिशत %)

$$\begin{array}{ccc} \text{PER} & + & \text{CENT} \\ \downarrow & & \downarrow \\ \text{divide by} & & 100 \end{array} \quad X\% = \frac{X}{100}$$

प्रतिशत वह चिह्न है जिसका हर 100 हो।

PERCENT VS PERCENTAGE

↓ ↓

जब % किसी known/unknown value के साथ आता है तो उसे PERCENT बुलमा जाता है।
 e.g. $X\% = X \text{ PERCENT}$
 $20\% = 20 \text{ PERCENT}$

↓ ↓

जब % अकेला आता है तो उसे % से बुलमा जाता है।
 $\% = \text{PERCENTAGE}$

why we study % (आखिर क्यों)?

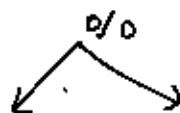
e.g. There are three students A, B & C
 by looking at table I it looks like C is better but wait and see Table II

	I MARKS
A	45
B	80
C	90

	TABLE II		
	marks	mn	%
A	45	50	90%
B	80	100	80%
C	90	150	60%

Note:- Table I was incomplete. Now in Table II we calculate marks of A, B, C out of 100 (%)

IND Table देखने के बाद हमने सबसे marks 100 में से निकाल लिए (%). अब हम compare कर सकते हैं और बता सकते हैं कि A is better.



divide by 100 OR Compare with 100

PERCENTAGE CHANGE

$$\% \text{ CHANGE} = \frac{\text{WHAT WE COMPARE (जिसको)}}{\text{TO WHOM WE COMPARE (जिसके साथ)}} \times 100$$

Let take Two No A = 120, B = 150



(i) A is what % of B

OR what % of B is A $= \frac{A}{B} \times 100 = \frac{120}{150} \times 100 = 80\%$

OR what % is A of B $\rightarrow$ [जिसके साथ of आता है व नीचे आता है]

(ii) B is what % of A

OR what % of A is B $= \frac{B}{A} \times 100 = \frac{150}{120} \times 100$
 OR what % is B of A $= 125\% \text{ Ans.}$

(iii) By how much % A is less than B $= \frac{B-A}{B} \times 100$
 $= \frac{30}{150} \times 100 = 20\%$

(iv) By how much % B is more than A $= \frac{30}{120} \times 100 = 25\%$

CONCEPT OF BY and TO1) A NUMBER is decrease TO 40%. [उतना हो जायगा]

$$\Rightarrow \frac{\text{INITIAL}}{100} \quad \text{FINAL} \quad 40$$

2) A NUMBER is decreased by (उतना से) 40%

$$\Rightarrow \frac{\text{INITIAL}}{100} \quad \text{FINAL} \quad 60 \quad \text{बच जायगा}$$

PERCENTAGE CHANGE VS PERCENTAGE POINT CHANGE

$$\begin{array}{ccc} \downarrow & A = 10\% & B = 20\% & \downarrow \\ \% \text{ Change} = \frac{5\%}{10\%} \times 100 = \underline{\underline{50\%}} & & 15\% - 10\% = \underline{\underline{5\%}} \end{array}$$

% TO FRACTION
[divide by 100]

$$20\% = \frac{20}{100} = \frac{1}{5}$$

$$12.5\% = \frac{12.5}{100} = \frac{1}{8}$$

$$40\% = \frac{40}{100} = \frac{2}{5}$$

FRACTION TO %
[multiply by 100]

$$\frac{1}{5} \Rightarrow \frac{1}{5} \times 100 = 20\%$$

$$\frac{1}{6} \Rightarrow \frac{1}{6} \times 100 = 16\frac{2}{3}\%$$

$$\frac{3}{8} \Rightarrow \frac{3}{8} \times 100 = 37.5\%$$

How To calculate % QUICKLY

$$\begin{array}{lcl}
 520 & \longrightarrow & 100\% \\
 \swarrow & & \\
 52 & \longrightarrow & 10\% = 52 \\
 \swarrow & & \\
 5.2 & \longrightarrow & 1\% = 5.2 \\
 \swarrow & & \\
 .52 & \longrightarrow & .1\% = .52
 \end{array}$$

21% of 520

$$\begin{aligned}
 &= 10\% + 10\% + 1\% \\
 &= 52 + 52 + 5.2 \\
 &= 109.2 \text{ Ans.}
 \end{aligned}$$

27% of 520

$$\begin{aligned}
 &= 25\% + 1\% + 1\% \\
 &= \frac{1}{4} \times 520 + 5.2 + 5.2 \\
 &= 130 + 10.4 = 140.4
 \end{aligned}$$

57% of 520

$$\begin{aligned}
 &= 50\% + 5\% + 2\% \\
 &= 260 + 26 + 5.2 + 5.2 \\
 &= 286 + 10.4 \\
 &= 296.4 \text{ Ans.}
 \end{aligned}$$

144 is what % of 450

$$\begin{aligned}
 &= \frac{144}{450} \times 100 \\
 &= \frac{(135+9)}{450} \times 100 \\
 &= \frac{135}{450} \times 100 + \frac{9}{450} \times 100 \\
 &= 30\% + 2\% = 32\% \text{ Ans.}
 \end{aligned}$$

126 is what % of 450

$$\begin{aligned}
 &= \frac{126}{450} \times 100 \\
 &= \frac{135-9}{450} \times 100 \\
 &= \frac{135}{450} \times 100 - \frac{9}{450} \times 100 \\
 &= 30\% - 2\% = 28\%
 \end{aligned}$$

138 is what % of 476

$$\begin{aligned}
 &= \frac{138}{476} \times 100 \\
 &= \frac{(119+19)}{476} \times 100 \\
 &= \frac{119}{476} \times 100 + \frac{19}{476} \times 100 \\
 &= 25\% + 4\% \approx \underline{29\%}
 \end{aligned}$$

63% of 560

$$\begin{aligned}
 &= (62.5\% + .5\%) \text{ of } 560 \\
 &= \frac{5}{8} \times 560 + 2.8 \\
 &= 350 + 2.8 \\
 &= 352.8 \text{ Ans.}
 \end{aligned}$$

FRACTION TO PERCENTAGE CONVERSION Table

Fraction	Percentage		Fraction	Percentage	
	I	II		I	II
1	100%	100%	$\frac{1}{14}$	7.14%	$7\frac{1}{7}\%$
$\frac{1}{2}$	50%	50%	$\frac{1}{15}$	6.67%	$6\frac{2}{3}\%$
$\frac{1}{3}$	33.33%	$33\frac{1}{3}\%$	$\frac{1}{16}$	6.25%	$6\frac{1}{4}\%$
$\frac{1}{4}$	25%	25%	$\frac{1}{17}$	5.88%	$5\frac{15}{17}\%$
$\frac{1}{5}$	20%	20%	$\frac{1}{18}$	5.56%	$5\frac{5}{9}\%$
$\frac{1}{6}$	16.67%	$16\frac{2}{3}\%$	$\frac{1}{19}$	5.26%	$5\frac{5}{19}\%$
$\frac{1}{7}$	14.28%	$14\frac{2}{7}\%$	$\frac{1}{20}$	5%	5%
$\frac{1}{8}$	12.5%	$12\frac{1}{2}\%$	$\frac{1}{24}$	4.16%	$4\frac{1}{6}\%$
$\frac{1}{9}$	11.11%	$11\frac{1}{9}\%$	$\frac{1}{25}$	4%	4%
$\frac{1}{10}$	10%	10%	$\frac{3}{4}$	75%	75%
$\frac{1}{11}$	9.09%	$9\frac{1}{11}\%$	$\frac{2}{5}$	40%	40%
$\frac{1}{12}$	8.33%	$8\frac{1}{3}\%$			
$\frac{1}{13}$	7.69%	$7\frac{9}{13}\%$			



DERIVED FRACTION TO CONVERSION Table

Fraction	Percentage		Fraction	Percentage	
	I	II		I	II
$\frac{1}{8}$	12.5%	$12\frac{1}{2}\%$	$\frac{2}{3}$	66.67%	$66\frac{2}{3}\%$
$\frac{3}{8}$	37.5%	$37\frac{1}{2}\%$	$\frac{1}{12}$	8.33%	$8\frac{1}{3}\%$
$\frac{5}{8}$	62.5%	$62\frac{1}{2}\%$	$\frac{7}{12}$	58.33%	$58\frac{1}{3}\%$
$\frac{7}{8}$	87.5%	$87\frac{1}{2}\%$	$\frac{11}{12}$	91.67%	$91\frac{2}{3}\%$
$\frac{1}{6}$	16.67%	$16\frac{2}{3}\%$	$\frac{5}{6}$	83.33%	$83\frac{1}{3}\%$

Larger FRACTION/PERCENTAGES

$$1) 108.33\% \text{ or } 108\frac{1}{3}\% = 100\% + 8.33\% = 1 + \frac{1}{12} = \frac{13}{12}$$

$$2) 362.5\% \text{ or } 362\frac{1}{2}\% = 300\% + 62.5\% = 3 + \frac{5}{8} = \frac{29}{8}$$

$$3) 191.67\% \text{ or } 191\frac{2}{3}\% = 200\% - 8\frac{1}{3}\% = 2 - \frac{1}{12} = \frac{23}{12}$$

$$4) 393.33\% \text{ or } 393\frac{1}{3}\% = 400\% - 6\frac{2}{3}\% = 4 - \frac{1}{15} = \frac{59}{15}$$

$$5) 283.33\% \text{ or } 283\frac{1}{3}\% = 200\% + 83\frac{1}{3}\% = 2 + \frac{5}{6} = \frac{17}{6}$$

$$\text{OR} = 300 - 16\frac{2}{3}\% = 3 - \frac{1}{6} = \frac{17}{6}$$

PERCENTage IS INTERCHANGable

$$A\% \text{ of } B = B\% \text{ of } A = \frac{A \times B}{100}$$

e.g. $64\% \text{ of } 62.5 = ?$
 $= 62.5\% \text{ of } 64$ [$\because \% \text{ is interchangable}$]
 $= \frac{5}{8} \times 64 = 40$ [$\because 62.5\% = \frac{5}{8}$]

e.g. $72\% \text{ of } 91\frac{2}{3} = ?$
 $= 91\frac{2}{3}\% \text{ of } 72$
 $= \frac{11}{12} \times 72 = 66 \text{ ANS.}$ [$\because 91\frac{2}{3}\% = \frac{11}{12}$]

$17\frac{1}{2}\% \text{ of } 84 = ?$
 $= 35\% \text{ of } 42$ [$\because a\% \text{ of } b = 2a\% \text{ of } \frac{b}{2}$]
 $= 70\% \text{ of } 21$
 $= 14.7 \text{ ANS.}$

$32\% \text{ of } 250 + 12.5\% \text{ of } 640 = ?$
 $12.5\% \text{ of } 640 = 25\% \text{ of } 320$ [$\because a\% \text{ of } b = 2a\% \text{ of } \frac{b}{2}$]
 $\therefore 32\% \text{ of } 250 + 25\% \text{ of } 320$
 $= 25\% \text{ of } 320 + 25\% \text{ of } 320$
 $= (25\% + 25\%) \text{ of } 320 = 50\% \text{ of } 320 = 160 \text{ ANS.}$

$$\# \quad 62.5\% \text{ of } 512 + 83\frac{1}{3}\% \text{ of } 216 = ?$$

$$\begin{aligned}
 &= \frac{5}{8} \times 512 + \frac{5}{6} \times 216 \quad \left[\because 62\frac{1}{2}\% = \frac{5}{8} \text{ P} \right. \\
 &= 5 \times 64 + 5 \times 36 = 5(100) \quad \left. 83\frac{1}{3}\% = \frac{5}{6} \right] \\
 &= 500 \text{ Ans.}
 \end{aligned}$$

$$\# \quad 193\frac{1}{3}\% \text{ of } 225 + 91\frac{2}{3}\% \text{ of } 144 = ?$$

$$\begin{aligned}
 &(200\% - 6\frac{2}{3}\%) \times 225 + (100\% - 8\frac{1}{3}\%) \text{ of } 144 \\
 &= 450 - \frac{1}{15} \times 225 + 144 - \frac{1}{12} \times 144 \\
 &= 435 + 132 = 567 \quad \left[\because 6\frac{2}{3}\% = \frac{1}{15}, \right. \\
 &\quad \left. 8\frac{1}{3}\% = \frac{1}{12} \right]
 \end{aligned}$$

Population of city beautiful Chandigarh is 490000 in 2016. If growth rate is $14\frac{2}{7}\%$. What will be population in 2017?

Solution:- $P_{2016} = 490000$

$$14\frac{2}{7}\% \text{ of } 490000 = \frac{1}{7} \times 490000 = 70000$$

$$P_{2017} = 490000 + 70000 = 560000$$

TRICKY CONCEPTS

1. Convert % In to fraction $\pm \frac{N}{D}$

where + means increase

& - means decrease.

2. Initial value = Denominator = D 3. Final Value = $D \pm N$ or $I \pm N$

+ for Increase

- for Decrease

e.g. 20% Increase = $+\frac{1}{5}$ $\begin{matrix} \nearrow \text{Increase} \\ \searrow \text{Initial} \end{matrix}$

$$\text{Final} = 5 + 1 = 6$$

20% decrease = $-\frac{1}{5}$ $\begin{matrix} \nearrow \text{decrease} \\ \searrow \text{Initial} \end{matrix}$

$$\text{Final} = 5 - 1 = 4$$

If $16\frac{2}{3}\%$ of a Number is added with itself, resultant Number becomes 3430. Find the Original Number?Solution :- $16\frac{2}{3}\% \uparrow = +\frac{1}{6}$ $\begin{matrix} \nearrow \\ \searrow \end{matrix}$

$$\Rightarrow F = 6 + 1 = 7$$

$$7 \rightarrow 3430$$

$$\Rightarrow 6 \rightarrow \frac{3430}{7} \times 6 = 2940 \text{ Ans.}$$



If $16\frac{2}{3}\%$ of a Number is subtracted from itself then the resultant Number becomes 225. Find the original Number?

Sol. $16\frac{2}{3}\% = \frac{-1}{6} \rightarrow \text{Dec.} \Rightarrow \text{Final} = 6 - 1 = 5$
 $6 \rightarrow \text{INITIAL}$

$5 \rightarrow 225$

$6 \rightarrow 45 \Rightarrow 6 \rightarrow 45 \times 6 = 270 \text{ Ans.}$

If 210 is added in a no. then the no. becomes 137.5% of itself. Find the original Number?

Solution :- $137\frac{1}{2}\% = 100\% + 37\frac{1}{2}\% = 1 + \frac{3}{8} = \frac{11}{8} \rightarrow F$
 $8 \rightarrow I$

Final $\leftarrow \frac{11}{8}$
 Initial $\leftarrow 8$
 $3 \rightarrow 210 \rightarrow \text{original} = 8 \rightarrow 8 \times 70 = 560 \text{ Ans.}$
 $1 \rightarrow 70$

Population of a city is 72900 in 2010. If population growth rate is $12\frac{1}{2}\%$. Find population in 2009.

Solution :- $12\frac{1}{2}\% = \frac{+1}{8} \rightarrow \text{Increase}$
 $8 \rightarrow 2009 \Rightarrow P_{2010} = 8 + 1 = 9$

$9 \rightarrow 72900$

$1 \rightarrow 8100$

$P_{2009} = 8 \rightarrow 8 \times 8100 = 64800$

SUCCESSIVE (क्रमगत/लगातार) % change



$$\% \text{ change} = X + Y + \frac{XY}{100}$$

IF $\uparrow$ take x, y +ve
 $\downarrow$ take x, y -ve.

L $\uparrow$ = 20% B $\downarrow$ 10%
 $A = ?$

$$= 20 - 10 - \frac{200}{100} \quad \boxed{A = L \times B} \quad B$$

$$= 8\% \uparrow$$

Square $A = a^2$

$a \uparrow 20\%$

$$A = 20 + 20 + \frac{400}{100}$$

$$= 44\% \uparrow$$

50% + 40% = ?

$$= -50 - 40 + \frac{2000}{100}$$

$$= -70\%$$

$\downarrow$

Show & dir count

Circle (वृत्त)

$$A = \pi R^2$$



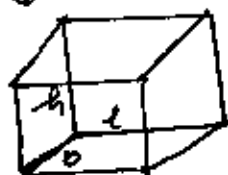
$R \downarrow 10\%$

$$A = -10 - 10 + \frac{100}{100}$$

$$= -19\% \downarrow$$

Cuboid

$$V = l b h$$



$l \uparrow 10\%$ $b \downarrow 10\%$

$h \uparrow 20\%$

$$V = 10 - 10 + 20 + \frac{-100 - 200 + 400}{100}$$

$$+ \frac{-10 \times 10 \times 20}{100^2}$$

$$= 20 - 1 - \frac{1}{5} = \frac{94}{5} = 18.8\% \uparrow$$

30% + 20% + 10% = ?

$$= -30 - 20 - 10 + \frac{600 + 200 + 300}{100}$$

$$- \frac{6000}{100^2}$$

$$= -60 + 11 - .6$$

$$= -49.6$$

Ans 49.6%

If the length of rectangle is increased by $12\frac{1}{2}\%$ and breadth decreased by 20% . Find the % change in Area?

Solution :-	L	x	B	=	A	$12\frac{1}{2}\% = +\frac{1}{8}$ $20\% \downarrow = -\frac{1}{5}$
original	8		5	=	40	
New	9		4	=	36	

$$\% \text{ change} = \frac{-4}{40} \times 100 = -10\% \text{ Ans}$$

- Shows Area is $\downarrow$

If radius of circle is decreased by 10% . Find the % change in Area of circle. ($A = \pi R^2$)

Solution :-	Radius	Area	$10\% \downarrow = -\frac{1}{10}$ $\text{New} = 10 - 1 = 9$
original	10	100	
New	9	81	

$$\% \text{ change} = \frac{-19}{100} \times 100 = 19\% \downarrow$$

If the price of the sugar is increased by $16\frac{2}{3}\%$ and consumption is decreased by 25% . Find the % change in Expenditure?

Solution	Price x	Consump.	=	Exp.	$16\frac{2}{3}\% \uparrow = +\frac{1}{6}$ $25\% \downarrow = -\frac{1}{4}$
original	6	4	=	24	
New	7	3	=	21	

$$\% \text{ change} = \frac{-3}{24} \times 100 = 12.5\% \downarrow \text{ Ans.}$$

The value of a machine depreciates at the rate of 10% per annum. If its present worth is 3645000. find its worth after 3 years

Solution $10\% \downarrow = -\frac{1}{10}$

$$\begin{array}{r} 10 \quad \text{---} \quad 9 \\ 10 \quad \text{---} \quad 9 \\ 10 \quad \text{---} \quad 9 \\ \hline 1000 \quad \text{---} \quad 729 \end{array}$$

$$1000 \rightarrow 3645000 \Rightarrow 729 \rightarrow 2657205 \text{ ANS.}$$

The present population of a town is 108000. During the 1st yr. population increases by $6\frac{2}{3}\%$, while decreases by $16\frac{2}{3}\%$ during second year. During 3rd year increases by $33\frac{1}{3}\%$. Find the population of town 3 yrs hence?

Sol $6\frac{2}{3}\% \uparrow = +\frac{2}{3}$, $16\frac{2}{3}\% \downarrow = -\frac{1}{6}$, $33\frac{1}{3}\% = +\frac{1}{3}$

$$\begin{array}{r} 3 \quad \quad 5 \\ 6 \quad \quad 5 \\ 3 \quad \quad 4 \\ \hline 54 \quad \quad 100 \end{array}$$

$$\begin{array}{l} 54 \rightarrow 10800 \\ \Rightarrow 100 \rightarrow \boxed{20000} \text{ ANS.} \end{array}$$

OR

$$\begin{array}{r} 3 \\ 6 \\ 3 \end{array}$$

The single discount which is equivalent to successive discounts of 50% & 40%?

Solution :- 50% 40%

$$\text{original } 2 \quad 5 = 10$$

$$\text{New } 1 \quad 3 = 3 \quad \text{---} 7$$

$$\text{equivalent \%} = -7/10 \times 100 = 70\% \quad \text{Ans.}$$

discount is always decrease

$$50\% \downarrow = -\frac{1}{2}$$

$$40\% \downarrow = -\frac{2}{5}$$

A single discount which is equivalent to successive discounts of 30%, 20% & 10%?

Solution 30% 20% 10%

$$\text{original } 10 \quad 5 \quad 10 = 500$$

$$\text{New } 7 \quad 4 \quad 9 = 252 \quad \text{---} 248$$

$$\text{equivalent discount \%} = \frac{248}{500} \times 100 = 49.6\%$$

If the price of cinema ticket is increased by $16\frac{2}{3}\%$, then the sale will be decreased by 20%. Find the % change in Revenue?

Solution :-

$$\text{Price} \times \text{Sale} = \text{Revenue}$$

$$\text{original } 6 \quad 5 = 30$$

$$\text{New } 7 \quad 4 = 28 \quad \text{---} 2$$

$$\% \text{ Change} = \frac{-2}{30} \times 100 = 6\frac{2}{3}\% \downarrow$$

Ans.

$$16\frac{2}{3}\% \uparrow = +\frac{1}{6}$$

$$20\% \downarrow = -\frac{1}{5}$$

Product Constancy Method

$$\text{Let } P = A \times B$$

If one increases by some %, in order to keep P as constant, he should decrease B or vice-versa.

or in simple way If $P = \text{constant}$

$$\Rightarrow A \propto \frac{1}{B} \quad [A \text{ \& B are inversely Proportional}]$$

e.g If $A = 2:3$, If $P = \text{constant}$

$$\Rightarrow B = 3:2$$

Application

(i) Area of Rectangle $A = l \times b$

(ii) Expenditure = Price $\times$ Consumption
 $E = P \times C$

(iii) Revenue = Price $\times$ Sales, $R = P \times S$

(iv) Distance = Speed $\times$ Time, $D = S \times T$

(v) WORK = Time $\times$ Efficiency etc

If the price of sugar is increased by 20%, then by how much % the consumption should be decreased so that expenditure will remain same?

Solution:- $20\% \uparrow = +\frac{1}{5} \rightarrow \text{Increase}$
 $5 \rightarrow \text{old Price}$

$$\text{New Price} = 5 + 1 = 6$$

$$\text{Also } E = P \times C \quad \text{as } E = \text{constant} \Rightarrow P \propto \frac{1}{C}$$

original New

$$\text{Price} \rightarrow 5 : 6$$

$$\Rightarrow \text{Consumption} \rightarrow 6 \text{ : } 5 \quad [\because P \propto \frac{1}{C}]$$

$$\% \downarrow = -\frac{1}{6} = 16\frac{2}{3}\% \text{ Ans.}$$

If the length of rectangle is decreased by 12.5%. By how much percentage breadth must be decreased in order to keep Area same?

Solution:- $12.5\% \downarrow = -\frac{1}{8} \rightarrow \text{old length}$

$$\text{New length} = 8 - 1 = 7$$

$$A = l \times b, \quad \text{as } A = \text{constant} \Rightarrow l \propto \frac{1}{b}$$

old New

$$\text{length} \quad 8 : 7$$

$$\text{Breadth} \quad 7 \text{ : } 8 \quad [\because l \propto \frac{1}{b}]$$

$$\% \uparrow = +\frac{1}{7} = 14\frac{2}{7}\% \text{ Ans.}$$

A reduction in of 25% in the Price of Sugar enables a Housewife to purchase 4 kg more for ₹ 800. Find original and current Price per kg?

Solution : $\rightarrow$ Here Expenditure = Constant = 800

$$\Rightarrow \text{Price} \propto \frac{1}{\text{Consumption}}$$

$$25\% \downarrow = -\frac{1}{4} \rightarrow \text{old} \Rightarrow \text{New} = 4 - 1 = 3$$

$$\begin{array}{ccc} \text{Price} & \text{old} & \text{New} \\ & 4 & : 3 \end{array}$$

$$\text{Consump} \quad 3 \quad : \quad 4$$

$$\quad \quad \quad +1 \rightarrow 4$$

$$\Rightarrow \text{old Consumption} = 3 \times 4 = 12 \text{ kg}$$

$$\text{New Consumption} = 4 \times 4 = 16 \text{ kg}$$

$$(i) \text{ old Price} = \frac{800}{12} = 66\frac{2}{3} \text{ ₹/kg} \quad (ii) \text{ New/current Price} = \frac{800}{16} = 50$$

OR DIRECT

$$25\% \text{ of } 800 = 200$$

$$\text{New/Reduced Price} = \frac{200}{4} = 50$$

$$\text{old} \Rightarrow 75\% \rightarrow 50$$

$$100\% \rightarrow 66\frac{2}{3}$$

Due to 30% increase in Price of apples, 6 apples are less available for ₹ 520. Find the old and new Price of a apple?

Solution : $30\% \uparrow = +\frac{3}{10}$

$$\begin{array}{ccc} \text{Price} & \text{old} & \text{New} \\ & 10 & : 13 \end{array}$$

$$\text{Consump.} \quad 13 \quad : \quad 10$$

$$\quad \quad \quad -3 \rightarrow 6$$

$$\quad \quad \quad 1 \rightarrow 2$$

OR DIRECT

$$30\% \text{ of } 520 = 156$$

$$\text{New Price} = \frac{156}{6} = 26$$

$$\text{old} \rightarrow 130\% = 26$$

$$100\% \rightarrow 20$$

$$\text{old Consum.} = 13 \times 2 = 26$$

$$\text{old Price} = \frac{520}{26} = 20 \text{ ₹}$$

$$\text{New Consumpt.} = 10 \times 2 = 20$$

$$\text{New Price} = \frac{520}{20} = ₹ 26$$

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A reduction of 20% in the price of sugar enables a man to buy 10kg more for ₹ 54. Find the original as well as reduced price per kg?

Solution $\Rightarrow$ Here Expenditure = 54 = constant
 $\Rightarrow$ Price $\propto \frac{1}{\text{Consumption}}$

$$20\% \downarrow = -\frac{1}{5} \quad \text{old} \quad \text{New}$$

$$\text{price} = 5 : 4$$

$$\text{Consumption} = 4 : 5$$

+1 $\rightarrow$ 10kg

Consumption

OR DIRECT

$$20\% \text{ of } 54 = 10.8$$

$$\frac{\text{New/Reduced Price}}{\text{Price}} = \frac{10.8}{10}$$

$$= 1.08 \text{ Ans.}$$

$$\Rightarrow \text{old Consum} = 10 \times 4 = 40$$

$$\text{New } = 10 \times 5 = 50$$

$$\text{Old Price} = \frac{54}{40} = 1.35 \text{ ₹/kg}$$

$$\text{New or Reduced} = \frac{54}{50} = 1.08 \text{ ₹/kg}$$

Due to an increase of 20% in the price of eggs, 2 eggs less available for ₹ 24. The present rate of egg per dozen?

Solution $\Rightarrow$ E = constant

$$20\% \uparrow = +\frac{1}{5}$$

old New

$$\text{price} = 5 : 6$$

$$\text{Consump.} = 6 : 5$$

-1 $\rightarrow$ 2

New Consumption

$$\Rightarrow 5 \rightarrow 5 \times 2 = 10 \text{ egg}$$

$$\text{Price per egg} = \frac{24}{10} = 2.4$$

$$\text{Price of dozen} = 2.4 \times 12 = 28.80 \text{ ₹ Ans.}$$

If the Price of sugar is decreased by 40%. By how much % consumption should be increased so that expenditure will decrease by 10%?

Solution :- METHOD 1

$$E = P \times C$$

$$\text{Exp.} = \text{Price} \times \text{Consumption}$$

$$\text{Let old } 100 = 10 \times 10$$

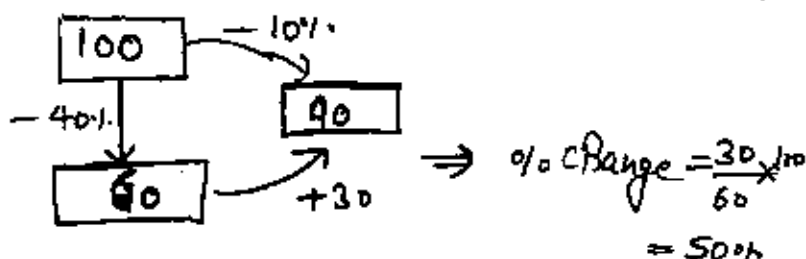
$$\text{New } 90 = 6 \times \text{New Consumption}$$

$$\Rightarrow \text{New Consumption} = \frac{90}{6} = 15$$

$$\% \uparrow \text{ in Consumption} = \frac{15-10}{10} \times 100 = 50\% \text{ Ans.}$$

$$\begin{aligned} \text{OR} \\ &= x + y + \frac{xy}{100} \\ -10 &= -40 + x \\ &\quad - \frac{40x}{100} \\ \Rightarrow 60x &= 3000 \\ \Rightarrow x &= 50\% \\ 50\% \uparrow \end{aligned}$$

METHOD 2:-



If the Price of Rice is decreased by 40%. By how much % consumption should be increased so that Expenditure also increased by 40%?

Solution :- 1st Method

$$E = P \times C$$

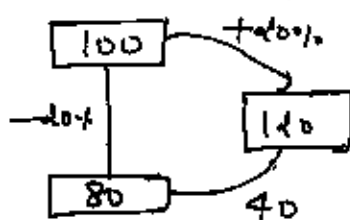
$$\text{Let old } 100 = 10 \times 10$$

$$\text{New } 140 = 8 \times N_c$$

$$\Rightarrow N_c = 15$$

$$\% \uparrow = \frac{5}{10} \times 100 = 50\%$$

2nd Method



% Increase

$$= \frac{40}{80} \times 100 = 50\%$$

Focus ON FIX $\Rightarrow$ The quantity that does not change in initial and final mixture make that equal.

e.g. If there is mixture of water and salt & water get evaporated. That means salt will not change in initial and final mixture. make salt equal. You will better understand by questions.

A vessel has 60 litres of solution of acid & water having 80% acid. How much water must be added to make it a solution in which acid is 60%?

Solution:- water is added $\Rightarrow$ acid qty is same

1st METHOD

let final mixture (after add water)
is = x ltr

equating Acid

$$\frac{80 \times 60}{100} = \frac{60 \times x}{100}$$

$$\Rightarrow x = 80 \text{ ltr}$$

$$\text{water added} = 80 - 60 = 20$$

2nd METHOD

by Ratio

	Acid	:	water
old	4x3	:	1x3

New	3x4	:	2x4
-----	-----	---	-----

make Acid equal

$$12 : 3$$

$$12 : 8 \rightarrow 5 \text{ water added}$$

$$15 \rightarrow 60 \Rightarrow 5 \rightarrow \boxed{20 \text{ ltr}}$$

75 gm of a solution has 30% sugar in it. Then the qty of sugar that should be added to the solution to make 70% sugar solution. ANS. 100 gm

FRESH grapes contains 90% water while dry grapes contains 20% water. What is the weight of dry grapes obtained from 20 kg FRESH GRAPES?

Solution 1st Method

The weight of non-water part called Pulp will remain same equate

$$\frac{20 \times 10}{100} = \frac{80 \times X}{100} \Rightarrow X = 2.5 \text{ kg}$$

$X \rightarrow$ dry grapes weight.

2nd METHOD Ratio

	PULP	:	WATER
Old	1×4	:	9×4
New	4	:	1

Make PULP same

$$\begin{array}{rcl} \cancel{5} : \cancel{45} & 4 : 36 & \rightarrow 40 \\ \cancel{5} : & 4 : 1 & \rightarrow 5 \end{array}$$

$$40 \rightarrow 20 \text{ kg} \Rightarrow 5 \rightarrow \boxed{2.5 \text{ kg}}$$

FRESH GRAPES contains 80% water, while dry grapes contains 10% water. If the weight of dry grapes is 500 kg. What is total wt. when it is FRESH?

Solution 2 - equate Non water Part (Pulp)

1st method

Let total wt. of fresh grapes = X kg.

Pulp₁ = Pulp₂

$$\frac{20 \times X}{100} = \frac{90 \times 500}{100}$$

$$\Rightarrow X = 2250 \text{ kg.}$$

2nd METHOD (Ratio)

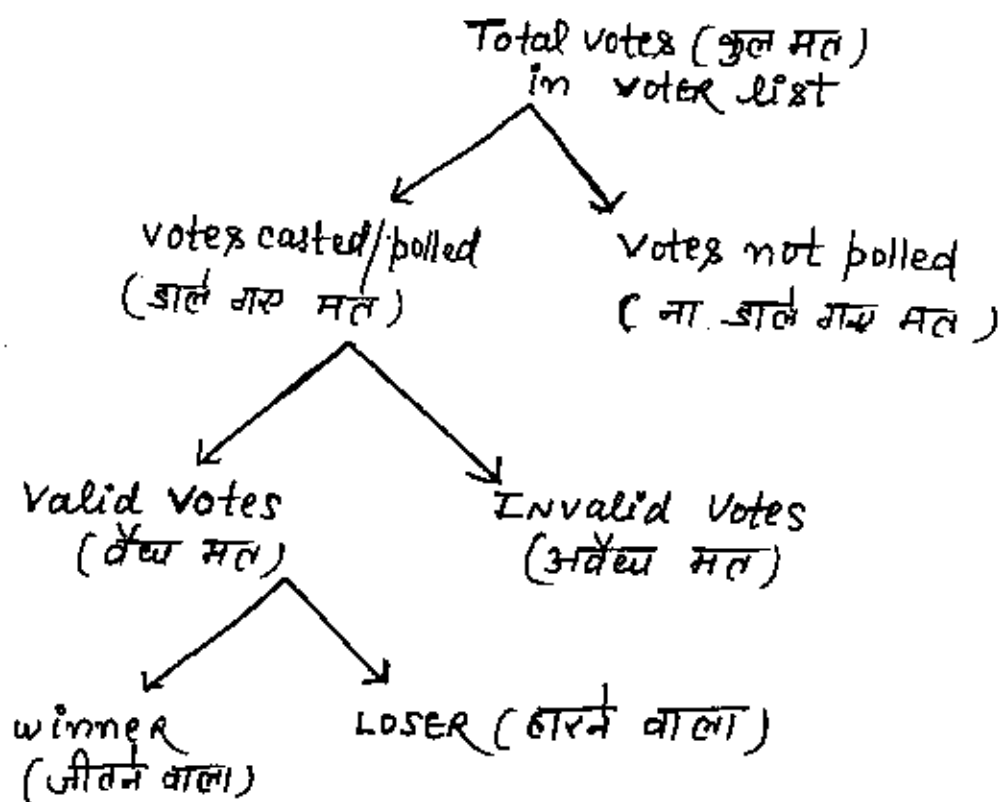
	PULP	:	WATER
Old	1×9	:	4×9
New	9	:	1

$$\begin{array}{rcl} \text{Dry } 10(9+1) & = & 500 \\ & 1 \rightarrow & 50 \text{ kg} \end{array}$$

$$\text{FRESH} = 9 + 36 = 45$$

$$\text{wt} = 45 \times 50 = 2250 \text{ kg}$$

ELECTIONS RELATED QUESTIONS



Note:- जो counting होगी व सिर्फ valid votes की होगी OR we can say that difference of winners and losers votes is of valid votes only (winner + LOSER votes = valid votes)

Note : अगर 1 के बोल Total votes की % में given है तो Total votes को 100 मान लो और अगर valid votes की % में given है तो valid votes को 100 मानने में फायदा होगा।

In an election between two candidates, the candidate getting 60% of votes polled is elected by a majority of 14000 votes. The No. of votes get by winner 1) 28000 2) 32000
3) 42000 4) 46,000

Solution :- winner LOSER
60% 100-60 = 40%
 $60\% - 40\% = 14000 \Rightarrow 20\% \rightarrow 14000$
 $\therefore 60\% \rightarrow \frac{14000}{20} \times 60$
 $= 42000 \text{ Ans}$

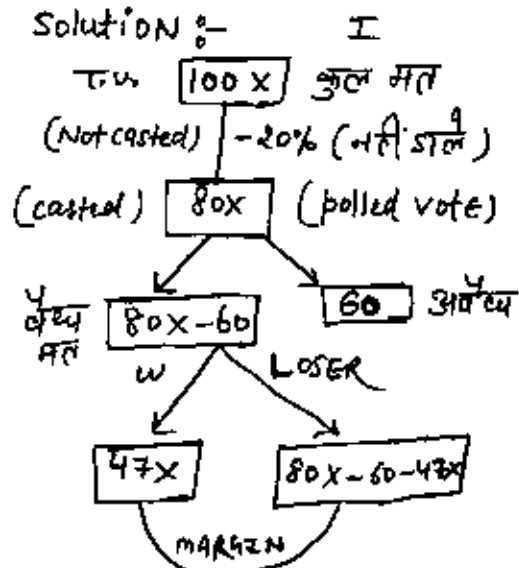
एक चुनाव में दो उम्मीदवार चुनाव लड़ रहे थे। जीतने वाले उम्मीदवार ने कुल मतों का 55% प्राप्त हुआ और वह 500 मतों से चुनाव जीत गया। तो इस चुनाव में कुल कितने वोट डाले गए।

Solution :- W L
55% 45%
(margin) 10% $\rightarrow 500$
 $\Rightarrow 100\% \rightarrow 5000 \text{ Ans.}$

In an election b/w BJP & CONGRESS, BJP SCORED 22% of total votes more than Cong. if Congress get 11700 votes and there is no invalid vote then find the total No. of votes. [30000 Ans.]

In an election there are two candidates. 20% of voters didn't cast their votes. And 60 votes were declared invalid. If winner gets 47% of total votes and win by 200 votes then find the total no. of votes?

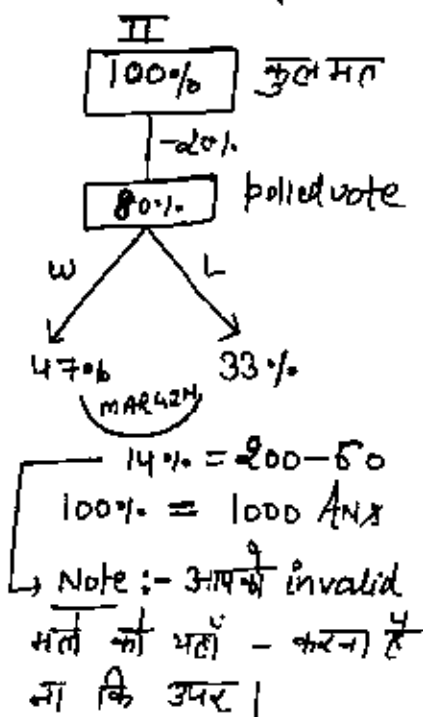
Solution :-



$$47x - 80x + 60 + 47x = 200$$

$$\Rightarrow 14x = 200 - 60 = 140$$

$$\Rightarrow x = 10 \quad \text{T.V.} = 100x = 1000 \text{ Ans.}$$

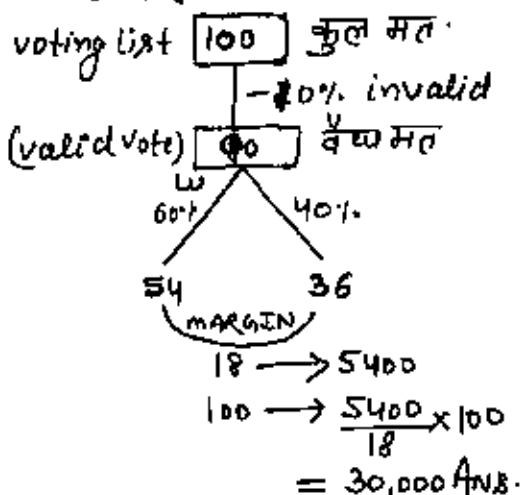


एक चुनाव में दो उम्मीदवार हैं। 20% मतदाताओं ने अपने मत का प्रयोग नहीं किया। और 200 मत अवैध पाये गये। जीतने वाले प्रत्याशी को कुल मतों का 47% प्राप्त हुआ और वह 600 मतों से चुनाव जीत गया तो मतदाता सूची में कितने मतदाता थे?

[40000 Ans.]

IN an election two candidate participated. 10% votes declared Invalid and the winner gets 60% of the valid votes and win by 5400 votes. find the total No. of votes in voting list?

Solution:- I



II

Let Total votes X

$$\therefore X \times \frac{9}{10} = \text{valid votes}$$

$$W = 60\% \text{ of valid votes}$$

$$\Rightarrow L = 40\% \text{ of valid votes}$$

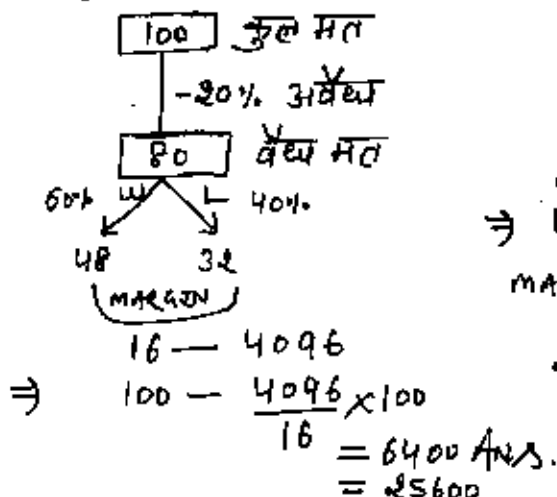
$$\text{margin} = 20\% = \frac{1}{5} \text{ of valid votes}$$

$$\therefore \boxed{X \times \frac{9}{10} \times \frac{1}{5} = 5400}$$

$$\Rightarrow X = 30,000 \text{ Ans.}$$

एक चुनाव में दो प्रतियोगी चुनाव लड़ रहे हैं। उसमें 20% मतों को अवैध घोषित किया गया। विजेता को कुल वैध मतों का 60% मत प्राप्त हुए और वह 4096 मतों से चुनाव जीत गया तो उस चुनाव में कुल कितने मत जल गए।

Solution:- I



II

Let total votes = (X)

$$\therefore X \times \frac{80}{100} = X \times \frac{4}{5} = \text{वैध मत}$$

$$\text{विजेता } 60\% \text{ of वैध मत}$$

$$\Rightarrow \text{LOSER} = 40\% \text{ of वैध मत}$$

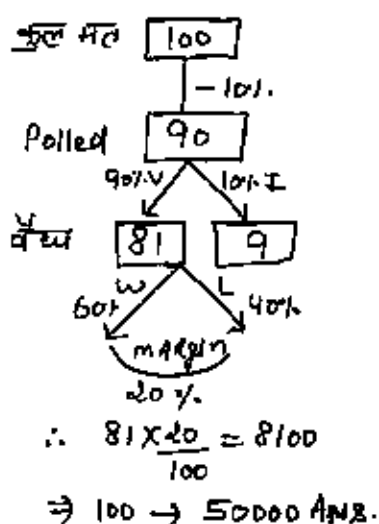
$$\text{margin} = 20\% \text{ of वैध मत}$$

$$\therefore X \times \frac{4}{5} \times \frac{1}{5} = 4096$$

$$X = 25600 \text{ Ans.}$$

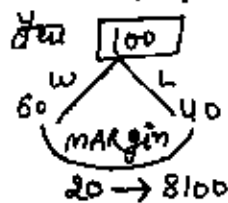
In an election two candidate participated. 10% voters did not vote, out of which 10% votes declared invalid and the winner got 60% of valid votes and win by 800 votes. Then find the total no of votes in voter list?

Solution :- I



II

Let valid vote = 100



$$\begin{aligned} \text{T.V.} &= 40500 \times \frac{10}{9} \times \frac{10}{9} \\ &= 50,000 \text{ Ans.} \end{aligned}$$

III

Let T.V. = X

$$\begin{aligned} \text{vote polled} &= X \times \frac{9}{10} \\ \text{valid votes} &= X \times \frac{9}{10} \times \frac{9}{10} \\ \text{margin} &= 20\% = \frac{1}{5} \\ \therefore X \times \frac{9}{10} \times \frac{9}{10} \times \frac{1}{5} &= 8100 \\ X &= 50,000 \text{ Ans.} \end{aligned}$$

एक चुनाव में दो प्रत्याशी थे। जिसमें 20% मतदाताओं ने अपने मत का प्रयोग नहीं किया। और 10% मत अवैध पाए गए। अगर विजेता को वैध मतों का 60% मिले तो वह 2160 मतों से चुनाव जीत जाता है। तो मतदाता सूची में कुल कितने मतदाता थे?

Solution : Let कुल मत = X

$$\text{poll कुल वोट} = X \times \frac{80}{100} = X \times \frac{4}{5}$$

$$\text{वैध मत} = X \times \frac{4}{5} \times \frac{9}{10}$$

$$\text{margin} = 40\% - 30\% = 10\% \text{ of वैध मत}$$

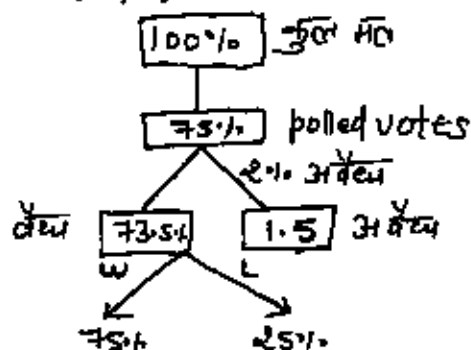
$$\therefore X \times \frac{4}{5} \times \frac{9}{10} \times \frac{1}{5} = 2160 \Rightarrow X = 7500 \text{ Ans.}$$

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In an election between two candidates 75% of the voters cast their vote, out of which 2% votes declared invalid. A candidate got 9261 votes which were 75% of valid votes. The total No. of voters in voting list was 1) 16000 2) 16400 3) 16800 4) 18000

Solution:- I



$$\therefore 75\% \text{ of } 73.5\% = 9261$$

$$\Rightarrow 100\% = 16800 \text{ Ans.}$$

II

$$\text{माना कुल मत} = X$$

$$\text{Polled votes} = X \times \frac{75}{100} = X \times \frac{3}{4}$$

$$\text{वैध मत (valid)} = X \times \frac{3}{4} \times \frac{98}{100}$$

winner get 75% of valid vote

$$\therefore \left[X \times \frac{3}{4} \times \frac{98}{100} \right] \times \frac{75}{100} = 9261$$

$$\Rightarrow X = 16800 \text{ Ans.}$$

एक चुनाव में बंसीलाल और देवीलाल ने भाग लिया। 37.5% मतदाताओं ने बंसीलाल को वोट करने का वादा किया और शेष ने देवीलाल को वादा किया। वोट वाले दिन 20% बंसीलाल ने वोट नहीं किया और 25% देवीलाल ने वोट नहीं किया अपना वादा वास्तव में (मुकदमा)। कुल मिलने वोट की ऊपर देवीलाल 8400 वोट से चुनाव जीत गया?

Solution:-

$$37.5\% = \frac{3}{8}$$

माना Total वोट 800

$$\Rightarrow \text{देवीलाल} \rightarrow 500$$

$$\text{बंसीलाल} \rightarrow 300$$

बंसीलाल देवीलाल

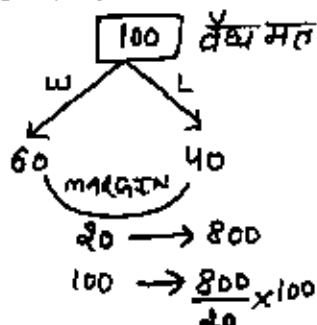
300	500
-60 (20%)	-125
240	375
+ 25	+60
365	435

$$\rightarrow 800 \rightarrow 96000 \text{ Ans}$$

किसी चुनाव में दो उम्मीदवार हैं। 10% मतदाताओं ने वोट नहीं डाले और 500 वोट अवैध घोषित कर दिए जाते हैं। जीतने वाला उम्मीदवार वैध मतों का 60% प्राप्त करता है और 800 वोटों से चुनाव जीत जाता है। पंजीकृत मतदाताओं की संख्या ज्ञात करें।

Solution :- I

Let valid vote = 100



वैध मत = 4000

Votes Polled = वैध + अवैध
= 4000 + 500 = 4500

Let $x = T.V. =$ कुल मत

$$\therefore x \times \frac{90}{100} = 4500 \Rightarrow x = 5000 \text{ Ans.}$$

II

Let कुल मत = x

$$\text{वोट पड़े} = x \times \frac{90}{100} = x \times \frac{9}{10}$$

$$\text{वैध मत} = x \times \frac{9}{100} - 500$$

$$\text{margin} = 60\% - 40\% = 20\% \\ = \frac{1}{5} \text{ of वैध मत}$$

$$\therefore \left[x \times \frac{9}{100} - 500 \right] \times \frac{1}{5} = 800$$

$$\Rightarrow x = 5000 \text{ Ans.}$$

In an election two candidate participated. 20% voters did not cast their votes out of which 1600 votes declared invalid and the winner get 62.5% of valid votes and wins by 2000 votes. Find the number of votes in voting list?

Solution :- let Total votes = x

$$\therefore \left[x \times \frac{4}{5} - 1600 \right] \times \frac{1}{4} = 2000$$

$$\Rightarrow x = 12000 \text{ Ans.}$$

$$\begin{array}{r} \textcircled{62.5} \quad \textcircled{37.5} \\ \hline 25\% = \frac{1}{4} \\ \text{margin} \end{array}$$



एक आदमी अपनी आय का 75% खर्च करता है। यदि उसकी आय में 20% की और खर्च में 10% की वृद्धि हो जाए तो उसकी बचत में कितने प्रतिशत की वृद्धि होगी?

Solution:- I (BASIC)

$$\text{Income} - \text{Exp} = \text{Saving}$$

$$100 - 75 = 25$$

$$\left(\begin{array}{cc} +20\% \uparrow & (10\% \uparrow \times) \end{array} \right)$$

$$120 - 82.5 = 37.5$$

$$X\% = \frac{17.5}{25} \times 100 = 50\%$$

II (Alligation)

खर्च : बचत

$$75\% : 25\% = 3:1$$

$$\begin{array}{cc} E & S \\ 10\% & X\% \end{array}$$

$$I(20\%)$$

$$3 : 1$$

$$\Rightarrow \frac{x-20}{20-10} = \frac{3}{1}$$

$$\Rightarrow x = 50\%$$

एक व्यक्ति अपनी आय का 20% बिराह पर शेष का 30% खाने पर और उसके शेष का 50% शिक्षा पर खर्च करता है। अन्त में उसकी बचत 6300/- है। तो उसकी आय क्या थी?

Solution:-

$$6300 \times \frac{100}{80} \times \frac{100}{70} \times \frac{100}{50} = 42500 \text{ ANS.}$$

कोई बिज़नेस कूल बिज़ी पर 12% कमिशन देता है। तथा 15000 रु से ऊपर की बिज़ी पर 10% बोनस देता है। यदि बिज़नेस की कूल आय 7650/- रु है तो कूल मिलने के की बिज़ी हुई?

Solution:- I (BASIC)

माना कूल बिज़ी = x रु

$$\therefore x \times \frac{12}{100} + (x - 15000) \frac{1}{100} = 7650$$

$$\Rightarrow \frac{12x}{100} + \frac{x}{100} - 150 = 7650$$

$$\Rightarrow x = 60,000 \text{ ANS.}$$

II (TRICKY)

$$\begin{array}{r|l} 15000 & + \\ \hline 12\% & 13\% \\ \hline 1.1 & \end{array}$$

$$15000 \times 1\% = 150$$

$$7650 + 150 = 7800$$

$$\Rightarrow 13\% \times 60000 = 7800$$

$$100\% \times 60000 = 60,000$$

A student multiplied a Number by $\frac{3}{5}$ instead of $\frac{5}{3}$. What is the Percentage error in the calculation?

Solution: Let the Number = 15 (LCM of 3 & 5)

$$\text{Wrong Result} = 15 \times \frac{3}{5} = 9, \quad \text{Actual Result} = 25$$

$$\% \text{ ERROR} = \frac{25-9}{25} \times 100 = 64\% \text{ Ans.}$$

Two Number are less than a third Number by 30% and 37% respectively. The Percent by which second Number is less by first is?

Solution:- Let third No = 100

$$\Rightarrow \text{I No} = 70, \text{ II No} = 63$$

$$\text{Req. Result} = \frac{7}{70} \times 100 = 10\% \text{ Ans.}$$

If the Numerator of a fraction is $\uparrow$ by 20% and denominator is $\downarrow$ by 10% the resultant fraction becomes $\frac{36}{45}$. Find the original fraction?

Solution:- Let original fraction = $\frac{x}{y}$

$$\therefore \frac{x [1.20]}{y [.90]} = \frac{36}{45} \Rightarrow \frac{x}{y} = \frac{36}{45} \times \frac{9}{12} = \frac{3}{5} \text{ Ans.}$$

IF the Income Tax is increased by 19%, then Net income will be decreased by 6%. Find the Rate of income tax?

Solution:- [CONCEPT] Net Income = Income - Tax

$$\Rightarrow \text{Net Income} + \text{Tax} = \text{Income}$$

in these type of questions Income is same

Note:- If Tax is ↑ by $\bar{x}\%$, Net Income will be decreased by $\bar{x}\%$. or vice-versa but % may vary.

$$\text{e.g. } \left. \begin{array}{l} \text{NE} \quad \text{T} \quad \text{I} \\ 50 + 30 = 80 \\ 45 + 35 = 80 \end{array} \right\} \text{Same}$$

TRICKY:- $\downarrow$ Income is same $\downarrow$
19% of Tax = 6% of Net Income

$$\Rightarrow \frac{19}{6} = \frac{\text{Net Income}}{\text{Tax}} \Rightarrow \text{Income} = 19 + 6 = 25$$

$$\text{Tax \%} = \frac{\text{Tax}}{\text{Income}} \times 100 = \frac{6}{25} \times 100 = 24\% \quad \underline{\text{Ans.}}$$

IF income Tax is increased by 19%, Net income get decreased by 1%. Find the Tax Rate?

Solution:- 19% of Tax = 1% of Net Income

$$\Rightarrow \frac{\text{Tax}}{\text{Net Income}} = \frac{1}{19} \Rightarrow \text{Income} = 1 + 19 = 20$$

$$\text{Tax \%} = \frac{1}{20} \times 100 = 5\% \quad \underline{\text{Ans.}}$$

A candidate got 25% marks and fail by 30 marks while another candidate who scores 50% marks gets 20 marks more than minimum required marks to pass the Examination. Find the maximum and passing marks for the Exam?

Solution :- equate passing marks

I

II

$$25\% + 30 = 50\% - 20$$

$$\Rightarrow 25\% = 50 \quad \therefore \text{mm} (100\%) = 2 \times 100 = 200$$

$$1\% = 2 \quad \text{Passing marks} = 25\% + 30$$

$$\# \text{ किसी परीक्षा में } 60\% \text{ छात्र अंग्रेजी में } = 80 \text{ Ans}$$

में उत्तीर्ण होते हैं। 40% हिन्दी में और 40% दोनों विषयों में उत्तीर्ण (पास) होते हैं। तो दोनों में फेल होने वाले छात्रों का प्रतिशत क्या होगा?

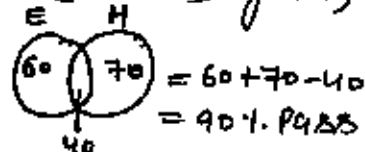
Solution :- I (Formula)

$$N(A \cup B) = N(A) + N(B) - N(A \cap B)$$

$$= 70 + 60 - 40 = 90$$

$$\text{Failed in both} = 100 - 90 = 10\%$$

II (Venn Diagram)



$$\text{Fail} = 100 - 90 = 10\%$$

किसी परीक्षा में 80% छात्र अंग्रेजी में और 85% गणित में उत्तीर्ण होते हैं। जबकि 75% दोनों में उत्तीर्ण होते हैं। यदि 45 छात्र इन दोनों विषयों में अनुत्तीर्ण हुए हो तो कुल छात्रों की संख्या ज्ञात करें?

Solution :-

$$E \rightarrow 80\%$$

$$H \rightarrow 85\%$$

$$\begin{array}{r} +165\% \\ -75\% \\ \hline 90\% \text{ — Pass} \end{array}$$

$$\text{Fail} = 100 - 90 = 10\%$$

$$10\% \rightarrow 45$$

$$100\% \rightarrow 450 \text{ Ans.}$$

रामकाण (जिसको माना जा नहीं आया)
जो बचगा वो आया

A total of ₹ 180 is to be divided among 100 students such that each girl student receives ₹ 2 and each boy student receives ₹ 1.5. Find the number of girl students?

Solution:- I

Let all are boys

money divided will be
 $= 100 \times 1.5$
 $= 150$

money Rem $\Rightarrow 180 - 150 = 30$

which was due to diff of 50

$$\# \text{ Girls} = \frac{30}{0.5} = 60 \text{ Ans.}$$

(Boys मान लें girls आया)

OR

II

माना सारी लड़कियां भी 100 की 100

$$\text{money} = 100 \times 2 = 200$$

$$\text{मिलना ज्यादा} = 200 - 180$$

क्योंकि लड़कियों को 0.5 रु कम मिल रहे थे।

$$\# \text{ लड़के} = \frac{20}{0.5} = 40$$

(Girls मान लें boys आया)

$$\Rightarrow \text{Girls} = 100 - 40 = 60$$

In a zoo, there are only pigeons and rabbits if there are total of 200 heads and 560 legs then find the number of pigeons and rabbits?

Solution:- Let all are pigeons (200) Let all are Rabbits (200)

$$\therefore \text{legs} = 200 \times 2 = 400$$

$$\text{extra} = 560 - 400 = 160$$

but extra 160 is due to 2 more legs of Rabbit

$$\# \text{ Rabbit} = \frac{160}{2} = 80 \text{ Ans}$$

(Pigeon माना लें Rabbit आया)

$$\# \text{ Pigeon} = 200 - 80 = 120 \text{ Ans.}$$

$$\text{OR } \therefore \text{legs} = 200 \times 4 = 800$$

$$\text{gap} = 800 - 560 = 240$$

240 ज्यादा coz pigeon has two less legs

$$\# \text{ pigeon} = \frac{240}{2} = 120$$

(Rabbit माना लें pigeon आया)

$$\# \text{ Rabbit} = 200 - 120 = 80$$

Each correct Answer fetches 4 marks and each wrong answer fetches a penalty of 1 mark. If a student attempts 48 questions and scores 132 marks. Then find the Number of correct and wrong answers?

Solution:- let he attempted all (48) correct

$$\text{Total marks} = 48 \times 4 = 192$$

but his actual score is 132. Difference of 60 is due to diff of 5 (4 - (-1)) b/w correct and wrong

$$\# \text{ wrong} = \frac{60}{5} = 12 \quad \left[\begin{array}{l} \text{Correct माना जा} \\ \text{wrong आया} \end{array} \right]$$

$$\# \text{ correct} = 48 - 12 = 36 \text{ Ans}$$

Population of a Town is 15000. If the Number of the males ↑ by 8% and that of females by 10%. Then the population would increase to 16300 after 1 yr. Find the Number of females?

Solution:- I

let all are male

$$\begin{aligned} \therefore \text{Population after 1yr} &= 15000 + \frac{15000 \times 8}{100} \\ &= 16200 \end{aligned}$$

$$\begin{aligned} \text{gap} &= 16300 - 16200 \\ &= 100 \text{ less} \end{aligned}$$

$$\# \text{ females} = \frac{100}{10\% - 8\%} \times 100 = 5000$$

(male माना जा female आया)

II

let all are females

$$\begin{aligned} \text{Population after 1yr} &= 15000 + \frac{15000 \times 10}{100} \\ &= 16500 \end{aligned}$$

$$\begin{aligned} \text{gap} &= 16500 - 16300 \\ &= 200 \text{ Extra} \end{aligned}$$

$$\begin{aligned} \# \text{ males} &= \frac{200}{2\%} \times 100 \\ &= 10,000 \end{aligned}$$

(female माना जा male आया)

$$\# \text{ female} = 15000 - \frac{10000}{2} = 5000$$

A man lent 2000 partly at 5% and balance at 4%. If he receives ₹ 92 as annual interest. Find the amount lent at 5%?

Solution:- Let whole amt is invested at 4%.

$$\text{SI after 1 yr} = \frac{2000 \times 4}{100} = 80 \text{ ₹}$$

$$\text{gap} = 92 - 80 = 12 \text{ ₹} \quad [\text{ coz Rate diff} = 5\% - 4\% = 1\%]$$

$$\text{amt lent at 5\%} = \frac{12}{1} \times 100 = 1200 \text{ Ans.}$$

[4% माना तो 5% वाला अंतरा।]

A man covers a total of 240 km in 6 hours partly at 30 km/hr and rest at 60 km/hr. What distance was covered at 30 km/hr.

Solution:- Note $\rightarrow$ Ratio of speed gives ratio of Time

Let he covered all distance at 60 km/hr

$$\therefore \text{in 6 hours } D = 60 \times 6 = 360 \text{ km}$$

$$\text{gap} = 360 - 240 = 120 \text{ km}$$

$$\text{Time for 30 km/hr journey} = \frac{120}{60-30} = 4 \text{ hr}$$

[60 का माना तो 30 का Time आया]

$$D = 4 \times 30 = 120 \text{ km Ans.}$$

PROFIT & LOSS

Profit, Loss & Discount
(लाभ, हानि और बट्टा)

Terminology :-

Cost Price (C.P.) क्रय मूल्य

The price at which article is bought or money that goes out of pocket.

Selling Price (S.P.) विक्रय मूल्य

The price at which article is sold or the amount of money that comes into pocket.

Note:- The Selling Price (S.P.) of the seller is the cost price of the buyer.

PROFIT (लाभ) if $SP > CP$

$$P = SP - CP = \text{वि० मू०} - \text{क्र० मू०}$$

LOSS (हानि) if $CP > SP$

$$L = CP - SP = \text{क्र० मू०} - \text{वि० मूल्य}$$

IF PROFIT on selling an article for ₹ 425 is same as loss on selling it for ₹ 355. Find the cost price of the Article?

Solution:- I (BASIC)

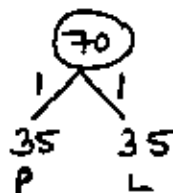
Let C.P. = X

$$\begin{array}{c} P \qquad \qquad L \\ \swarrow \qquad \searrow \\ 425 - X = X - 355 \end{array}$$

$$\Rightarrow 2X = 780 \Rightarrow X = 390 \text{ A.}$$

II (TRICKY)

$$\text{Find } SP_2 - SP_1 = 425 - 355 = 70$$



OR III $\frac{SP_1 + SP_2}{2} = \frac{425 + 355}{2}$
 $= 390 \text{ Ans.}$

$$\therefore CP = 425 - 35 = 390$$

$$\text{OR } CP = 355 + 35 = 390$$

किसी वस्तु को 1560 में बेचने पर जो लाभ होता है, वह उसी वस्तु को 760 में बेचने पर हुई हानि से 3 गुणा है। तो उस वस्तु का क्र. मू. क्या होगा?

Solution:- I (BASIC)

Let C.P. = X

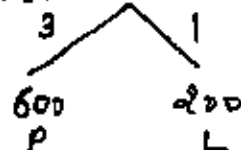
$$\begin{array}{c} P \qquad \qquad L \\ \swarrow \qquad \searrow \\ 1560 - X = 3(X - 760) \end{array}$$

$$\therefore 1560 - X = 3(X - 760)$$

$$\Rightarrow 4X = 3440 \Rightarrow X = 860$$

Ans.

II $1560 - 760 = 800$



$$CP = 1560 - 600 = 960$$

$$\text{OR } CP = 760 + 200 = 960 \text{ Ans.}$$

IF the Profit on selling an article for ₹ 480 is 25% more than loss on the selling it for ₹ 300. Find the cost price of article?

Solution:- P : L

$$125 : 100$$

$$5 : 4$$

$$SP_2 - SP_1 = 480 - 300$$

$$= 180$$

$$\begin{array}{c} 5 \quad 4 \\ \swarrow \quad \searrow \\ P \quad 100 \quad 80 \quad L \end{array}$$

$$\Rightarrow CP = 480 - 100 = 380$$

$$\text{OR } CP = 300 + 80 = 380 \text{ Ans.}$$



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Profit and Loss is always calculated/Reckoned on Cost Price unless otherwise stated.

लाभ और हानि प्रतिगत हमेशा क्रम मूल्य (C.P.) पर ज्ञात किए जाते हैं। परन्तु अगर Examiner बता दें कि आपकी वि० मू० पर निचालना है तो आप वि० मू० पर ही निकालें।

$$\# \text{ PROFIT \%} = \frac{\text{PROFIT}}{\text{COST PRICE}} \times 100 = \frac{P}{CP} \times 100$$

$$\text{LOSS \%} = \frac{\text{LOSS}}{\text{COST PRICE}} \times 100 = \frac{L}{CP} \times 100$$

e.g. $CP = 800$ $SP = 1000$

$$\therefore P\% = \frac{200}{800} \times 100 = 25\%$$

$CP = 800, SP = 400$

$$\therefore L\% = \frac{400}{800} \times 100 = 50\% \text{ LOSS}$$

$\text{PROFIT (P\%)}$

$$SP = CP \times \frac{(100 + P\%)}{100}$$

$$\text{OR } CP = \frac{SP \times 100}{(100 + P\%)}$$

$\text{LOSS (L\%)}$

$$SP = CP \times \frac{(100 - L\%)}{100}$$

$$\text{OR } CP = \frac{SP \times 100}{(100 - L\%)}$$

METHOD 1 (100% वाला तरीका)

आपकी फ़ॉर्मूला (CP) को हमेशा 100% लेना है।

e.g. let $P\% = 20\%$ $\rightarrow$ $CP = 100\%$.

$SP = 120\%$.

$L\% = 20\%$ $\rightarrow$ $CP = 100\%$.

$SP = 80\%$.

$CP = 600$, $P\% = 20\%$, $SP = ?$

$100\% \rightarrow 600$

$1\% \rightarrow \frac{600}{100}$

$SP = 120\% \rightarrow \frac{600}{100} \times 120 = 720$ ANS.

$SP = 5120$, $L\% = 20\%$, $CP = ?$

$L\% = 20\%$ $\rightarrow$ $CP = 100\%$.

$SP = 80\%$ $\rightarrow$ 5120

$1\% \rightarrow \frac{5120}{80}$

$CP = 100\% \rightarrow \frac{5120}{80} \times 100$
6400 ANS.

$CP = ?$, $P\% = 25\%$, $SP = 6250$

$P\% = 25\%$ $\rightarrow$ $CP = 100\%$.

$SP = 125\%$ $\rightarrow$ 6250

CP $100\% \rightarrow \frac{6250}{125} \times 100$
5000

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METHOD 2 (Ratio वाला तरीका)

इसमें सबसे पहले आपको P% या L% को fraction में convert करना है।

$$\text{e.g. } P\% = 20\% = \frac{1 \rightarrow P}{5 \rightarrow CP} \Rightarrow SP \rightarrow S+1=6$$

$$L\% = 20\% = \frac{1 \rightarrow L}{5 \rightarrow CP} \Rightarrow SP = 5-1=4$$

$$P\% = 12.5\% = \frac{1 \rightarrow P}{8 \rightarrow CP} \Rightarrow SP = 8+1=9$$

$$L\% = 16\frac{2}{3}\% = \frac{1 \rightarrow L}{6 \rightarrow CP} \Rightarrow SP = 6-1=5$$

$$\# P\% = 14\frac{2}{7}\%, CP = 490 \quad \# L\% = 10\% \quad SP = 7290$$

$$SP = ?$$

$$14\frac{2}{7}\% = \frac{1 \rightarrow P}{7 \rightarrow CP} \Rightarrow SP = 8$$

$$7 \rightarrow 490$$

$$1 \rightarrow 70$$

$$8(SP) \rightarrow 70 \times 8 = 560 \text{ Ans.}$$

$$CP = ?$$

$$10\% = \frac{1 \rightarrow L}{10 \rightarrow CP}$$

$$\Rightarrow SP \rightarrow 10-1=9$$

$$9 \rightarrow 7290$$

$$10 \rightarrow \frac{7290 \times 10}{9}$$

$$= 8100 \text{ Ans.}$$

$$\# P\% = 40\%, CP = 3000$$

$$SP = ?$$

$$40\% = \frac{2 \rightarrow P}{5 \rightarrow CP} \Rightarrow SP = 5+2=7$$

$$5 \rightarrow 3000$$

$$7 \rightarrow \frac{3000 \times 7}{5} = 4200 \text{ Ans.}$$

$$\# L\% = 8.33\%$$

$$SP = 1210, CP = ?$$

$$8.33\% = \frac{1 \rightarrow L}{12 \rightarrow CP}$$

$$SP \rightarrow 12-1=11$$

$$11 \rightarrow 1210$$

$$12 \rightarrow \frac{1210 \times 12}{11} = 1320 \text{ Ans.}$$

IF Ratio of cost price and selling price is 8:9. Find Profit/Loss %?

Solution:- $\frac{CP}{SP} = \frac{8}{9} \Rightarrow \text{Profit} = 1 \Rightarrow P\% = \frac{1}{8} \times 100 = 12.5\%$

IF SP is $\frac{5}{4}$ of CP. Find Profit %?

Solution:- $SP : CP = \frac{5}{4} : 1 \Rightarrow P\% = \frac{1}{4} \times 100 = 25\% \text{ Ans.}$
OR $5 : 4$

IF Profit is calculated on SP then $P\% = 20\%$. what is original $P\%$ (on CP)?

Solution:- $20\% = \frac{1 \rightarrow P}{5 \rightarrow SP} \Rightarrow CP = 4$

Actual $P\%$ on CP = $\frac{1}{4} \times 100 = 25\% \text{ Ans.}$

IF Loss % is calculated on SP, then $L\% = 10\%$. what is original Loss %? (on CP)

Solution:- $L\% = 10\% = \frac{1 \rightarrow L}{10 \rightarrow SP} \Rightarrow CP = 10 + 1 = 11$

Actual $L\% = \frac{1}{11} \times 100 = 9.09\% \text{ Ans.}$

$P\%$ on CP = 12.5% then $P\%$ on SP?

$12.5\% = \frac{1 \rightarrow P}{8 \rightarrow CP} \Rightarrow SP \rightarrow 9$

$P\%$ on SP = $\frac{1}{9} \times 100 = 11.11\% \text{ Ans.}$

By selling a mobile for ₹ 1045 a man loses 5%.
At what price should he sell the mobile to gain 5%?

Solution:- I (Basic)

$$CP = 100\% \text{ always}$$

$$SP_1 = 95\% [5\% \text{ Loss}]$$

$$SP_2 = 105\% [5\% \text{ gain}]$$

$$95\% \longrightarrow 1045$$

$$\Rightarrow 105\% \longrightarrow \frac{1045}{95} \times 105 = 1155$$

II (Tricky)

$$5\% \text{ Loss} = \frac{1 - L}{100} \times CP$$

$$SP_1 = 95$$

$$5\% \text{ gain} = \frac{1 - P}{100} \times CP$$

$$(SP_1) \quad SP_2 = 115$$

$$\begin{array}{r} 19 \\ 19 \times 55 \\ \hline 1045 \end{array}$$

$$\begin{array}{r} 115 \\ 115 \times 55 \\ \hline 1155 \end{array}$$

एक आदमी कोशक बेंच को 1920 में खरीदने पर 20% की हानि हुई। यदि वह उसे 2520 ₹ में बेचता तो कितने % की लाभ या हानि होती?

Solution:- I (BASIC)

$$1920 \longrightarrow 80\% [20\% L]$$

$$1 \longrightarrow \frac{80\%}{1920}$$

$$2520 \longrightarrow \frac{80\%}{1920} \times 2520$$

$$= 105\%$$

105% means 5% profit

II (TRICKY)

$$\frac{SP_1}{100 \pm P/L} = \frac{SP_2}{100 \pm P/L}$$

$$\frac{192}{80} = \frac{252}{100 \pm P/L}$$

$$\Rightarrow 100 \pm P/L = 105$$

$$\Rightarrow \pm P/L = 5$$

$$\Rightarrow P\% = 5\%$$

An Article is sold at 8% loss. Had the shopkeeper sold it for ₹ 650 more he would have gain 5%. Find the cost price?

Solution :- I (Basic)

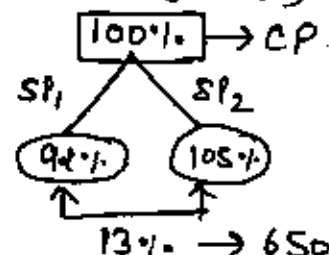
Let CP = ₹ X

$$\begin{array}{cc} SP_1 & SP_2 \\ X[.92] + 650 = X[1.05] \end{array}$$

$$\Rightarrow X[.13] = 650$$

$$\Rightarrow X = 5000 \text{ Ans.}$$

II (TRICKY)



$$\therefore 100\% \rightarrow 5000 \text{ Ans}$$

एक दुकानदार ने एक मोबाइल 5% हानि पर बेचा। अगर वो पहले से इसको ₹ 3375 ज्यादा में बेचता तो उसको अब 10% का लाभ होता। वह उस मोबाइल को कितने ₹ में बेच कि अब उसको 20% का लाभ हो?

Solution :- I (Basic)

माना CP = ₹ X

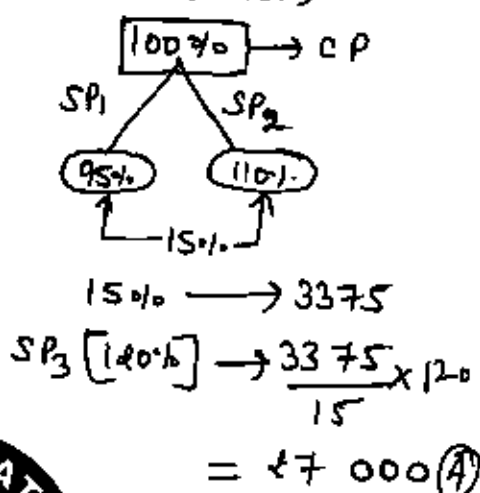
$$\begin{array}{cc} SP_1 & SP_2 \\ \downarrow & \downarrow \\ X[.95] + 3375 = X[1.10] \end{array}$$

$$\Rightarrow X[.15] = 3375$$

$$\Rightarrow X = 22500 \rightarrow CP$$

$$\begin{aligned} SP_3 &= 22500[1.20] \\ &= 27000 \text{ ₹ Ans.} \end{aligned}$$

II (TRICKY)



A shopkeeper sells an article at 12% Profit. Had he bought it at a price 10% Less than earlier and sold for ₹ 250 more. He would have gain 30%. Find the cost price of Article?

Solution:- I (BASIC)

Let CP = ₹ X

$$\begin{array}{ccc} SP_1 & & SP_2 \\ \downarrow & & \downarrow \\ X[1.12] + 250 & = & X[.90][1.30] \end{array}$$

$$\Rightarrow X[1.12] + 250 = X[1.17]$$

$$\Rightarrow X[.05] = 250 \Rightarrow X = 5000$$

II (TRICKY)

$$\begin{array}{ccc} \text{old} & & \text{NEW} \\ \text{let CP} = 100 & & 90 \\ \downarrow 12\% P & & \downarrow 30\% P \\ SP \rightarrow 112 & & 117 \\ & \nearrow +5 & \\ & 5 \rightarrow 250 & \\ \therefore CP(100) \rightarrow 5000 \text{ Ans.} & & \end{array}$$

एक दुकानदार एक वस्तु को 20% लाभ पर बेचता है। अगर वह उस वस्तु को 20% पहले से कम पर खरीदे और ₹ 75 कम पर बेचे तो उसे अब 25% का लाभ होगा है। तो उस वस्तु का मूल्य बताओ?

Solution:- I (BASIC)

माना CP = ₹ X

$$\begin{array}{ccc} SP_1 & & SP_2 \\ \downarrow & & \downarrow \\ X[1.20] - 75 & = & X[.80][1.25] \end{array}$$

$$\Rightarrow X[1.20] - 75 = X[1]$$

$$\Rightarrow X[.20] = 75$$

$$X = ₹ 375 \text{ Ans.}$$

II (TRICKY)

$$\begin{array}{ccc} \text{old} & -20\% & \text{NEW} \\ CP = 100 & & 80 \\ \downarrow 20\% P & & \downarrow 25\% P \\ SP = 120 & & 100 \\ & \nearrow -20 & \\ & 20 \rightarrow 75 & \\ CP = 100 \rightarrow 75 \times 5 & & \\ & & 375 ₹ \text{ Ans} \end{array}$$

A shopkeeper sells his goods at 25% Profit. Had he purchased it for ₹ 900 less and sold for ₹ 900 less he would have gain 5% more. Find the original cost price?

Solution :- I (BASIC) II

$$125\% - 900 = \frac{100\% - 900}{\text{New SP}} (1.30)$$

$$1250\% - 9000 = 1300\% - 11700$$

$$\Rightarrow 50\% = 2700$$

$$\therefore 100\% = \text{CP} = 5400 \text{ Ans.}$$

III (TRICKY)

$$\text{TRICK} = \frac{\text{New Profit}}{P_2 - P_1} \times \text{Gap}$$

$$= \frac{30\%}{5\%} \times 900$$

$$= 5400 \text{ Ans.}$$

एक दुकानदार अपनी वस्तु को 20% लाभ पर बेचता है। यदि वह 5% मुल्य और विक्रम मुल्य दोनों को 100 रु कम कर दे तो उसका लाभ 4% बढ़ जाता है। तो वस्तु का मूल मूल्य क्या होगा?

Solution :- I (BASIC)

$$20\% P = \frac{1-P}{5-CP} \quad SP=6$$

$$CP_1 : SP_1 = 5 : 6$$

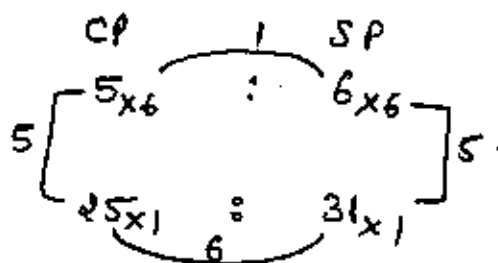
$$24\% = \frac{6-P}{25-CP} \quad SP \rightarrow 31$$

$$CP_2 : SP_2 = 25 : 31$$

$$\therefore \frac{5x-100}{6x-100} = \frac{25}{31}$$

$$\Rightarrow 5x = 600 \text{ Ans.}$$

II TRICKY



$$54 \text{ unit} \rightarrow 100$$

$$30 \text{ unit CP} \rightarrow 600 \text{ Ans}$$

A shopkeeper sells his goods at a profit of 20%. If he reduces both CP and SP by ₹10 and ₹5 respectively then the profit % ↑ by 10%. Find the cost price of article?

Solution:-

$$\text{I} \quad 20\% P = \frac{1 - P}{5 - CP} \rightarrow SP - 6$$

$$\text{let } CP = 5x, SP = 6x$$

$$\text{Now } 30\% P = \frac{3 - P}{10 - CP} \rightarrow SP - 13$$

$$\therefore \frac{5x - 10}{6x - 5} = \frac{10}{13}$$

$$\text{OR } 65x - 130 = 60x - 50$$

$$\Rightarrow 5x = 80 = CP \text{ Ans}$$

II

$$(120\% - 5) = (100\% - 10)(130)$$

$$\text{OR } 1200\% - 50 = 1300\% - 130$$

$$\Rightarrow 100\% = 80 \text{ Ans.}$$

एक दुकानदार एक वस्तु को 25% के लाभ पर बेचता है। यदि वह अपने क्र. मूल्य और बि. मूल्य क्रमशः 20 ₹ और 5 ₹ बढ़ा दे तो उसका लाभ 15% बढ़ जाता है। तो वस्तु का क्रय मूल्य ज्ञात करो?

Solution:-

$$(125\% + 4) = (100\% + 20) \times \frac{110}{100}$$

$$\Rightarrow 1250\% + 40 = 1100\% + 220$$

$$150\% = 180 \quad \therefore 100\% = CP = \frac{180}{150} \times 100$$

$$= 120 \text{ ₹ Ans.}$$

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ARTICLE BASED QUESTIONS

NOTE:- In this type of question आपको Articles (वस्तुओं) को बराबर रखना होगा।
eg if you have 20 articles CP then you must have 20 article SP

Articles were bought at 6 for ₹ 5 and sold for 5 for ₹ 6. Gain/loss percent is ?

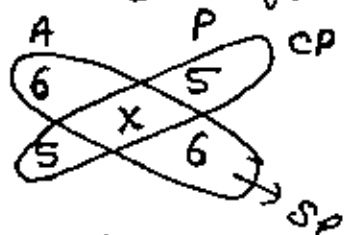
Solution:- I (Basic)
MAKE the Article equal
at Lcm of 6, 5 = 30

Buy 5×6 $5 \times 5 = 25$ CP

Sold 6×5 $6 \times 6 = 36$ SP

$$\text{gain \%} = \frac{11}{25} \times 100 = 44\% \text{ Ans.}$$

II (Tricky)



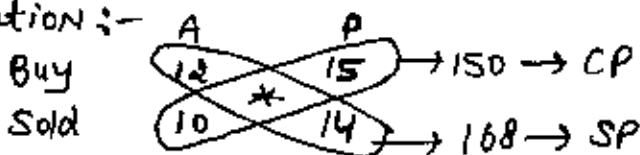
$$CP = 25$$

$$SP = 36$$

$$P\% = \frac{11}{25} \times 100 = 44\%$$

एक दुकानदार कुछ वस्तुओं को 15 रु में 12 वस्तुओं में बर सँ खरीदता है और सनी वस्तुओं को 14 रु में 10 वस्तुओं के हिसाब सँ बच देता है। तो कितने प्रतिशत लाभ या हानि हुई ?

Solution:-



$$\Rightarrow P\% = \frac{18}{150} \times 100$$

$$= 12\% \text{ Ans.}$$

IF COST PRICE OF 18 articles is equal to selling price of 15 articles. The gain % is?

Solution :- I

$$\text{Let CP of 18} = \text{SP of 15} = 90$$

$$\Rightarrow \text{CP of 1 article} = 5$$

$$\Rightarrow \text{SP of 1 article} = 6$$

$$\text{gain \%} = \frac{1}{5} \times 100 = 20\%$$

II

$$\text{CP of 18} = \text{SP of 15}$$

$$\Rightarrow \frac{\text{CP}}{\text{SP}} = \frac{15}{18} = \frac{5}{6}$$

$$\Rightarrow \text{gain \%} = \frac{1}{5} \times 100 = 20\%$$

अगर 9 वस्तुओं का क्रय मूल्य 12 वस्तुओं के बिक्री मूल्य के बराबर हो तो लाभ/हानि % बताइए

Solution :- I

$$\text{माना CP of 9} = \text{SP of 12} = 36$$

$$\text{CP of 1 article} = 4$$

$$\text{SP of 1 article} = 3$$

$$\text{loss \%} = \frac{1}{4} \times 100 = 25\%$$

II

$$\text{CP of 9} = \text{SP of 12}$$

$$\Rightarrow \frac{\text{CP}}{\text{SP}} = \frac{12}{9} = \frac{4}{3}$$

$$\Rightarrow \text{loss \%} = \frac{1}{4} \times 100 = 25\%$$

$$\# \text{ III RD METHOD} = \frac{\# \text{ Good LEFT}}{\# \text{ Goods sold}} \times 100$$

$$\text{CP of 18} = \text{SP of 15}$$

$$\begin{aligned} \% \text{ Profit} &= \frac{3}{15} \times 100 \\ &= 20\% \text{ Ans.} \end{aligned}$$

$$\text{CP of 9} = \text{SP of 12}$$

$$\begin{aligned} \% \text{ Loss} &= \frac{3}{12} \times 100 \\ &= 25\% \text{ Ans.} \end{aligned}$$

इस type के questions में आपको focus करना है कि Profit/Loss SP या CP किसके रूप में given है। अगर Profit/Loss SP के रूप में given है तो 1 article का SP = 1 ₹ मान लें और अगर Profit/Loss CP के रूप में given है तो 1 article का CP = 1 ₹ मान लें।

e.g. Profit = SP of 11 articles $\Rightarrow$ let SP of 1 article = 1 ₹
 Loss = CP of 11 articles $\Rightarrow$ let CP of 1 = 1 ₹

By selling 33m of cloth a shopkeeper gains CP of 11m. find gain %?

Solution :- I

$$\text{gain} = \text{SP} - \text{CP}$$

$$\therefore \text{SP of 33m} - \text{CP of 33m} = \text{CP of 11m}$$

$$\Rightarrow \text{SP of 33m} = \text{CP of 44m}$$

$$\Rightarrow \frac{\text{CP}}{\text{SP}} = \frac{33}{44}$$

$$\Rightarrow \text{gain \%} = \frac{11}{33} \times 100 = 33\frac{1}{3}\%$$

II

$$\text{let CP of 1m} = 1 ₹$$

$$\Rightarrow \text{CP of 33m} = 33 ₹$$

$$\text{gain} = \text{CP of 11m} = 11 ₹$$

$$\text{gain \%} = \frac{11}{33} \times 100$$

$$= 33.33\% \text{ Ans}$$

By selling 33m of cloth a shopkeeper gains SP of 11m. find gain %?

Solution :- I

$$\text{SP of 33m} - \text{CP of 33m}$$

$$= \text{SP of 11m}$$

$$\Rightarrow \text{SP of 22m} = \text{CP of 33m}$$

$$\Rightarrow \frac{\text{CP}}{\text{SP}} = \frac{22}{33} \Rightarrow \text{gain \%} = \frac{11}{22} \times 100$$

$$= 50\%$$

II

$$\text{let SP of 1m} = 1 ₹$$

$$\Rightarrow \text{SP of 33m} = 33 ₹$$

$$\text{gain} = \text{SP of 11m} = 11 ₹$$

$$\text{CP of 11m} = 33 - 11 = 22$$

$$\text{gain \%} = \frac{11}{22} \times 100$$

$$= 50\%$$

By selling 33m of cloth a shopkeeper loses CP of 11m. find loss%?

Solution:- I

$$\text{Loss} = \text{CP} - \text{SP}$$

$$\text{CP of 33m} - \text{SP of 33m} \\ = \text{CP of 11m}$$

$$\Rightarrow \text{CP of 22m} = \text{SP of 33m}$$

$$\Rightarrow \frac{\text{CP}}{\text{SP}} = \frac{33}{22}$$

$$\text{loss \%} = \frac{11}{33} \times 100 = 33\frac{1}{3}\%$$

II

$$\text{let CP of 1m} = 1 \text{ ₹}$$

$$\text{CP of 33m} = 33 \text{ ₹}$$

$$\text{loss} = \text{CP of 11m} = 11 \text{ ₹}$$

$$\text{loss \%} = \frac{11}{33} \times 100 \\ = 33\frac{1}{3}\% \text{ Ans}$$

By selling 33m of cloth a shopkeeper loses SP of 11m. find loss%?

Solution:- I

$$\text{CP of 33m} - \text{SP of 33m} \\ = \text{SP of 11m}$$

$$\Rightarrow \text{CP of 33m} = \text{SP of 44m}$$

$$\Rightarrow \frac{\text{CP}}{\text{SP}} = \frac{44}{33}$$

$$\text{loss \%} = \frac{11}{44} \times 100 = 25\%$$

II

$$\text{let SP of 1m} = 1 \text{ ₹}$$

$$\text{SP of 33m} = 33 \text{ ₹}$$

$$\text{loss} = \text{SP of 11m} = 11 \text{ ₹}$$

$$\Rightarrow \text{CP of 33m} = 33 + 11 \\ = 44$$

$$\therefore \text{loss \%} = \frac{11}{44} \times 100 = 25\%$$

36 संतरों को 9 रुपये में बेचकर 4 रुपये का फायदा हुआ है। तो लाभ %?

Solution:- I

$$\text{SP of 36} - \text{CP of 36} \\ = \text{CP of 4}$$

$$\Rightarrow \text{SP of 36} = \text{CP of 40}$$

$$\Rightarrow \frac{\text{SP}}{\text{CP}} = \frac{36}{40} = \frac{9}{10}$$

$$\Rightarrow \text{P \%} = \frac{1}{9} \times 100 = 11\frac{1}{9}\%$$

II

$$\text{माना 1 संतर का क्र० मूल्य} = 1 \text{ ₹}$$

$$\text{लाभ} = 4 संतरों का क्र० मूल्य = 4$$

$$\text{लाभ \%} = \frac{4}{36} \times 100$$

$$= 11\frac{1}{9}\% \text{ Ans.}$$

36 संतरों के चयन पर एक दुकानदार को 4 संतरों के विक्रय मूल्य के बराबर लाभ होता है। लाभ % क्या होगा?

Solution:- I (Basic) OR

$$\begin{aligned} \text{SP of 36} - \text{CP of 36} \\ = \text{SP of 4} \end{aligned}$$

$$\Rightarrow \text{SP of 32} = \text{CP of 36}$$

$$\text{OR } \frac{\text{CP}}{\text{SP}} = \frac{32}{36} = \frac{8}{9}$$

$$\text{लाभ \%} = \frac{1}{8} \times 100 = 12.5\%$$

II (Tricky)

$$\text{माना 1 संतर का वि० मू०} = 1 \text{ ₹}$$

$$\text{लाभ} = 4 \text{ संतरों का वि० मू०} = 4$$

$$36 \text{ संतरों का वि० मू०} = 36$$

$$\Rightarrow 36 \text{ संतरों का क्र० मू०} = 36 - 4 = 32$$

$$\therefore \text{लाभ \%} = \frac{4}{32} \times 100 = 12.5$$

36 संतरों के चयन पर एक दुकानदार को 4 संतरों के क्र० मू० के बराबर हानि होती है। हानि प्रतिशत क्या होगा?

Solution:- I (Basic)

$$\begin{aligned} \text{CP of 36} - \text{SP of 36} \\ = \text{CP of 4} \end{aligned}$$

$$\Rightarrow \text{CP of 32} = \text{SP of 36}$$

$$\Rightarrow \frac{\text{CP}}{\text{SP}} = \frac{36}{32} = \frac{9}{8}$$

$$\text{Loss \%} = \frac{1}{9} \times 100 = 11\frac{1}{9}\%$$

II (Tricky)

$$\text{माना 1 संतर का CP} = 1 \text{ ₹}$$

$$\text{हानि} = 4 \text{ संतरों का CP} = 4 \text{ ₹}$$

$$36 \text{ संतरों का CP} = 36 \text{ ₹}$$

$$\text{हानि \%} = \frac{4}{36} \times 100 = 11\frac{1}{9}\%$$

36 संतरों के चयन पर एक दुकानदार को 4 संतरों के विक्रय मूल्यों के बराबर हानि होती है। हानि % क्या होगा?

Solution:- I (Basic)

$$\begin{aligned} \text{CP of 36} - \text{SP of 36} \\ = \text{SP of 4} \end{aligned}$$

$$\Rightarrow \text{CP of 36} = \text{SP of 40}$$

$$\text{OR } \frac{\text{CP}}{\text{SP}} = \frac{40}{36}$$

$$\therefore \text{हानि \%} = \frac{4}{40} \times 100$$

$$= 10\% \text{ Ans.}$$

II (Tricky)

$$\text{माना 1 संतर का क्र० मू०} = 1 \text{ ₹}$$

$$\text{हानि} = 4 \text{ संतरों का SP} = 4$$

$$36 \text{ संतरों का SP} = 36$$

$$\Rightarrow 36 \text{ संतरों का CP} = 36 + 4 = 40$$

$$\text{हानि \%} = \frac{4}{40} \times 100 = 10\%$$

A man buys some apples at the rate of 1 Apple for ₹ 2. and equal no. of apples at the rate of 2 apple for ₹ 1. He sells all of them @ 4 apples for ₹ 3 find the profit/loss %?

Solution:-

	Article	Price
मूल्य समान होने-वाला	2 x 1	2 ₹ x 2 = 4
	2	1 ₹ = 1

$$CP \text{ of } 4 = 4 + 1 = ₹ 5$$

$$SP \text{ of } 4 = ₹ 3$$

$$\Rightarrow \text{loss \%} = \frac{2}{5} \times 100 = 40 \% \text{ Ans}$$

1 दुकानदार कुछ वस्तुओं को 4 ₹ में 3 वस्तुओं की दर से खरीदा है और वह उतनी ही वस्तु 5 ₹ में 4 वस्तु की दर से खरीदा है। इन सभी वस्तुओं को 3 ₹ में 2 वस्तु की दर से बच देता है। उसे कितने % Profit या loss हुआ?

Solution:-

	Article	Price
	4 x 3	4 x 4 = 16
	3 x 4	5 x 3 = 15

$$24 \text{ वस्तुओं का } CP = 31$$

$$2 \text{ वस्तुओं का } SP = 3 ₹$$

$$24 \text{ वस्तुओं का } SP = 3 \times 12 = 36$$

$$\text{Profit \%} = \frac{5}{31} \times 100 = \frac{500}{31} \%$$

A shopkeeper purchases oranges 30 for ₹ 100.
How many should he sell in ₹ 100 to gain 20%?

Solution:- I (Basic)

CP of 30 oranges = 100

SP of 30 = 100×1.20
= 120 ₹

Now 120 ₹ में — 30 orange

⇒ 100 ₹ में — $\frac{30}{120} \times 100$
= 25 orange
अथ Ans.

II (Tricky)

$$E = P \times Q$$

$$E = 100 = \text{Same}$$

$$\Rightarrow P \propto \frac{1}{Q}$$

	Old	NEW
Price	100	120
Qty	120	100

$$\times 5 \left(\begin{array}{c} 6 \\ \downarrow \\ 30 \end{array} \right) : 5 \times 5$$

(25) Ans.

एक दुकानदार ने 1 ₹ में 25 लॉफियाँ खरीदी। वह 1 ₹ में कितनी लॉफियाँ बेचें कि उसे 25% का लाभ हो?

Solution:- I (BASIC)

25 लॉफियों का क्र. मू. = 1

25 लॉफियों का बि. मू. = 1.25

⇒ 1.25 ₹ में बेचें 25

⇒ 1 ₹ में बेचें $\frac{25}{1.25} = 20$

II (TRICKY)

$$E = \text{Same} = 1 ₹$$

$$P \propto \frac{1}{Q}$$

	Old	New
Price	100	125
Qty	125	100

$$5 \times \left(\begin{array}{c} 5 \\ | \\ 25 \end{array} \right) : 4 \times \left(\begin{array}{c} 1 \\ | \\ 1 \end{array} \right) \times 5$$

(20) Ans.

Note: → अगर Sale / Purchase की Price Same है तो Price और Qty का Ratio एक दूसरे का उल्टा होगा। अगर नहीं Same है तो starting में Same रख लेंगे और बाद में Unitary method का इस्तेमाल करेंगे।

By selling 15 oranges for ₹ 1 a man loss 20%.

How many for ₹ 1 should he sell to gain 20%?

Solution:- I (Basic)

$$SP_1 \text{ of 15 oranges} = ₹ 1$$

$$CP \text{ of 15 oranges} = \frac{1}{.80} = \frac{5}{4}$$

$$SP_2 \text{ of 15 oranges} = \frac{5}{4} \times 1.20$$

$$= 1.5 ₹$$

$$\therefore 1.5 ₹ \text{ में } 15 = 15$$

$$\text{त } 1 ₹ \text{ में } 15 = \frac{15}{1.5} = \textcircled{10}$$

II (Tricky)

$$Exp = Price \times Qty$$

$$E = Same = ₹ 1$$

$$Price \propto \frac{1}{Qty}$$

	old	New
Price	80	120
Qty	120	80
	3	2
	15	10

$$5 \times \left(\frac{1}{15} \right) = \frac{2}{10} \times 5$$

(10) ANS.

By selling 45 oranges for ₹ 40 a man loss 20%.

How many should he sell for ₹ 24 to earn 20%?

Solution:- I (Basic)

$$SP_1 \text{ of 45 oranges} = 40$$

$$CP \text{ of 45 } = \frac{40}{.80} = 50$$

$$SP_2 \text{ of 45 oranges} = 50 \times 1.20$$

$$= 60 ₹$$

$$\therefore 60 ₹ \text{ में } 45 \text{ संख्या}$$

$$\text{त } 1 ₹ \text{ में } 45 = \frac{45}{60}$$

$$\text{और } 24 ₹ \text{ में } 45 = \frac{45}{60} \times 24$$

$$= \textcircled{18}$$

II (Tricky)

$$\text{Sale Price} \neq \text{Purchase/Sale}$$

(but हम पहले 40 का निकालेंगे (same रखें) बाद में 24 का unitary से)

	old	New
Price	80	120
Qty	120	80

$$15 \times \left(\frac{3}{45} \right) = \frac{2}{30} \times 15$$

$$40 ₹ \text{ में } - 30$$

$$24 ₹ \text{ में } - \frac{30}{40} \times 24 = \textcircled{18}$$



MARKED PRICE / LIST PRICE / TAG PRICE / LABELLED PRICE
(अंकित मूल्य) M.R.P / Print Price
Price that is printed on the tag of article

Discount (बटुका / छूट)

$$\text{Discount} = \text{MP} - \text{SP} = \text{अं. मू.} - \text{बि. मू.}$$

Discount % हमेशा अंकित मू. पर ज्ञात किया जाता है।

D% is always calculated/Reckoned ON MP unless otherwise stated.

$$D\% = \frac{D}{\text{MP}} \times 100$$

$$\# \quad \text{Selling Price (S.P.)} = \text{M.P.} \times \frac{(100 - D\%)}{100}$$

$$\text{OR } \text{M.P.} = \text{SP} \times \frac{100}{(100 - D\%)}$$

Note :- discount is always given by shopkeeper.

and if no discount $\Rightarrow \text{MP} = \text{SP}$

MARK UP %

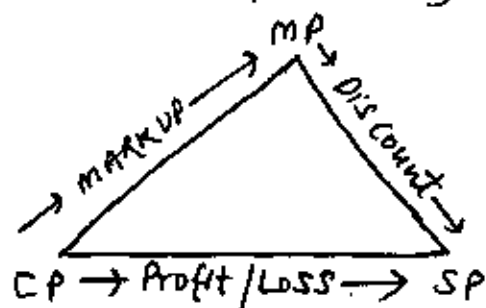
$$\text{MARK UP} = \text{MP} - \text{CP}$$

Note: → MARK UP % is always calculated on Cost Price.

$$\text{MARK UP \%} = \frac{\text{MARK UP}}{\text{CP}} \times 100$$

e.g. $\text{CP} = 100$, $\text{MP} = 150 \Rightarrow \text{MARK UP \%} = 50\%$

Relationship b/w CP, SP & MP



Let $P\% = 20\%$ and $D = 25\%$

$$\therefore \text{CP} \times (1.20) = \text{MP} \quad \text{--- (1)}$$

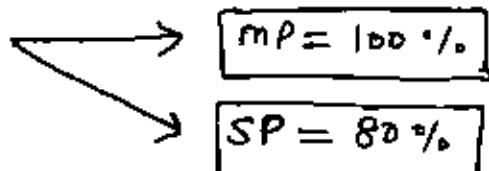
$$\text{also } \text{MP} \times (.75) = \text{SP} \quad \text{--- (2)}$$

from (1) and (2), we get

$$\text{CP} (1.20) = \text{MP} (.75)$$

100% METHOD [TAKE MP = 100%]

$$D = 20\%$$



e.g. $SP = ₹ 720$, $D\% = 10\%$, $MP = ?$

$$D = 10\% \Rightarrow MP = 100\%$$

$$SP = 90\%$$

$$90\% \longrightarrow ₹ 720$$

$$\Rightarrow 100\% \longrightarrow \frac{720}{90} \times 100 = 800 \text{ Ans.}$$

Ratio METHOD

$$D = 10\% = \frac{1}{10} \begin{matrix} \xrightarrow{\text{discount}} \\ \xrightarrow{\text{MP}} \end{matrix} \Rightarrow SP = 10 - 1 = 9$$

e.g. $D = 10\%$, $SP = ₹ 720$, $MP = ?$

$$D = 10\% = \frac{1}{10} \begin{matrix} \xrightarrow{D} \\ \xrightarrow{MP} \end{matrix} \Rightarrow SP = 10 - 1 = 9$$

$$9 \text{ units} \longrightarrow ₹ 720$$

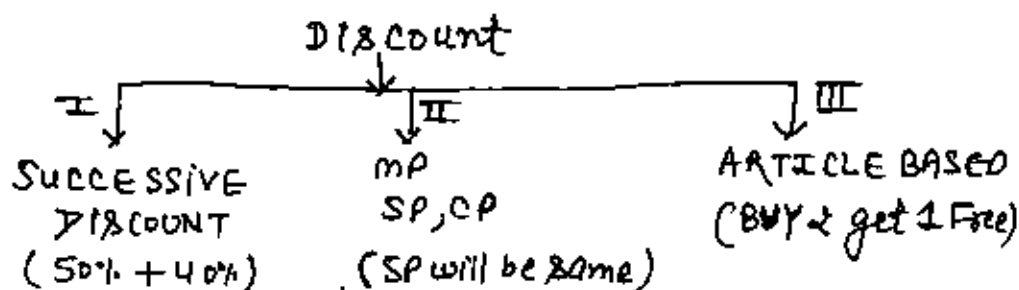
$$MP = 10 \text{ units} \longrightarrow \frac{720}{9} \times 10 = 800 \text{ Ans.}$$

$D = 12.5\%$, $MP = 1600$, $SP = ?$

$$D = 12.5\% = \frac{1}{8} \begin{matrix} \xrightarrow{D} \\ \xrightarrow{MP} \end{matrix} \Rightarrow SP = 7$$

$$8 \text{ units} \longrightarrow 1600$$

$$SP = 7 \text{ units} \longrightarrow 1400 \text{ Ans.}$$



Find the equivalent discount for 50% + 40%?

Solution: I

$$\begin{aligned}
 & X + Y + \frac{XY}{100} \\
 & X = -50, Y = -40 \\
 & = -50 - 40 + \frac{2000}{100} \\
 & = -70 \quad 70\% \text{ Ans.}
 \end{aligned}$$

II

Let MP = 100

70% $\left(\begin{array}{l} \downarrow -50\% \\ 50 \\ \downarrow -40\% \\ 30 \end{array} \right)$

III

50% = $-\frac{1}{2}$
40% = $-\frac{2}{5}$

old	New
2	1
5	3
10	3

70% D

If two successive discount of 60% + X% equivalent discount is 72%. Find X?

Solution → I

$$-72 = -60 - X + \frac{60X}{100}$$

$$X = 30\%$$

II

MP

100

60% D →

40

↓ -72%
28

$\frac{12 \times 100}{40} = 30\% \text{ Ans.}$

which offer is better (i) 60% + 30%
(ii) 70% + 20%

Solution → Total is same = 60 + 30 = 70 + 20 = 90

जिसकी numericle value उसी की ही रही ज्यादा discount मिले। (ii) is better coz of 70%

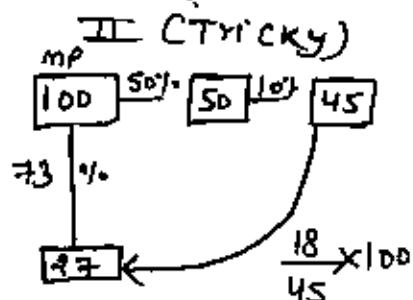
एक दुकानदार अंकिता मूल्य पर तीन क्रमागत/लगातार छूट देता है। पहला discount 50%, दूसरा 10% और तीसरा $x\%$ है और तीन का equivalent (समतुल्य) छूट 73% है तो x का मान बताइए?

Solution:- I (BASIC)

$$SP = MP \left(\frac{100 - D_1}{100} \right) \left(\frac{100 - D_2}{100} \right)$$

$$27 = 100 \times \frac{50}{100} \times \frac{90}{100} \times \frac{100 - D_3}{100}$$

$$\Rightarrow D_3 = 40\% \text{ Ans.}$$

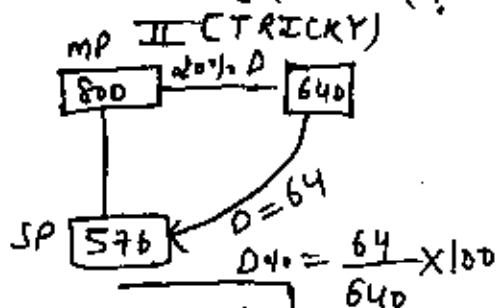


marked price of an article is 800 ₹. If after giving two successive discount selling price is 576. If 1st discount is 20%. Find second?

Solution:- I

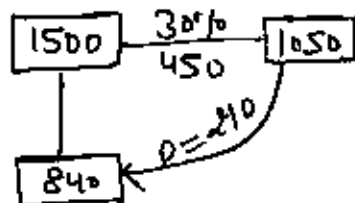
$$576 = 800 \times \frac{80}{100} \times \frac{100 - D}{100}$$

$$\Rightarrow D = 10\% \text{ Ans}$$



किसी वस्तु का अंकिता मूल्य (MP) 1500 ₹ है। एक दुकानदार अंकिता मूल्य पर दो लगातार छूट देता है। यदि एक छूट 30% है और दूसरा $x\%$ है। दुकानदार वस्तु को अन्त में 840 में बेचता है। x का मान क्या होगा?

Solution:- MP



$$D\% = \frac{210}{1050} \times 100 = 20\% \text{ उत्तर}$$

CONCEPT/TRICK [SP = same]

$$SP = CP \times \left[\frac{100 \pm P/L\%}{100} \right] = MP \times \left[\frac{100 - D\%}{100} \right]$$

$$\therefore CP \left[\frac{100 \pm P/L\%}{100} \right] = MP \times \left[\frac{100 - D\%}{100} \right]$$

$$\Rightarrow \begin{array}{cc} CP & : & MP \\ 100 - D\% & : & 100 \pm P/L\% \end{array}$$

e.g. $P\% = 20\%$, $D = 10\%$
 $CP : MP = 90 : 120$
 $= 3 : 4$

A shopkeeper after allowing a discount of 10% still gains 20%. If marked price is 800 find cost price of article?

Solution: $CP : MP$

$$100 - D\% : 100 + P\%$$

$$90\% : 120\%$$

$$3 : 4$$

$$4 \rightarrow 800$$

$$1 \rightarrow \frac{800}{4}$$

$$3 = CP \rightarrow \frac{800}{4} \times 3$$

$$= 600 \text{ ₹ ANS.}$$

एक दुकानदार अगर 20% की छूट दे तो उसे 10% की हानि होती है। IF $CP = 640$, $MP = ?$

Solution: — I

$$CP : SP : MP$$

$$10 \times 4 \quad 9 \times 4$$

$$4 \times 9$$

$$5 \times 9$$

$$16 \begin{pmatrix} 40 \\ 1 \\ 640 \end{pmatrix}$$

$$\begin{pmatrix} 45 \\ 16 \end{pmatrix} \times 16 = 720 \text{ ANS.}$$

II

$$CP : MP$$

$$100 - D : 100 - L\%$$

$$80 : 90$$

$$8 : 9$$

$$80 \begin{pmatrix} 1 \\ 640 \end{pmatrix}$$

$$9 \begin{pmatrix} 1 \\ 720 \end{pmatrix} \times 80 = 720 \text{ ANS.}$$

एक दुकानदार को अपनी वस्तुओं के ऊपर कितना % अधिक अंकित करना पड़ेगा कि 20% छूट देने के बाद भी 10% का लाभ हो ?

Solution :- I

$$CP : MP$$

$$100 - D\% : 100 + P\%$$

$$100 - 20 : 100 + 10$$

$$80 : 110$$

$$\frac{30}{80} \times 100 = 37.5\%$$

II

$$P\% = 10\% = \frac{1}{10} \Rightarrow \frac{CP}{SP} = \frac{10}{11}$$

$$D\% = 20\% = \frac{1}{5} \Rightarrow \frac{SP}{MP} = \frac{4}{5}$$

$$CP : SP : MP$$

$$10 \times 4 : 11 \times 4$$

$$4 \times 11 : 5 \times 11$$

$$40 : 55$$

$$= \frac{15}{40} \times 100 = 37\frac{1}{2}\%$$

III Direct

$$\frac{P + D \times \frac{100}{100 - D}}{100 - D}$$

$$= \frac{30 \times 100}{80} = 37.5\%$$

By how much percentage a shopkeeper marks his goods above its CP so that after allowing 10% discount he may gain 30% P.

Solution :- I

$$CP : MP$$

$$100 - D\% : 100 + P\%$$

$$100 - 10 : 100 + 30$$

$$90 : 130$$

$$\frac{40}{90} \times 100 = 44\frac{4}{9}\%$$

II (DIRECT)

$$(P + D) \times \frac{100}{100 - D}$$

$$= 40 \times \frac{100}{90}$$

$$= 44\frac{4}{9}\%$$

By selling an article for ₹ 750 & allowing 20% discount a shopkeeper still earns 40% profit. If the article is sold at 0% discount then profit would be?

Solution :-	CP : MP	No disc count (MP = SP)
	100 - 20 : 100 + 40	CP : SP
	80 : 140	4 : 7
	4 : 7	P% = $\frac{3}{4} \times 100 = 75\%$

एक दुकानदार अपनी वस्तु को अंशित मूल्य पर 15% छूट देकर 425 रु में बेचता है। यदि वह छूट न देता तो उसे 25% का लाभ होता है। उस वस्तु का मूल्य कितना होगा?

Solution :- 85% (SP) $\rightarrow$ 425

100% (MP) $\rightarrow 425 / 85 \times 100 = 500$ रु

IF No discount MP = SP $\Rightarrow$ SP = 500

P% = 25% $\Rightarrow$ CP = $500 \times \frac{100}{125} = 400$ रु ANS.

एक दुकानदार अपनी वस्तु को 10% छूट देने के बाद भी 20% लाभ कमाता है। यदि वह छूट / बढ़ते को बढ़ाकर 20% कर देता तो उसे अब कितने % का लाभ होगा?

Solution :- CP : MP = (100 - 10) : (100 + 20)

= 3 : 4

Now discount 20% CP : SP 3 : 4 $\times \frac{80}{100} = 15 : 16$

P% = $\frac{1}{15} \times 100 = 6.66\%$ ANS.

On Purchase of 3 shirts, 1 shirt is given free. Find the discount %?

Solution:- Let Price of 1 shirt = ₹

$$\text{MP of 4 shirts} = 4 \quad [\because 1 \text{ is given free.}]$$

$$\text{SP of 4 shirts} = 3$$

$$\% = \frac{1}{4} \times 100 = 25\% \text{ Ans.}$$

6 पुस्तकें खरीदने पर 4 पुस्तकें फ्री दी जाती हैं। तो बटल (discount) प्रतिशत क्या होगा।

Solution:- माना एक पुस्तक का मूल्य = ₹

$$10 \text{ पुस्तकें का अंकित मूल्य} = 10 \text{ ₹} \quad [\text{क्योंकि 4 लीं}]$$

$$10 \text{ पुस्तकें का विजय मूल्य} = 6 \quad [\text{फ्री हैं}]$$

$$\% = \frac{4}{10} \times 100 = 40\% \text{ Answer}$$

4 वन खरीदने पर एक वन फ्री दिया जाता है और साथ में 20% आतिरेक बटल (discount) दिया जाता है। तो समतुल्य बटल (equivalent discount) क्या होगा?

Solution:- माना एक वस्तु का मूल्य = ₹ 100

$$5 \text{ वस्तुओं का अंकित मूल्य} = 500 \text{ ₹}$$

$$5 \text{ वस्तुओं का विजय मूल्य} = 4 \times 100 = 400 \text{ ₹}$$

$$20\% \text{ Extra discount } \therefore \text{ नया विजय SP} = 400 \times \frac{80}{100}$$

$$\% = \frac{500 - 320}{500} \times 100 = 36\% \text{ Ans.} \quad = 320$$

एक दुकानदार अपनी वस्तु के अंकित मूल्य पर 4% की छूट (discount) देता है। और 15 वस्तुओं खरीदने पर 1 वस्तु फ्री भी देता है। फिर भी उसने 35% का लाभ होता है। तो दुकानदार ने वस्तु का कितना अंकित मूल्य से कितने प्रतिशत अधिक रखा था?

Solution :- I (BASIC)

16 वस्तुओं का अंकित मूल्य = 1600

16 " " " " " " " " = 1500

4% D extra

$$\text{नया बि. मू.} = 1500 \times \frac{96}{100} \\ = 1440 \text{ ₹}$$

35% लाभ
135% $\rightarrow$ 1440

$$100\% \text{ (CP)} \rightarrow \frac{1440 \times 100}{135} = \frac{3200}{3}$$

$$\text{CP : MP} \\ \frac{3200}{3} : 1600 \Rightarrow 2 : 3 \\ \text{50\% Ans.}$$

II (TRICKY)

CP : MP

100 - D% : 100 + P%

100 - 4% : 100 + 35

$$\frac{96}{16} : \frac{135}{15}$$

$$6 : 9$$

$$2 : 3$$

$$\text{MARKUP \%} = \frac{1}{2} \times 100 \\ = 50\%$$

A shopkeeper gives 3 articles free on purchase of 5 articles. He also allows a discount of 20% & still earns 25% profit. Find the MARKUP %?

Solution :- I (BASIC)

CP : MP

$$5 \times \frac{80}{100} \times \frac{100}{125} : 8 \\ 32 : 80 \\ \text{150\% Ans.}$$

II (TRICKY)

CP : MP

100 - 20 : 100 + 25

$$\frac{80}{8} : \frac{125}{5}$$

$$10 : 25 \\ \text{150\%}$$

सुमन एक दुकानदार से 18 रु प्रति दर्जन की दर से 6 पेंसिल खरीदता है। और दुकानदार सुमन को 1 पेंसिल मुफ्त में देता है। तो सुमन का लाभ प्रतिशत बताओ?

Solution :- I (BASIC)

$$12 \text{ पेंसिल} \rightarrow 18 \text{ रु.}$$

$$6 \text{ पेंसिल} \rightarrow 9 \text{ रु.}$$

$$7 \text{ पेंसिल} \rightarrow 7 \times \frac{3}{2} \text{ रु.}$$

$$7 \text{ पेंसिल} \rightarrow \frac{18}{12} \times 7 = \frac{21}{2}$$

$$\text{लाभ \%} = \frac{3/2}{9} \times 100 = 16\frac{2}{3} \%$$

II (TRICK)

$$\text{लाभ \%} = \frac{\text{मुफ्त वस्तु}}{\text{खरीदी गई वस्तु}} \times 100$$

$$= \frac{1}{6} \times 100$$

$$= 16\frac{2}{3} \%$$

एक दुकानदार अपनी वस्तु का मूल्य CP से 20% अधिक अंकित करण है। वह आधे वस्तु को अंकित मूल्य पर बेचता है। और $\frac{1}{4}$ भाग को 20% छूट पर तथा शेष को 40% की छूट पर बेचता है। उसका लाभ % क्या होगा?

Solution :- Let CP = 100 रु

$$MP = 120 \text{ रु}$$

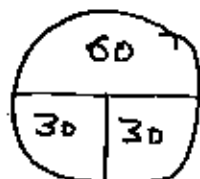
$$SP_1 = 60 \text{ [No dis count]}$$

$$SP_2 = 30 \times \frac{80}{100} = 24 \text{ [20\% dis count]}$$

$$SP_3 = 30 \times \frac{60}{100} = 18 \text{ [40\% dis count]}$$

$$\text{Total SP} = 60 + 24 + 18 = 102 \text{ रु}$$

$$P\% = 2\% \text{ ANS.}$$



TO PURCHASE
NOTES 4

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FALSE WEIGHT / DISHONEST SHOPKEEPER [वैईमान दुकानदार]

$$\text{PROFIT/LOSS \%} = \frac{\text{ERROR}}{\text{FALSE WT}} \times 100$$

where Error = True wt - False wt.

IF Ans is + $\Rightarrow$ Profit otherwise (-) Loss

A dishonest shopkeeper promises to sell his goods at its CP. But he uses 800 gm wt. instead of 1 kg. Find Profit %?

Solution:- I (Basic)

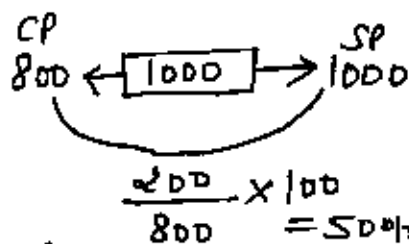
Let CP of 1 gm = ₹

CP of 800 gm = 800

but SP of 800 gm = 1000

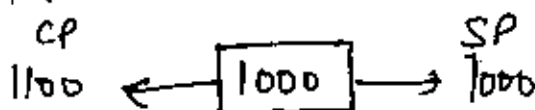
$$\Rightarrow P.\% = \frac{200}{800} \times 100 = 25\%$$

II (TRICKY)



एक दुकानदार अपनी वस्तुओं को क्रम सूचक पर ही बेचने का वायदा करता है। लेकिन वह सामान देते वस्तु 1 kg के भार के स्थान पर 1100 gm का भार गलती से इस्तेमाल कर लेता है। तो लाभ % बताओ?

Solution:- I



$$\text{Loss \%} = \frac{100}{1100} \times 100 = 9.09\%$$

II

By formula

$$\text{Loss \%} = \frac{-100}{1100} \times 100 = -9.09\%$$

A shopkeeper Promises To sell at CP. But by means of false weight he gains 25%. Find the weight he uses in place of 1kg?

Solution:- I (BASIC)

Let he cheats by x gm

$$\Rightarrow \frac{x}{1000-x} \times 100 = 25$$

$$\Rightarrow 4x = 1000 - x \Rightarrow x = 200$$

$$\text{True wt} = 1000 - 200 = 800 \text{ Ans.}$$

II (TRICKY)

$$25\% = \frac{1-p}{4-cp} \Rightarrow sp=5$$

$$\text{if } 5 \rightarrow 1000 \text{ gm}$$

$$\therefore 4 \rightarrow 800 \text{ gm Ans}$$

A shopkeeper Promises to sell his goods at 10% Profit but he uses 20% less weight. Find the Profit %?

Solution:- I

$$\begin{array}{c} \text{CP} \quad \boxed{1000} \quad \text{SP} \\ \leftarrow 20\% \quad \quad \quad \rightarrow 10\% \end{array}$$

$$\uparrow \quad \frac{300}{800} \times 100 = 37.5\%$$

$$\text{II } P\% = \frac{1-p}{10-cp} \Rightarrow sp=11$$

$$\begin{array}{ccc} \text{CP} & & \text{SP} \\ \downarrow 10 & : & \downarrow 11 \end{array}$$

$$\begin{array}{ccc} 4 & : & 5 \end{array}$$

$$\frac{40}{55} \times 100 = 72.7\%$$

एक दुकानदार अपने ग्राहक से वादा करता है कि प्रत्येक वस्तु को 44% की छानि पर बेचेगा लेकिन वह भार में 30% कम कर देता है। तो उसकी वास्तविक छानि % बताओ?

Solution:- I

$$\begin{array}{c} \leftarrow 30\% \quad \boxed{1000} \quad \leftarrow 44\% \end{array}$$

$$\uparrow \quad \frac{140}{700} \times 100 = 20\% \text{ Ans.}$$

II

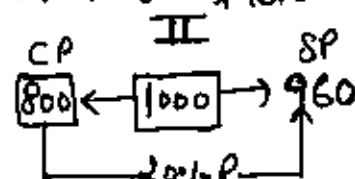
$$\begin{array}{ccc} \text{CP} & : & \text{SP} \\ 44\% \rightarrow 100 & : & 56 \\ 30\% भार \rightarrow 70 & : & 100 \\ \hline 70 & : & 56 \\ \uparrow \quad \frac{14}{70} \times 100 = 20\% \end{array}$$

एक दुकानदार अपने ग्राहक से वादा करता है कि वह प्रत्येक वस्तु को $x\%$ की छानि पर बेचेंगा लेकिन वह भार में 20% कम भार का प्रयोग करता है। तथा 20% का लाभ कमाल है तो x का मान ज्ञात कीजिए

Solution:- I

20% कम भार CP : SP
 $\downarrow 4 \quad \downarrow 5$
 $\text{loss } \% = x \rightarrow \frac{a}{b} : \frac{b}{a}$
 कुल लाभ $4a \rightarrow (5) \quad (6) - 5b$

$$\frac{4a}{5b} = \frac{5}{6} \Rightarrow \frac{a}{b} = \frac{25}{24} \Rightarrow \frac{1}{24} \times 100 = 4\%$$



$$x\% = \frac{40}{1000} \times 100 = 4\% \text{ Ans.}$$

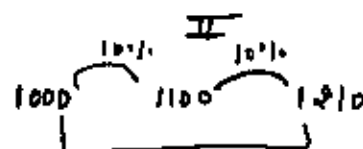
एक दुकानदार सामान खरीदते वन्त 10% बड़िमानी तथा सामान बेचते वन्त भी 10% की बड़िमानी करता है। तो कुल लाभ % ज्ञात करें?

Solution:- I

$$x + y + \frac{xy}{100}$$

$$10 + 10 + \frac{100}{100} = 21\% \text{ लाभ}$$

OR



$$\frac{210}{1000} \times 100 = 21\% \text{ लाभ}$$

MATHS

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9728435915

A shopkeeper sells an article for ₹ 96 such that P% and CP are equal. Find CP?

Solution:- I (BASIC)

$$\text{Let CP} = \text{P}\% = x$$

$$\Rightarrow x \times \frac{(100+x)}{100} = 96$$

$$\Rightarrow x^2 + 100x = 9600$$

$$\Rightarrow x^2 + 100x - 9600 = 0$$

$$(x+160)(x-60) = 0$$

$$x = -160 \text{ or } \boxed{x = 60}$$

II (TRICKY)

$$\text{SP} = 96$$

अन्तर होगा 10 रु. चाहिए

$$16 - 6 = 10$$

$$\times 10$$

$$\boxed{60} \text{ Ans.}$$

[$a \times b = 96$
 $a - b = 10$] then add 0 to smaller No. to get Ans.

III) Try to do these Q. by options

$$\text{CP} = 60, \text{P}\% = 60 \Rightarrow \text{SP} = 96$$

एक दुकानदार एक वस्तु को 16 रु में इस प्रकार से बेचता है कि लाभ % और क्रय मूल्य बराबर हो तो उस वस्तु का क्रय मूल्य बताइए ? a) 60 b) 40 c) 80 d) 90

Solution:- I (BASIC)

$$\text{Let CP} = \text{L}\% = x$$

$$\Rightarrow x \times \frac{(100-x)}{100} = 16$$

$$\Rightarrow 100x - x^2 = 1600$$

$$\Rightarrow x^2 - 100x + 1600 = 0$$

$$\Rightarrow (x-80)(x-20) = 0$$

$$\boxed{x = 80, 20} \text{ both are } \checkmark$$

II (TRICKY)

$$\text{SP} = 16$$

$$a \times b = 16$$

$$a + b = 10$$

$$8 \quad 2$$

$$\times 10 \quad \times 10$$

$$\boxed{80} \quad \boxed{20}$$

both are ✓

III) By options check (c) CP = 80, L% = 80%

$$\Rightarrow \text{SP} = 16 \text{ Ans (C)}$$

A man buys some Articles for ₹ 4800. He sells $\frac{2}{5}$ th of the articles at a profit of 25%. At what Price should he sell the Remaining articles so that he gains 19% on a whole?

Solution :- I

$$\begin{array}{ccc} \text{Let Total Articles} & = & 5 \\ \frac{2}{5} & \frac{25\% \text{ पर}}{2} & \frac{3}{5} \end{array}$$

$$\times 19 \left(\begin{array}{c} x\% \\ (x\% \div 3) \end{array} \right) = 15\% \text{ Ans.}$$

$$95\% - 50\% = 45\%$$

II (Alligation)

$$\begin{array}{cc} x\% & 25\% \\ & \swarrow \searrow \\ & 19\% \\ & \swarrow \searrow \\ 3 & : & 2 \end{array}$$

$$\Rightarrow \frac{25-19}{19-x} = \frac{3}{2}$$

$$\Rightarrow x = 15\% \text{ Ans.}$$

$$\text{IIIrd) } \frac{2}{5}(25\%) + \frac{3}{5}(x\%) = 19\%$$

$$\Rightarrow 50\% + 3x\% = 95\% \Rightarrow x = 15\%$$

IN what Ratio must water be mixed with milk to gain 20% on selling the mixture at cost Price?

Solution :- Alligation (I)

Let CP of 1ltr of milk = ₹ 1

CP of 1ltr of water = 0

SP of 1ltr of mixture = 1

Now CP of 1ltr of mix = $\frac{100}{120} \times 1 = \frac{5}{6}$

$$\begin{array}{cc} w & m \\ 0 & 1 \\ & \swarrow \searrow \\ & \frac{5}{6} \\ & \swarrow \searrow \\ 1 & : & 5 \end{array}$$

$$w:m = 1:5$$

II (TRICKY)

$$20\% = \frac{1}{5} \rightarrow w$$

$$5 \rightarrow m$$

$$w:m = 1:5$$

[क्योंकि जो Profit होगा वो पानी के खर्च ही होगा]



IF selling price of an article gets doubled profit will get triple? In original P%?

Solution:- BASIC (I)

$$\text{Let } CP = X, SP = Y$$

$$\text{Profit} = (Y - X)$$

$$\text{New } SP = 2Y \quad \text{New } P = 2Y - X$$

$$\therefore 2Y - X = 3(Y - X)$$

$$\Rightarrow Y = 2X \quad \text{OR } X:Y = 1:2$$

$$P\% = \frac{1}{1} \times 100 = 100\% \text{ Ans.}$$

TRICKY (II)

$$CP = \text{same} = \text{constant}$$

$$SP \quad \begin{array}{c} \text{old} \quad \text{NEW} \\ 1 \times 2 \quad : \quad 2 \times 2 \end{array}$$

$$P \quad \begin{array}{c} 1 \times 1 \quad : \quad 3 \times 1 \\ \quad \quad \quad 2 \end{array}$$

$$SP \quad 2 \quad] CP=1 \quad 4 \quad] CP=1$$

$$P \quad 1 \quad] CP=1 \quad 3 \quad] CP=1$$

$$CP:SP = 1:2$$

$$\Rightarrow P\% = 100\%$$

III) By options $P\% = 100\%$

$$\text{Let } CP = 100 \Rightarrow SP = 200 \quad SB = 400 \Rightarrow P = 300 \text{ Ans } 100\%$$

एक दुकानदार ने एक वस्तु को 20% के लाभ पर B को बेचा, B ने उसको 25% के लाभ पर C को बेचा दिया। अगर C का CP 225 है तो A का CP ज्ञात करो?

Solution:- Note: जो B का CP है वो A का SP है।

I

$$CP(A:B) = 100:120$$

$$= 5:6$$

$$CP(B:C) = 100:125$$

$$= 4:5$$

$$CP(A:B:C) = 20:24:30$$

$$30 \rightarrow 225$$

$$\Rightarrow 20 \rightarrow 225/30 \times 20 = 150 \text{ ₹ Ans.}$$

II

$$CP(A) = 225 \times \frac{100}{120} \times \frac{100}{125}$$

$$= 150 \text{ ₹ Ans.}$$

III

$$CP(A) = 225 \times \frac{5}{6} \times \frac{4}{5}$$

$$= 150 \text{ ₹ Ans.}$$

Selling Price Same

A) IF $SP_1 = SP_2 \Rightarrow$ Always $\Rightarrow$ $Loss\% = \frac{x^2}{100}$
 $P\% = L\% = x\% \text{ Loss}$

B) IF $CP_1 = CP_2 \Rightarrow$ No Profit, No Loss
 $P\% = L\%$

Two Books are sold for ₹ 960 each. One is sold at 20% Profit and other at 20% Loss. Find Profit and Loss in ₹?

Solution :- I (BASIC)

$$SP_1 + SP_2 = 960 + 960 = 1920$$

$$CP_1 = 960 \times \frac{100}{120} = 800 \text{ ₹}$$

$$CP_2 = 960 \times \frac{100}{80} = 1200 \text{ ₹}$$

$$CP_1 + CP_2 = 2000 \Rightarrow \text{Loss} = 80 \text{ ₹}$$

II (TRICKY)

$$P\% = L\% = 20\%$$

$$SP_1 = SP_2 = 960$$

$$\Rightarrow \text{Loss}\% = \frac{(20)^2}{100} = 4\%$$

$$\text{Loss} = 1920 \times \frac{4}{100} = 76.8 \text{ ₹}$$

राम के पास दो बाइ हैं। जिनमें से वह प्रत्येक के समान विक्रम मूल्य 275000 में बेचता है। अगर वह पहले को 20% लाभ पर और दूसरे को 10% की हानि पर बेचता है तो दोनों बाइ के क्रय मूल्यों के अनुपात ज्ञात करें?

Solution :- Trick

$$\frac{CP_1}{CP_2} = \frac{100 \pm x_2}{100 \pm x_1} \Rightarrow \frac{CP_1}{CP_2} = \frac{100 - 10}{100 + 20} = \frac{3}{4}$$

$$CP_1 : CP_2 = 3 : 4$$

Two horses are sold for ₹ 72 EACH
IF one is sold at 20% Profit and
other at 10% loss. Find the OVERALL
PROFIT and loss % ?

Solution :- I 20% P $\rightarrow$ 120% of CP = SP
10% L $\rightarrow$ 90% of CP = SP

Let $SP_1 = SP_2 = 36$ (LCM of 12 & 9)

$$CP_1 = \frac{36}{1.20} = 30, \quad CP_2 = \frac{36}{.90} = 40$$

$$CP_1 + CP_2 = 70$$

$$SP_1 + SP_2 = 72$$

$$P\% = \frac{2}{70} \times 100$$

$$= 2.85\% \text{ Ans.}$$

$$\text{II} \quad 20\% P = \frac{1 \rightarrow P}{5 \rightarrow CP}$$

$$SP_1 = 6 \times 3 = 18$$

$$10\% L = \frac{1 \rightarrow L}{10 \rightarrow CP}$$

$$SP_2 = 9 \times 2 = 18$$

$$\begin{array}{l} CP_1 \rightarrow 5 \times 3 = 15 \\ CP_2 \rightarrow 10 \times 2 = 20 \end{array} \quad \left. \vphantom{\begin{array}{l} CP_1 \\ CP_2 \end{array}} \right\} 35 \quad SP = 36$$

$$P\% = \frac{1}{35} \times 100 = 2.85\% \text{ Ans.}$$

एक आदमी दो वस्तुओं को 1710 ₹ में बेचता है अगर पहली वस्तु का क्रय मूल्य दूसरी वस्तु के विक्रय मूल्य के बराबर है तो और पहली को 10% की हानि पर और दूसरी को 25% के लाभ पर बेचा जाय तो उस कुल कितने ₹ की लाभ या हानि होगी?

Solution:- I

1st	2nd	
CP $\frac{100}{100}$	80 = 180	
SP 90	125% P $\frac{100}{100}$ = 190	10 ₹ P
190	180	

$$SP_1 + SP_2 \rightarrow 190 \rightarrow 1710$$

$$1 \rightarrow 9$$

$$10 \rightarrow 90 ₹ \text{ Ans}$$

II (Ratio)

$$10\% L \quad CP_1 \quad SP_1$$

$$10 : 9$$

$$25\% P \quad CP_2 : SP_2$$

$$4 : 5$$

$$\text{but } CP_1 = SP_2 = 10$$

$$10 : 9 \quad SP_1 + SP_2 = 10 + 9$$

$$4x_2 : 5x_2 = 19$$

$$18 - 1710 \Rightarrow 1 \rightarrow 90 ₹ \text{ Ans.}$$

A man sells two articles first at 15% loss and second at 19% profit. If during the whole transaction he bears a loss of 90 ₹. If he sells both article at same price Find the cost price of each?

Solution:- 15% L $\Rightarrow \frac{-3}{20}$

$$CP_1 : SP_1$$

$$20 : 17$$

$$19\% P \rightarrow \frac{19}{100}$$

$$CP_2 : SP_2$$

$$100 : 119$$

$$\text{but } SP_1 = SP_2$$

I

$$CP \quad 20 \times 7$$

$$P/L \quad 3 \times 7$$

$$SP \quad 17 \times 7$$

$$2 \rightarrow 90$$

$$1 \rightarrow 45$$

$$CP_1 (140) \rightarrow 45 \times 140 \quad 6300$$

$$CP_2 (100) \rightarrow 45 \times 100 \quad 4500$$

II

$$100 = 240$$

$$19$$

$$119 = 238$$

A man purchased two watches for ₹ 840. If he sells one at 16% profit and other at 12% loss then he get no profit no loss in whole Transaction. Find the CP of each watch?

Solution :- I (BASIC)

$$\text{Let } CP_1 = x \Rightarrow CP_2 = 840 - x.$$

$$x \times \frac{88}{100} + (840 - x) \times \frac{116}{100} = 840$$

$$x = 480 = CP_1 \Rightarrow CP_2 = 360$$

II (Alligation)

$$\begin{array}{ccc} -12 & & 16 \\ & \searrow & \swarrow \\ & 0 & \\ & \swarrow & \searrow \\ 16 & : & 12 \\ 4 & : & 3 \end{array}$$

$$CP_1 = \frac{840 \times 4}{7} = 480$$

$$CP_2 = 360$$

III (TRICKY)

$$16 CP_1 = 12 CP_2$$

$$\Rightarrow \frac{CP_1}{CP_2} = \frac{3}{4}$$

$$7 \rightarrow 840$$

$$1 \rightarrow 120$$

$$CP_1 (3) = 360$$

$$CP_2 (4) = 480$$

एक व्यक्ति दो वस्तुओं को 1250 ₹ में खरीदता है। वह एक को 5% की हानि पर और दूसरे को 20% के लाभ पर बेचता है। अगर उसे पूरे व्यवसाय में न लाभ और न ही हानि होती है। तो दोनों वस्तुओं का अलग-अलग क्रय मूल्य ज्ञात करें?

Solution I (BASIC)

$$\text{Let } CP_1 (5\% L) = x$$

$$\Rightarrow CP_2 (20\% P) = 1250 - x$$

$$x \times \frac{95}{100} + (1250 - x) \times \frac{120}{100} = 1250$$

$$\Rightarrow \frac{-25x}{100} = \frac{-1250 \times 20}{100}$$

$$x = 1000$$

$$CP_2 = 1250 - 1000 = 250$$

II (Alligation)

$$\begin{array}{ccc} -5\% & & 20\% \\ & \searrow & \swarrow \\ & 0 & \\ & \swarrow & \searrow \\ 20 & : & 5 \\ 4 & : & 1 \end{array}$$

$$5 \rightarrow 1250$$

$$4 \rightarrow 1000 (CP_1)$$

$$1 \rightarrow 250 (CP_2)$$

III (TRICKY)

$$5 CP_1 = 20 CP_2$$

$$\frac{CP_1}{CP_2} = \frac{4}{1}$$

$$5 \rightarrow 1250$$

$$4 \rightarrow 1000 (CP_1)$$

$$1 \rightarrow 250 (CP_2)$$

A shopkeeper sells table at 15% profit and chair at 20% loss then he gains 18 ₹. But if he sells table at 20% loss and chair at 15% then there is no profit no loss situation. Find the cost price of chair and table?

Solution:- No Profit No loss

$$20T = 15C$$

$$T:C = 3:4$$

$$CP \rightarrow 500:400$$

$$\left[\begin{array}{l} 400 \times \frac{20}{100} = 80 \text{ ₹ loss} \\ 500 \times \frac{15}{100} = 125 \text{ Profit} \end{array} \right] 40 \text{ Profit}$$

CP of Table

$$\frac{45}{1} = \frac{18}{15}$$

$$C \rightarrow 500 \times \frac{2}{5} = 200$$

CP of chair

$$400 \rightarrow 400 \times \frac{2}{5} = 160 \text{ ₹}$$

एक पेन की 5% की छानि पर और एक कितान की 15% की लाभ पर बेचना पर एक दुकानदार को 7 ₹ का फायदा होता है। यदि वह पेन को 5% लाभ पर और कितान को 10% लाभ पर बेचता तो उसे 13 ₹ का लाभ होता। कितान का वास्तविक हल्व ज्ञात करो?

Solution

पेन	कितान	
5% (L)	15% P	= 7 ₹ P
5% (P)	10% P	= 13 ₹ P

$$25\% \rightarrow 20$$

$$4 \times \left(\frac{\quad}{100\%} \right) \times 4 = 80 \text{ Ans.}$$

Total cost of 8 books and 5 pens is 92. And cost of 5 books and 8 pens is 77. Then find the cost of 3 books and 2 pens?

Solution:-

$$8B + 5P = 92 \text{ --- (I)}$$

$$5B + 8P = 77 \text{ --- (II)}$$

Add (I) + (II), we get

$$13B + 13P = 169 \Rightarrow B + P = 13 \text{ --- (3)}$$

adding 3 and (4) $B = 9, P = 4 \Rightarrow 3B + 2P$
 $= 3 \times 9 + 2 \times 4 = 35$

Subtract (I) and (II)

$$3B - 3P = 15$$

$$B - P = 5 \text{ --- (4)}$$

The Selling Price of X and Y are ₹ 100. X calculates his profit on SP while Y on CP. Find the difference between their CP if both earn same P.

Solution:-

	X	Y
CP	4×6	$5 \times 5 \Rightarrow 0 = 1$
P/L	1×6	1×5
SP	5×6	6×5

$$30 \rightarrow \text{₹} 100$$

$$1 \rightarrow \text{₹} 70 \text{ Ans.}$$

A shopkeeper bought two bicycles in ₹ 1600. If he sold 1st bicycle at 10% profit and 2nd at 20% profit he earns a certain profit. But if he sold 1st at 20% P and 2nd at 10% P he will get ₹ 5 more. Find the CP of bicycles?

Solution:-

$$B_1 + B_2 = 1600$$

$$B_1 - B_2 = 50 \rightarrow$$

$$2B_1 = 1650$$

$$\Rightarrow B_1 = 825 \Rightarrow B_2 = 775$$

$$\left[\begin{array}{l} 101 \cdot B_1 + 201 \cdot B_2 = P \\ 201 \cdot B_1 + 101 \cdot B_2 = P + 5 \\ \hline B_1 - B_2 = \frac{5 \times 100}{10} = 50 \end{array} \right]$$

CP of 12 oranges is equal to SP of 9 oranges and discount on 10 oranges is equal to profit on 5 oranges. What is the percent profit difference between the profit % and discount %?

- a) $16\frac{2}{3}\%$ b) 25% c) 22.22% d) 20%

Solution: — 12 CP = 9 SP

$$\Rightarrow \frac{CP}{SP} = \frac{3}{4} \Rightarrow P\% = \frac{1}{3} \times 100 = 33.33\%$$

Now 10 D = 5 P

$$\frac{D}{P} = \frac{1}{2} \Rightarrow 1 + P \Rightarrow 1$$

$$0 \rightarrow \frac{1}{2}$$

CP		SP		MP
3	$\xrightarrow{+1}$	4	$\xrightarrow{+1/2}$	4.5

$$D\% = \frac{4.5 - 4}{4.5} \times 100 = 11\frac{1}{3}\%$$

$$D - P\% = \frac{100}{3} - \frac{100}{9} = \frac{200}{9} = 22.22\% \text{ Ans.}$$

The ratio of cost price and marked price of an article is 2:3 and the ratio of percentage profit and percentage discount is 3:2 what is D%?

- a) 16.66% b) 20% c) 25% d) 33.33%

Solution: Let $P\% = 3x$ and $D\% = 2x$ [$\because 3:2$]

$$CP : MP$$

$$100 - D\% : 100 + P\%$$

$$100 - 2x : 100 + 3x$$

$$\frac{100 - 2x}{100 + 3x} = \frac{2}{3} \Rightarrow 300 - 6x = 200 + 6x$$

$$\Rightarrow 12x = 100 \Rightarrow x = \frac{25}{3}$$

$$D\% = 2x = 2 \times \frac{25}{3} = \frac{50}{3} = 16.66\%$$

10 सामंजस्यताओं को बेचने पर एक व्यक्ति को 36% के विक्रम मूल्य के बराबर लाभ होता है। जबकि 10 कलम को बेचने पर 4 सामंजस्यताओं के विक्रम मूल्य के बराबर हानि होती है। यदि लाभ 4 और हानि 1. संरचनात्मक बराबर है और सामंजस्यताओं का क्रम मूल्य कलम के क्रम मूल्य का आध्म है तो सामंजस्यता तथा कलम के विक्रम मूल्य का अनुपात ज्ञात करें?

Solution :- सामंजस्यता कलम

$$CP \quad 1 \quad : \quad 2$$

$$SP \quad a \quad : \quad b$$

$$CP = 10$$

$$P = 36$$

$$P\% = \frac{36}{10} \times 100$$

$$CP = 10 \times 2$$

$$= 20$$

$$L = 4a$$

$$L\% = \frac{4a}{20} \times 100$$

$$\frac{36 \times 100}{10} = \frac{4a \times 100}{20}$$

$$\frac{36}{10} = \frac{4a}{20}$$

$$a:b = 3:2 \text{ Ans.}$$

एक व्यक्ति तीन वस्तुओं को समान विक्रम मूल्य पर बेचता है। वह पहली वस्तु को 20% के लाभ पर, दूसरी 10% की हानि पर तथा तीसरी को 25% की हानि पर बेचता है। इस प्रकार प्रत्येक वस्तु में उसे कुल मिलाकर 40% की हानि होती है। तो क्रम मूल्य ज्ञात करें?

Solution :- 20% ↑ 10% ↓ 25% ↓

$$+\frac{1}{5} \quad -\frac{1}{10} \quad -\frac{1}{4}$$

$$SP_1 = 6 \quad SP_2 = 9 \quad SP_3 = 3$$

$$\text{But } SP_1 = SP_2 = SP_3 = 18 \text{ (LCM)}$$

$$CP$$

$$SP$$

$$15 = 5 \times 3$$

$$6 \times 3 = 18$$

$$5 \rightarrow 120$$

$$20 = 10 \times 2$$

$$9 \times 2 = 18$$

$$1 \rightarrow 24$$

$$24 = 4 \times 6$$

$$3 \times 6 = 18$$

$$CP(59) \times 24 = 1416$$

$$\frac{59}{54} \xrightarrow{-5} \frac{54}{54}$$

$$CP_1 \rightarrow 15 \times 24 = 360$$

$$CP_2 \rightarrow 20 \times 24 = 480$$

A sells an article to B at a profit of 10%.
and B sells the article back to A at a loss of 10%.
find the overall profit and loss % of A.

a] A neither gains nor loses b] A loss 1%

c] A makes a profit of 20% d] A makes a profit of 11%

Solution:- I

Let CP of A = 100

CP for B = $100 + 10 = 110$

Now B sells it at ~~gain~~ loss
of 10% to A

Now CP of A this time = 99

Total Profit of A = 11

%

II

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MATHS.

By **SUNIL SIR**

SSC

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Near Guru Nanak School, Nagori Gate, Hissar-125001

97284-35915

Simple Interest

SIMPLE INTEREST

(साधारण व्याज)

SI/CI are Extension of Percentages.

"Credit The giver, Debit the Receiver"

$$\# \text{ SI (साधारण व्याज) } = \frac{P \times R \times T}{100}$$

where $P \rightarrow$ Principle (मूलधन)

$R \rightarrow$ Rate of Interest (दर)

$T \rightarrow$ Time (समय)

Amount (मिफाद्यन)

$$A = P + \text{SI/CI}$$

Rate (R%) $\Rightarrow$ Interest for 1 year for every 100 ₹ is called Rate of Interest.

5% Rate means $\rightarrow$ ₹5 interest on 100 for 1 yr

In Simple Interest Principle (मूलधन) हमेशा Same रहता है। OR

Simple Interest remains same for same period of time. e.g. If SI of 2 yrs = 20

$$\Rightarrow \text{SI of 4 yrs} = 40$$

$$\text{SI of 8 yrs} = 80$$

Note:— whenever it is not mentioned whether we have to take CI or SI we should take SI.

Principle is always 100%.

if rate of interest is $R\%$.

$\Rightarrow$ interest for 1 yr is $R\%$ of P

" " 2 yrs is $2R\%$ of P

" " 10 yrs is $10R\%$ of P

" " T yrs is $TR\%$ of P

generally $SI = (RT\%)$ of P

$\rightarrow$ Effective Rate

e.g.

IF $R = 5\%$.

$T = 6$ years

$P = 100\%$

$SI = 30\%$

$A = 130\%$

IF $P = 15625$, $R = 2.5\%$ p.a., $T = 4$ yrs, $SI = ?$

Solution: $\rightarrow$ effective Rate $= RT\% = 2.5 \times 4 = 10\%$

P (Principal) $\rightarrow 100\% \rightarrow 15625$

$SI \rightarrow RT\% \rightarrow 10\% \rightarrow 1562.5 \text{ ₹ Ans.}$

$R = 4\%$, $T = 5$ yrs, $SI = 80$, $P = ?$

Solution: $\rightarrow$ effective Rate $= RT\% = 4 \times 5 = 20\%$

$SI \rightarrow 20\% \rightarrow 80$

$P \rightarrow 100\% \rightarrow \frac{80}{20} \times 100 = 400 \text{ ₹ Ans.}$

$$\begin{aligned}
 \# \text{ Amount (मिक्ताधन)} &= P + SI \\
 &= 100\% + RT\% \\
 &= (100 + RT)\%
 \end{aligned}$$

$$\# R = 5\%, T = 4 \text{ yrs}, P = 2000, A = ?$$

$$\text{Solution :- } P = 100\%$$

$$RT\% = 5 \times 4 = 20\% \Rightarrow A = 120\%$$

$$(P) 100\% \longrightarrow 2000$$

$$(A) 120\% \longrightarrow \frac{2000}{100} \times 120 = 2400 \text{ ₹ ANS.}$$

$$\# R = 5\%, T = 4 \text{ yrs}, A = 480, P = ?$$

$$\text{Solution :- } P = 100\% \quad RT\% = 5 \times 4 = 20\%$$

$$\Rightarrow A = 120\%$$

$$A (120\%) \longrightarrow 480$$

$$P (100\%) \longrightarrow \frac{480}{120} \times 100 = 400 \text{ ₹ ANS.}$$

IN how many years ₹ 500 will Amount to ₹ 625 at 5% p.a.

$$\text{Solution :- } A = P + SI$$

$$\downarrow$$

$$100\%$$

$$P (500) \longrightarrow 100\%$$

$$A (625) \longrightarrow \frac{100\%}{500} \times 625 = 125\%$$

$$\Rightarrow SI = 125\% - 100\% = 25\% = RT\%$$

$$\text{but } R = 5\% \Rightarrow T = \frac{25}{5} = 5 \text{ yrs ANS.}$$

एक निश्चित राशि 4 साल में 360 और 10 साल में 450 हो जाती है। तो मूलधन और व्याज की दर बताओ ?

Solution:-

$$\begin{array}{rcl} 10 \text{ years} & = & 450 \\ 4 \text{ year} & = & 360 \\ \hline 6 \text{ year} & = & 90 \Rightarrow 1 \text{ year} = 15 \end{array}$$

Principal = $360 - 4 \times 15 = 300$

$\frac{300 \times R \times 1}{100} = 15 \Rightarrow R = 5\%$

एक निश्चित राशि 2 years में 720 रु और 5 साल में 1020 रु हो जाती है तो मूलधन और व्याज की दर बताओ ?

Solution:-

$$\begin{array}{rcl} 5 \text{ साल में व्याज} & = & 1020 \\ 2 \text{ साल में व्याज} & = & 720 \\ \hline 3 \text{ साल में व्याज} & = & 300 \\ 1 \text{ साल में व्याज} & = & \frac{300}{3} = 60 \end{array}$$

Principal = $720 - 60 \times 2 = 600$

$\frac{600 \times R \times 2}{100} = 60 \Rightarrow R = 10\%$

कोई धन राशि साधारण व्याज की दर से 4 साल में मिल जाती है। यदि व्याज की दर 5% कम होता तो 400 रु कम व्याज मिलता तो मूलधन ज्ञात करो ?

Solution:-

20%	—	400
1%	—	20
100%	—	<u>2000 Ans.</u>



₹ 425 amounts to ₹ 476 in 3 yrs at certain rate of interest. Find rate of interest?

Solution:- $A = P + SI$

↓
100%

$P \quad 425 \rightarrow 100\%$

$A \quad 476 \rightarrow \frac{100\%}{425} \times 476 \rightarrow 112\%$

$SI = A - P = 112\% - 100\% = 12\% = RT\%$

but $T = 3 \text{ yrs} \Rightarrow R\% = \frac{12}{3} = 4\% \text{ Ans.}$

A certain Principle in four years will become 3 times. Find the Rate?

Solution:- I (Formula)

$P = x \quad A = 3x$
 $\quad \quad \quad \downarrow$
 $\quad \quad \quad SI$

$2x = \frac{x \times R \times T}{100}$

$\Rightarrow R = 50\% \text{ Ans}$

II (CONCEPT)

$A = P + SI$

$P = ₹ 100 \quad A = ₹ 300$

$\Rightarrow SI = ₹ 200 = RT$

but $T = 4 \text{ yrs}$

$\therefore R = \frac{200}{4} = 50\%$

III (DIRECT)

$R = \frac{(N-1) \times 100}{T}$

$= \frac{3-1}{4} \times 100$

$= 50\%$

कोई राशी 5% वार्षिक साधारण व्याज की दर से एक निश्चित समय में दुगुनी हो जाती है। समय ज्ञात कीजिए

Solution:- I (Formula)

$P = x \quad A = 2x$
 $\quad \quad \quad \downarrow$
 $\quad \quad \quad SI$

$x = \frac{x \times 5 \times T}{100}$

$\Rightarrow T = 20 \text{ yrs Ans}$

II (CONCEPTUAL)

$A = P + SI$

$P = ₹ 100 \quad A = ₹ 200$

$SI = ₹ 100 = RT\%$

but $R = 5\%$

$\Rightarrow T = \frac{100}{5} = 20 \text{ yrs}$

III (DIRECT)

$R = \frac{N-1}{T} \times 100$

$5 = \frac{2-1}{T} \times 100$

$\Rightarrow T = 20 \text{ yrs}$

अगर समय महीना में या दिन में दिया गया हो तो उसको वर्ष में बदलना होगा।

e.g. 6 months = $\frac{6}{12} = \frac{1}{2}$ वर्ष 3 months = $\frac{3}{12} = \frac{1}{4}$ वर्ष

73 days = $\frac{73}{365} = \frac{1}{5}$ वर्ष, 146 days = $\frac{146}{365} = \frac{2}{5}$ वर्ष

219 days = $\frac{219}{365} = \frac{3}{5}$ वर्ष, 292 days = $\frac{292}{365} = \frac{4}{5}$ वर्ष

6000 रु. पर 6% वार्षिक दर से 8 माह का साधारण व्याज तथा सिफाईशन ज्ञात करो?

Solution:- I (FORMULA)

$$SI = \frac{P \times R \times T}{100}$$

$$= \frac{6000 \times 6 \times 8}{100 \times 12} = 240$$

$$A (\text{सिफाईशन}) = P + SI$$

$$= 6000 + 240$$

$$= 6240/-$$

II (CONCEPTUAL)

$$T = \frac{8}{12} = \frac{2}{3} \text{ yrs}, R = 6\%$$

$$RT\% = \frac{2}{3} \times 6 = 4\%$$

$$SI \rightarrow 4\% \text{ of } 6000$$

$$= 240$$

$$A = 6000 + 240$$

$$= 6240$$

3200 रु. पर 4 अप्रैल 2012 से 16 जून 2012 तक का 5% वार्षिक दर पर साधारण व्याज कितना होगा?

Solution:- Days = April May June

$$= 26 + 31 + 16$$

$$= 73 \text{ दिन}$$

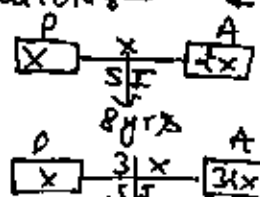
$$\text{Time} = \frac{73}{365} = \frac{1}{5} \text{ वर्ष}$$

Note:- पहले दिन का व्याज नहीं लगता लेकिन अंतिम दिन पर लगता है।

$$SI = \frac{3200 \times 5 \times \frac{1}{5}}{100} = 32 \text{ रु. Ans.}$$

A sum of money doubled in 8 yrs at the rate of simple interest. In how much time it will become $3\frac{1}{2}$ times?

Solution:- I (Simple)



$$3\frac{1}{2} \times 8 = 28 \text{ yrs Ans.}$$

II (TRICKY)

$$\frac{(N_1 - 1)}{(N_2 - 1)} = \frac{T_1}{T_2}$$

$$\frac{2 - 1}{3\frac{1}{2} - 1} = \frac{8}{T_2} \Rightarrow T_2 = 28$$

कोई राशी 4 साल में 3 गुनी हो जाती है। वही राशी साधारण व्याज की दर से 7 गुनी होने में कितना समय लगेगा?

Solution:- I (Simple)

माना मूलधन = 1 रु

मिश्रधन = 3 रु

$$SI = 3 - 1 = 2 \text{ रु}$$

2 रु SI $\rightarrow$ 4 साल

7 गुना में 1 अपना है।

$$SI \rightarrow 7 - 1 \rightarrow 6 \text{ रु}$$

2 $\rightarrow$ 4 साल

$$6 \rightarrow \frac{4}{2} \times 6 = 12 \text{ साल उत्तर}$$

II (TRICKY)

$$\frac{(N_1 - 1)}{(N_2 - 1)} = \frac{T_1}{T_2}$$

$$\frac{3 - 1}{7 - 1} = \frac{4}{T_2}$$

$$\Rightarrow T_2 = 12 \text{ yrs}$$

कोई राशी 5 साल में साधारण व्याज की दर से 5 गुनी हो जाती है। तो वही राशी कितने समय में 7 गुना हो जाएगी?

$$\text{Solution:- } \frac{(N_1 - 1)}{(N_2 - 1)} = \frac{T_1}{T_2} \Rightarrow \frac{5 - 1}{7 - 1} = \frac{5}{T_2}$$

$$\Rightarrow T_2 = 7.5 \text{ साल}$$

A sum of money was invested at SI at a certain rate for 3 years. Had it been invested at 4% higher rate. It would have fetched ₹ 480 more. Find the principal?

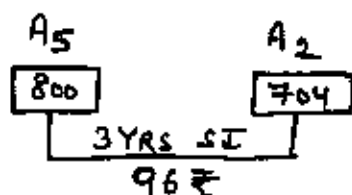
Solution:- Additional money obtained
 $= 4\% \text{ for } 3 \text{ yrs} = 12\% \text{ of } P$

$$12\% \longrightarrow 480$$

$$100\% (P) \longrightarrow \frac{480}{12} \times 100 = 4000 \text{ ₹ Ans.}$$

A certain sum of money amounts to ₹ 704 in 2 yrs and ₹ 800 in 5 yrs. Find the Principal and rate?

Solution:-



$$\text{Interest for } 3 \text{ yr} = 96$$

$$) \quad) \quad 1 \text{ yr} = \frac{96}{3} = 32$$

$$) \quad) \quad 2 \text{ yrs} = 32 \times 2 = 64$$

$$\therefore P = A_2 - SI_2 = 704 - 64 = 640 \text{ Ans.}$$

$$\text{Rate (R) \%} = \frac{32 \times 100}{640 \times 1} = 5\% \text{ Ans}$$

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If the Rate of Interest for first 3 years is 6% p.a. for the next 4 years is 7% p.a. and for the period beyond 7 years is 7.5% p.a. If a man lent out ₹ 1200 for 11 years. Find the total Interest earned by him?

Solution:- 3 yrs — $6\% \times 3 = 18\%$

4 yrs — $7\% \times 4 = 28\%$

4 yrs — $7.5\% \times 4 = 30\%$

76% of P

Simple Interest = 76% of P = 76% of 1200

= $(75\% + 1\%)$ of 1200

= $\frac{3}{4} \times 1200 + 12 = 912$ Ans.

एक व्यक्ति एक निश्चित धन राशि बैंक में जमा करता है। यदि व्याज की दर प्रथम दो वर्ष 3% वार्षिक और अगले तीन वर्ष तक 8% वार्षिक और अन्तिम 1 वर्ष 10% वार्षिक साधारण व्याज हो तो और अगर 6 साल में कुल ₹ 20 रु. साधारण व्याज प्राप्त हुआ हो तो हलचल बताओ ?

Solution:- 2 yrs → $2 \times 3\% = 6\%$

3 yrs → $3 \times 8\% = 24\%$

1 yrs → $1 \times 10\% = 10\%$

SR → 40% of P

40% → 1520

100% (P) → $\frac{1520}{40} \times 100 = 3800$ Ans.

A man lent 2000 partially at 5% p.a. and balance at 4% p.a. If he receives ₹ 92 as annual interest. Find the amount lent at 5%?

Solution:- I (BASIC)

Let ₹ invested @ 5%.

$$\Rightarrow 5\% \text{ of } x + (2000 - x)4\% = 92$$

$$\Rightarrow 5x - 4x - 8000 = 9200$$

$$\Rightarrow x = 1200 \text{ Ans}$$

II (Alligation)

$$R\% = \frac{92}{2000} \times 100 = 4.6\%$$

$$\begin{array}{ccc} 4\% & & 5\% \\ & \swarrow \quad \searrow & \\ & 4.6\% & \\ & \swarrow \quad \searrow & \\ .4 & & .6 \\ \times & & \times 3 \\ 2 & & 1800 \end{array}$$

III (रामबाग)

Let whole amt @ 4%

$$SI = \frac{2000 \times 4}{100} = 80$$

$$\text{gap} = 92 - 80 = 12$$

$$5\% - 4\% \rightarrow 12$$

$$1\% \rightarrow 12$$

$$100\% \rightarrow 1200$$

A man lent ₹ 1600 part of which he lent at 4% and rest at 5% p.a. SI. If the total interest received was ₹ 700 in one year, the money lent at 4% was?

Solution:- I (Alligation)

$$\text{overall } R\% = \frac{700}{1600} \times 100$$

$$= \frac{350}{80}$$

$$\begin{array}{ccc} 4\% & & 5\% \\ & \swarrow \quad \searrow & \\ & \frac{350}{80} & \\ & \swarrow \quad \searrow & \\ 50 & & 30 \end{array}$$

$$\begin{array}{ccc} 50 & & 30 \\ 5 & & 3 \end{array}$$

at 4%.

$$= \frac{16000 \times 5}{8} = 10,000 \text{ Ans.}$$

II (रामबाग)

Let all (16000) @ 5%.

$$SI = \frac{16000 \times 5}{100} = 800$$

$$\text{gap } 800 - 700 = 100$$

$$5\% - 4\% \rightarrow 100$$

$$1\% \rightarrow 100$$

$$100\% \rightarrow 10000$$

(जिसको मानेंगे वो नहीं आसगा दूसरा आसगा)

Sunil borrowed a sum of 39,000. He took a part of it at 12% and remaining at 10%. At the end of two years he returned 36480. What amount was borrowed at 12% Rate?

Solution:- SI for 2yr = 6480

$$SI \text{ for } 1\text{yr} = 3240$$

Let all (39000) invested at 10%.

$$SI = 39,000 \times 10\% = 3900$$

$$\text{gap} = 240$$

$$12\% - 10\% = 2\%$$

$$2\% \rightarrow 240$$

$$100\% \rightarrow 12000$$

(जिसको मानेंगे वो नहीं आसगा)

A sum of ₹ 14400 is divided into 3 parts such that 1st part is invested at 2% p.a. for 3 years and part ② 3% p.a. for 4 years and 3rd part at 4% p.a. for 5 years. If simple interest of all the 3 parts are equal find Principal of each part?

Solution:- let principal of 3 parts is P_1, P_2 & P_3

SI₁

SI₂

SI₃

$$\text{let } 6\% \text{ of } P_1 = 12\% \text{ of } P_2 = 20\% \text{ of } P_3 = 60 \text{ [Arbit]} \\ \Rightarrow P_1 : P_2 : P_3 = 10 : 5 : 3$$

$$P_1 = \frac{14400}{18} \times 10 = 8000, P_2 = \frac{14400}{18} \times 5 = 4000, P_3 = 2400$$

A sum of ₹ 18750 is left by will by a father to be divided between two sons of 12 and 14 years of age so that when they attain maturity at 18, the amount received by each at 5% p.a. SI will be same. Find the sum allotted to both at present.

Solution:- A_1

A_2

$$(100 + 5\% \times 6) \text{ of } P_1 = (100 + 5\% \times 4) \text{ of } P_2$$

$$130\% \text{ of } P_1 = 120\% \text{ of } P_2$$

$$13P_1 = 12P_2 \Rightarrow \frac{P_1}{P_2} = \frac{12}{13} \text{ OR } P_1 : P_2 = 12 : 13$$

$$\therefore P_1 = \frac{18750}{25} \times 12 = 9000, P_2 = \frac{18750}{25} \times 13 = 9750$$

with a given Rate of interest, the Ratio of the Principal and Amount for a certain period of the time is 4:5. After 3 years with the same Rate of interest the ratio of Principal and Amount becomes 5:7. The Rate of interest is?

Solution:- (Principal is always same in SI)

P	A	I	Time
4	5	1	T yrs
5	7	2	T+3 yr
we will make P same [Lcm 20]			
P	A	I	
20 = 4 × 5	5 × 5 = 25	5	T
20 = 5 × 4	7 × 4 = 28	8	T+3

$$I \text{ on } ₹ 20 \text{ for } 3 \text{ yr} = 8 - 5 = 3$$

$$R\% = \frac{3 \times 100}{20 \times 3} = 5\%$$

if a person invested some amount at the rate of 12% p.a. SI and remaining at 10% p.a. SI, he receives yearly interest ₹ 130. Had he interchanged the amount invested he would have received ₹ 134. How much money did he invest at different rates?

a) 500 @ 10%, 800 @ 12%

c) 800 @ 10%, 400 @ 12%

b) 700 @ 10%, 600 @ 12%

d) 700 @ 10%, 500 @ 12%

Solution:- I (BASIC)

$$12\% \text{ of } X + 10\% \text{ of } Y = 130$$

$$10\% \text{ of } X + 12\% \text{ of } Y = 134$$

$$\Rightarrow X = 500, Y = 700$$

Option D is CORRECT

II (TRICKY)

$$\text{Diff in 2 rates} = 2\%$$

$$\text{Diff in Interest} = ₹ 4$$

$$\text{Diff in Amount} = \frac{4}{2} \times 100 = 200$$

only diff of 200 in option D

Das ✓

एक निश्चित धन राशि, एक निश्चित समय के लिए, एक निश्चित साधारण व्याज की दर पर निवेश किया जाता है। यदि हितद्वयन से 20% बढ़ा दिया जाय और उसने दर को $\frac{6}{5}$ गुना कर दिया जाय और समय को $\frac{2}{3}$ कर दिया जाय तो नया साधारण व्याज 2500 हो जाता है। प्रारंभिक साधारण व्याज ज्ञात करें।

Solution:- [जो Effect Principal, Rate, Time पर पड़ेगा वही effect व्याज पर पड़ेगा]

$$\therefore \text{New SI} = \text{Old SI} \times \frac{120}{100} \times \frac{6}{5} \times \frac{2}{3}$$

$$2500 = \text{Old SI} \times \frac{120}{100} \times \frac{6}{5} \times \frac{2}{3}$$

$$\Rightarrow \text{Old SI} = 2500 \text{ ₹ Ans.}$$

एक निश्चित धन राशि एक निश्चित समय के लिए के लिए निवेश की जाती है। यदि साधारण व्याज की दर 5% वार्षिक होता तो मिफावदन 80 रु होता, जब व्याज की दर 2% होता तो मिफावदन 40 रु होता तो समय क्या था?

Solution:- I (BASIC)

$$\text{Let } P = 100 \%$$

$$SI = RT\% = St\% \cdot P\%$$

$$\frac{100 + St}{100 + 2t} = \frac{80}{40}$$

$$\Rightarrow 100 + St = 200 + 4t$$

$$\Rightarrow \boxed{t = 100 \text{ yrs}}$$

II (TRICKY)

$$t = \frac{A_1 - A_2}{A_1 R_2 - A_2 R_1} \times 100$$

$$= \frac{80 - 40}{200 - 100} \times 100$$

$$= \frac{40}{100} \times 100$$

$$= 100 \text{ yrs}$$

INSTALLMENT [किरात]

$$\# \quad X = \frac{100 D}{100T + \frac{R \times T(T-1)}{2}}$$

where $X \rightarrow$ installment [equal]

$T \rightarrow$ Time $R \rightarrow$ Rate (दर)

$D \rightarrow$ Debt (ऋण) Amount

OR Equate Future Value (Amount) सिद्धिमान

FOR MONTHLY you can use it

$$P + \frac{P \times R \times T}{12 \times 100} = X \times T + \frac{R \times X}{12 \times 100} [1 + 2 + \dots + (T-1)]$$

what annual installment will discharge a debt of ₹ 2360 due in four years at 12% p.a. Simple interest?

Solution:- I (FORMULA)

$$\begin{aligned} X &= \frac{100 D}{100T + \frac{RT(T-1)}{2}} \\ &= \frac{100 \times 2360}{100 \times 4 + \frac{12 \times 4 \times 3}{2}} \\ &= \frac{2360 \times 100}{472} = 500 \text{ ₹ Ans.} \end{aligned}$$

II Let each I = 100
[48वीं किरात पर 3 साल, दूसरी पर 2 साल, तीसरी पर 1 साल और चौथी पर 0 साल का ब्याज लगेगा]

$$100 + 12 \times 3 = 136\%$$

$$100 + 12 \times 2 = 124\%$$

$$100 + 12 \times 1 = 112\%$$

$$100 + 12 \times 0 = 100\%$$

$$\frac{472\%}{100\%}$$

$$\therefore 472\% \rightarrow 2360$$

$$100\% \rightarrow 500 \text{ Ans.}$$

1740 ₹ की राशि 5 परावर वार्षिक किरतों में देम है। यदि प्रत्येक किरत पर शेष समय के लिए 8% की दर से साधारण व्याज देम हो तो प्रत्येक किरत की राशि ज्ञात करो?

Solution:— I (Formula)

$$X = \frac{100 \times 1740}{100 \times 5 + \frac{8 \times 5 \times 4}{2}}$$

$$= \frac{100 \times 1740}{580} = 300 \text{ ₹}$$

IIIrd

$$5x + \frac{x \times 8}{100} [1+2+3+4] = 1740$$

$$\Rightarrow 58x = 17400$$

$$\Rightarrow x = 300 \text{ ₹}$$

II माना प्रत्येक किरत = 100%

$$\begin{aligned} 100 + 4 \times 8\% &= 132\% \\ 100 + 3 \times 8\% &= 124\% \\ 100 + 2 \times 8\% &= 116\% \\ 100 + 1 \times 8\% &= 108\% \\ 100 + 0 \times 8\% &= 100\% \end{aligned}$$

$$580\% \longrightarrow 1740$$

$$\Rightarrow 100\% \longrightarrow \frac{1740}{580} \times 100 = 300$$

12800 की राशि 3 वार्षिक किरतों में देम है। पहली किरत की राशि दूसरी किरत की राशि की आधी और तीसरी की $\frac{1}{3}$ है। यदि प्रत्येक किरत समान समय के लिए 10% वार्षिक व्याज देम हो तो प्रत्येक किरत की राशि ज्ञात करो?

Solution:— $P_1 : P_2 : P_3$ we will find first

$$P_1 = \frac{P_2}{2} = \frac{P_3}{3} \Rightarrow P_1 : P_2 : P_3 = 1 : 2 : 3$$

$$100 + \frac{100 \times 10 \times 2}{100} = 120\%$$

$$200 + \frac{200 \times 10 \times 1}{100} = 220\%$$

$$300 + 0 = 300\%$$

$$640\%$$

$$\text{OR } 100 \quad 200 \quad 300$$

[पहली किरत पर 2 वर्ष का,
दूसरी पर 1 वर्ष का और
तीसरी पर 0 वर्ष का व्याज
व्याज]

$$640\% \longrightarrow 12800$$

$$100\% \longrightarrow 2000$$

$$200\% \longrightarrow 4000$$

$$300\% \longrightarrow 6000$$

cash price of a mobile is ₹ 1500. but a man purchased for ₹ 350 cash down payment and 3 equal monthly installment of ₹ 400 each. Find the Rate of Interest?

Solution:- $Rem = 1500 - 350 = 1150$

Find equivalent 1 month P

$$\begin{array}{r|l} 1150 - 400 & \text{also } 400 \times 3 = 1200 \\ 750 - 400 & \text{but } Rem = 1150 \\ 350 & \Rightarrow SI = 50 \\ \hline 2250 & \end{array}$$

$$\therefore \frac{2250 \times R \times 1}{100 \times 12} = 50 \Rightarrow R = 26\frac{2}{3}\%$$

OR

$$P + \frac{PRT}{12 \times 100} = XT + \frac{RX}{12 \times 100} [1 + 2 + \dots + (n-1)]$$

$$1150 + \frac{1150 \times R \times 3}{12 \times 100} = 400 \times 3 + \frac{400R}{1200} [1+2+3]$$

$$1150 + \frac{115R}{40} = 1200 + R$$

$$\Rightarrow 75R = 2000$$

$$R = 26\frac{2}{3}\% \text{ Ans.}$$

1 व्यक्ति 10 रु. प्रति महीना लेता है और वह 1 रु. मासिक बचत में 11 महीने में जुटाता है। व्याज की दर क्या होगी?

Solution:- equivalent principle for 1 month

पहले महीने 10 रु. पर, दूसरे महीने 9 रु. पर and so on
व्याज लगगा क्योंकि वह हर महीने 1 रु. बचा रहा है।

$$SI = \frac{55 \times R \times 1}{100 \times 12} = 1 [1+0]$$

$$\Rightarrow R = \frac{240}{11}\% = 21\frac{9}{11}\%$$

$$II \quad P + \frac{PTR}{12 \times 100} = NX + \frac{RX}{12 \times 100} [1 + 2 + \dots + (n-1)]$$

$$X = 1, N = 11, P = 10, T = 11$$

$$\Rightarrow R = \frac{240}{11}\%$$

$$10 + \frac{10 \times 11 \times R}{100} = 1 \times 11 + \frac{R \times 1}{12 \times 100} [1 + 2 + \dots + 10] = 21\frac{9}{11}\%$$



Compound Interest

COMPOUND INTEREST

$$\# \text{ Amount (A)} = P \left[1 + \frac{R}{100} \right]^T$$

$$\text{COMPOUND Interest (CI)} = A - P$$

$$= P \left[1 + \frac{R}{100} \right]^T - P = P \left\{ \left(1 + \frac{R}{100} \right)^T - 1 \right\}$$

IN Compound Interest principle keep on changing / compounding / updating after specific period of time according to term specified like yearly / half yearly / Quarterly etc

IN simple words we can say that compound interest is interest on interest or principle keep on compounding (updating)

FIRST year Simple Interest and compound interest are equal.

$$[SI_{\text{first yr}} = CI_{\text{first year}}]$$

After one year $CI > SI$

If Rate of Interest ARE Different

$$A = P \left(1 + \frac{R_1}{100} \right) \left(1 + \frac{R_2}{100} \right) \left(1 + \frac{R_3}{100} \right) \dots \dots$$

# Interest Compounded	New Rate	New Time	Amount
Half yearly	$R/2$	$2T$	$P \left[1 + \frac{R}{2 \times 100} \right]^{2T}$
Quarterly	$R/4$	$4T$	$P \left[1 + \frac{R}{4 \times 100} \right]^{4T}$
Monthly	$R/12$	$12T$	$P \left[1 + \frac{R}{12 \times 100} \right]^{12T}$

when Interest is compounded annually but time is in fraction. e.g. $T = 2\frac{2}{5}$ yrs, then

$$\text{Amount (A)} = P \left[1 + \frac{R}{100} \right]^2 \left[1 + \frac{\frac{2}{5}R}{100} \right]$$

Difference B/w Simple and Compound Interest

e.g. $P = 1000$, $R = 10\%$, $T = 3$ yrs

	SIMPLE INTEREST		COMPOUND INTEREST	
	P	I	P	I
1st yr.	1000	100	1000	100 → Same
2nd yr.	1000	100	1100	110
3rd yr.	1000	100	1210	121
		300		331

So we can see that In Simple Interest P as well as I is same for all years but in case of CI both P & I keep on updating coz in CI Interest also become principle for Next year

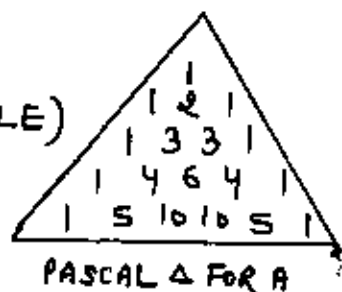
Different Techniques For Calculating CI

If $P = 20000$, $R = 10\%$ Per Annum, $T = 3$ yrs
find the compound interest $CI = ?$

Technique 1 (FORMULA METHOD)

$$CI = P \left[\left(1 + \frac{R}{100} \right)^T - 1 \right] = 20000 \left[\left(1 + \frac{10}{100} \right)^3 - 1 \right] = 6620$$

Technique 2 (PASCAL'S TRIANGLE)



$$CI \text{ FOR } 2 \text{ yrs} = 2A + B$$

$$CI \text{ FOR } 3 \text{ yrs} = 3A + 3B + C$$

$$CI \text{ FOR } 4 \text{ yrs} = 4A + 6B + 4C + D$$

$$CI \text{ FOR } 5 \text{ yrs} = 5A + 10B + 10C + 5D + E$$

where $A = R\%$ of Principle [$R\%$ is Rate]

$B = R\%$ of A

$C = R\%$ of B , $D = R\%$ of C , $E = R\%$ of D

$$\text{For Amount } A_2 = P + 2A + B$$

$$A_3 = P + 3A + 3B + C$$

$$A_4 = P + 4A + 6B + 4C + D$$

$$\text{Here } CI_3 = 3 \left[\underset{\substack{\downarrow \\ A = 2000}}{10\% \text{ of } 20000} \right] + 3 \left[\underset{\substack{\downarrow \\ B = 200}}{10\% \text{ of } 2000} \right] + \left[\underset{\substack{\downarrow \\ C}}{10\% \text{ of } 200} \right]$$

$$= 3 \times 2000 + 3 \times 200 + 20 = 6620 \text{ Ans.}$$

EXAMPLE SOLVED BY PASCAL TRIANGLE

$$\# \quad P = 5000, R = 5\%, T = 3 \text{ yrs}, CI = ?$$

$$CI_3 = 3A + 3B + C$$

$$A = 5\% \text{ of } 5000 = 250$$

$$B = 5\% \text{ of } A = 5\% \text{ of } 250 = 12.5$$

$$C = 5\% \text{ of } B = 5\% \text{ of } 12.5 = .625$$

$$\Rightarrow CI_3 = 3 \times 250 + 3 \times 12.5 + .625 = 788.125 \text{ ANS.}$$

$$\# \quad P = 25000, R = 12\%, T = 3 \text{ yrs}, CI = ?$$

$$CI = 3(A) + 3(B) + 1(D)$$

$$A = 12\% \text{ of } 25000 = 3000$$

$$B = 12\% \text{ of } A = 12\% \text{ of } 3000 = 360$$

$$C = 12\% \text{ of } B = 12\% \text{ of } 360 = 43.2$$

$$\Rightarrow CI = 3 \times 3000 + 3 \times 360 + 43.2 = 10123.2$$

$$\# \quad P = 25000, R = 10\%, T = 4 \text{ yrs}, CI = ?$$

$$CI_4 = 4(A) + 6(B) + 4(C) + 1(D)$$

$$A = 10\% \text{ of } 25000 = 2500$$

$$B = 10\% \text{ of } A = 10\% \text{ of } 2500 = 250$$

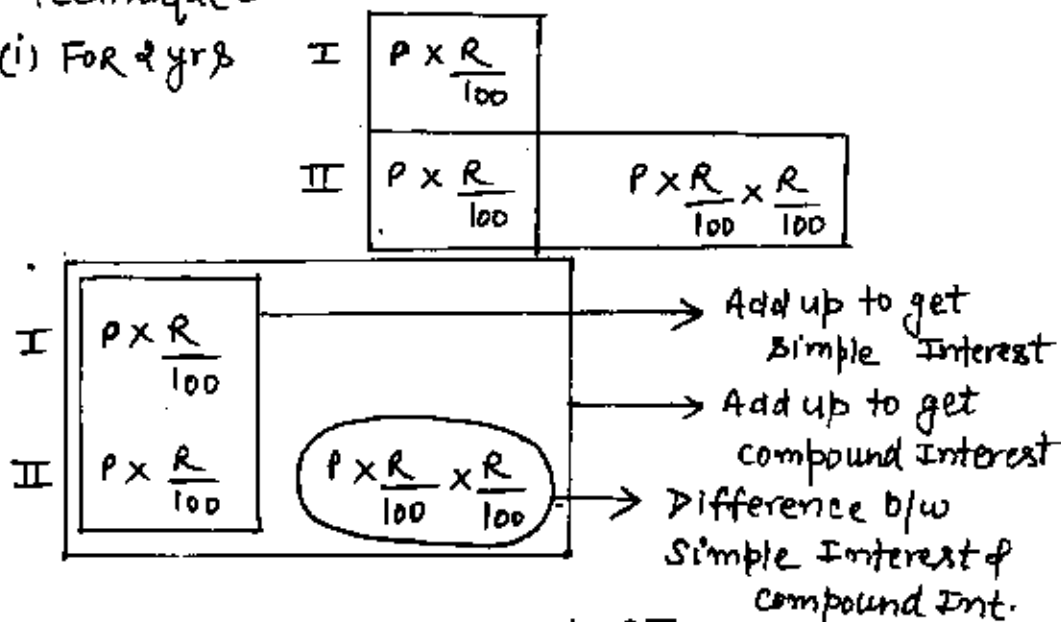
$$C = 10\% \text{ of } B = 10\% \text{ of } 250 = 25$$

$$D = 10\% \text{ of } C = 10\% \text{ of } 25 = 2.5$$

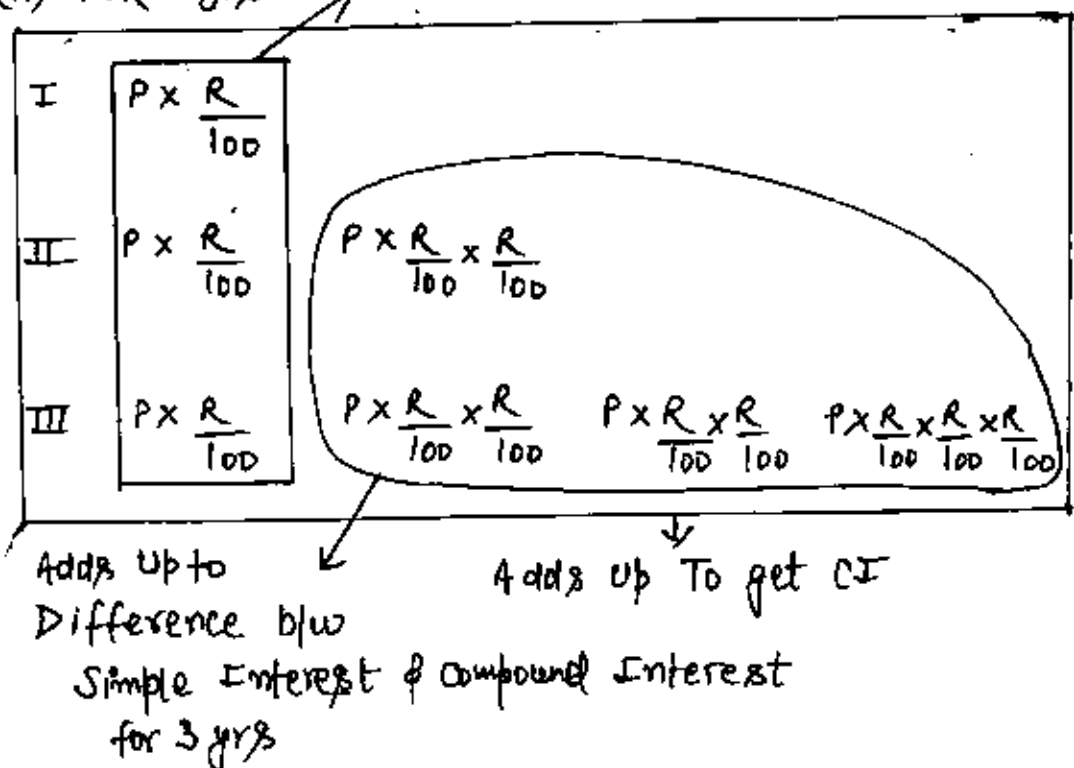
$$CI_4 = 4 \times 2500 + 6 \times 250 + 4 \times 25 + 2.5 = 11602.5 \text{ ANS.}$$

Technique 3

(i) For 2 yrs



(ii) For 3 yrs



$P = 12000$, $R = 10\%$, $T = 2 \text{ yrs}$, $CI = ?$

I YR 1200 $\xrightarrow{10\%}$
 II YR 1200 120

$$CI_2 = 1200 + 1200 + 120 = 2520 \text{ ANS.}$$

Explanation:-

$$1\text{st yr } CI = 1\text{st yr } SI = 10\% \text{ of } 12000 = 1200$$

$$\begin{aligned} 2\text{nd yr } CI &= 10\% \text{ of } 12000 + 10\% \text{ of } 1\text{st yr } CI (1200) \\ &= 1200 + 120 = 1320 \end{aligned}$$

$$CI_2 = CI \text{ 1st YR} + CI \text{ 2nd YR} = 1200 + 1320 = 2520$$

$P = 25000$, $R = 10\%$, $T = 3 \text{ yrs}$, $CI = ?$

1st yr CI 2500

2nd YR CI 2500 250

3rd YR CI 2500 250 250 25

$$\begin{aligned} CI \text{ for 3 year} &= \text{Add All} = 2500 + 2500 + 1500 + \\ &\quad 250 + 250 + 250 + 25 \\ &= 8275 \text{ ANS.} \end{aligned}$$

$$\# \quad P = 20000, R = 10\% \text{ p.a.}, T = 3 \text{ yrs}, CI = ?$$

$$\text{I} \quad 2000$$

$$\text{II} \quad 2000 \quad 200$$

$$\text{III} \quad 2000 \quad 200 \quad 200 \quad 20$$

Add. all to get CI

$$\begin{aligned} CI_3 &= 2000 + 2000 + 2000 + \\ &\quad 200 + 200 + 200 + 20 \\ &= 6620 \end{aligned}$$

Technique 4 (Ratio/Fraction Method)

$$R = 10\% = \frac{1 \rightarrow I}{10 \rightarrow P} \Rightarrow A = P + I = 11$$

Try to assume Principle according to denominator of fraction and no. of years.

Here Assumed Principle = $(P)^n$, where

$$P = 10 \quad \{R = \frac{1}{10}\}$$

P = Denominator

n = No of years

$$\text{Principle}_{\text{New/Assumed}} = (10)^3 = 1000$$

$$P = 1000, R = 10\%, T = 3 \text{ yrs}$$

$$\text{I} \quad 100$$

$$\text{II} \quad 100 + 10$$

$$\text{III} \quad 100 + 10 + 10 + 1$$

$$\Rightarrow CI_3 = 331$$

By Unitary method

$$\text{if } P = 1000 \Rightarrow CI = 331$$

$$\begin{aligned} \text{but } P = 20,000 \Rightarrow CI_3 &= \frac{331}{1000} \times 20,000 \\ &= 6620 \end{aligned}$$



Technique 5 (SUCCESSIVE PERCENTAGE)

$$R = 10\% = \frac{1}{10} \rightarrow I \Rightarrow A = 11$$

	P	A
I	10	11
II	10	11
III	10	11

$$\frac{1000}{331} \rightarrow 1331$$

↓
CI

$$\text{If } P = 1000$$

$$\Rightarrow CI = 331$$

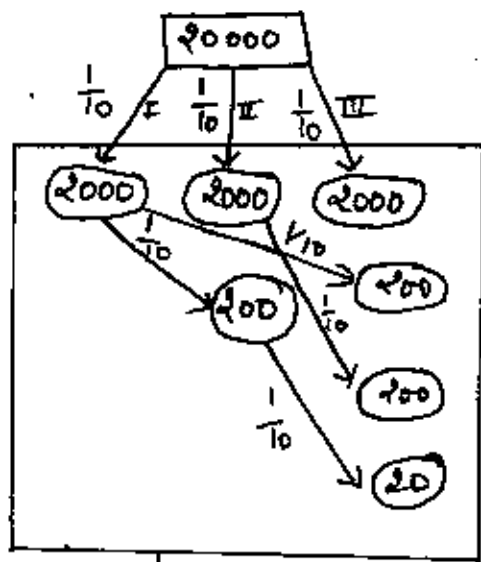
$$\text{but } P = 20000$$

$$\Rightarrow CI = \frac{331}{1000} \times 20000 = 6620 \text{ ANS.}$$

$$\frac{P}{A} = \frac{10 \times 10 \times 10}{11 \times 11 \times 11}$$

$$P:A = 1000:1331$$

Technique 6 (TREE METHOD)

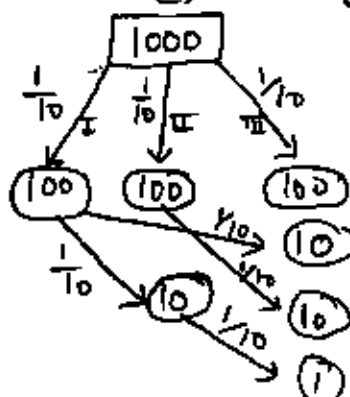


Add all these to get CI

$$CI = 2000 + 2000 + 2000 + 200 + 200 + 200 + 20 + 20 + 20 = 6620 \text{ ANS.}$$

$$\text{OR } R = 10\% = \frac{1}{10} \rightarrow I \Rightarrow A = 11$$

$$\text{Let } P = (10)^3 = 1000 \{T=3 \text{ yrs}\}$$



Add all present in Circle

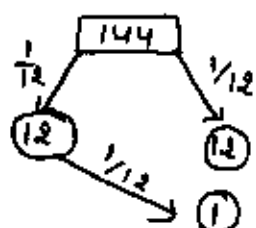
$$CI = 100 + 100 + 100 + 10 + 10 + 10 + 1 + 1 + 1 = 331$$

$$\text{If } P = 1000 \Rightarrow CI = 331$$

$$\text{but } P = 20000 \Rightarrow CI = \frac{331}{1000} \times 20000 = 6620 \text{ ANS.}$$

$P = 28800$, $R = 8\frac{1}{3}\%$, $T = 2Yr$, $CI = ?$

$$R = 8\frac{1}{3}\% = \frac{1}{12} \rightarrow I \quad \text{let } P = (12)^2 = 144 \quad \{ \text{ coz } t=2 \}$$



OR	I	12	
	II	12	1
	$CI_1 = 12 + 12 + 1 = 25$		

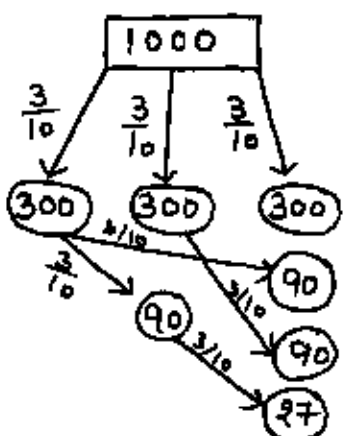
Add all in circle to get $CI = 12 + 12 + 1 = 25$

$$\text{If } P = 144 \Rightarrow CI = 25$$

$$\text{but } P = 28800 \Rightarrow CI_2 = \frac{25}{144} \times 28800 = 5000 \text{ Ans.}$$

$P = 25000$, $R = 30\%$, $T = 3Yrs$, $CI = ?$

$$R = 30\% = \frac{3}{10} \rightarrow I \Rightarrow \text{let } P = (10)^3 = 1000 \quad \{ t=3 \}$$



OR	I	300		
	II	300	90	
	III	300	90	27
	$CI = 1197$			

OR	$R = \frac{3}{10} \rightarrow I \Rightarrow A = 13$												
	<table border="0"> <tr> <td></td> <td>P</td> <td>A</td> </tr> <tr> <td>I</td> <td>10</td> <td>13</td> </tr> <tr> <td>II</td> <td>10</td> <td>13</td> </tr> <tr> <td>III</td> <td>10</td> <td>13</td> </tr> </table>		P	A	I	10	13	II	10	13	III	10	13
	P	A											
I	10	13											
II	10	13											
III	10	13											
(P)	1000 2197 (A)												
	$1197 \rightarrow CI$												

add all in circle
 $CI = 1197$

$$\text{Now If } P = 1000 \Rightarrow CI_3 = 1197$$

$$\text{but } P = 25000 \Rightarrow CI_3 = \frac{1197}{1000} \times 25000 = 29925 \text{ Ans.}$$

$$\# P = 4000, R = 5\%, T = 3 \text{ yrs}, CI = ?$$

$$\text{I } 200$$

$$\text{II } 200 \quad 10$$

$$\text{III } 200 \quad 10 \quad 10 \quad .5$$

$$CI = 200 + 200 + 200 + 10 + 10 + 10 + .5 = 630.5 \text{ Ans}$$

$$\text{OR } R = 5\% = \frac{1}{20} = \frac{21}{20} \rightarrow A$$

$$P \quad A \quad 20 \quad 20 \rightarrow P$$

$$\text{I } 20 \quad 21$$

$$\text{II } 20 \quad 21$$

$$\text{III } 20 \quad 21$$

$$(P) \quad 8000 - 9161 (A)$$

$$1261 \rightarrow CI$$

$$\text{If } P = 8000 \Rightarrow CI = 1261$$

$$\text{but } P = 4000 \Rightarrow CI = \frac{1261}{2} = 630.5$$

$$\# P = 3600, R = 16\frac{2}{3}\%, T = 3 \text{ yrs}, CI = ?$$

$$R = 16\frac{2}{3}\% = \frac{1}{6} \rightarrow \frac{I}{P} \Rightarrow A = 7 \quad \text{OR } \frac{A}{P} = \frac{7}{6}$$

$$\begin{array}{cc} P & A \\ \text{I} & 6 & 7 \end{array}$$

$$\text{II } 6 \quad 7$$

$$\text{III } 6 \quad 7$$

$$(P) \quad 216 \quad 343 (A)$$

$$127 \rightarrow CI$$

$$\text{If } P = 216 \Rightarrow CI = 127$$

$$\text{but } P = 3600 \Rightarrow CI = \frac{127 \times 3600}{216} = 2116.67 \text{ Ans.}$$

$$\# A = 4096, R = 14\frac{2}{7}\%, T = 2 \text{ yrs}, P = ?$$

$$R = 14\frac{2}{7}\% = \frac{1}{7} \rightarrow \frac{I}{P} \Rightarrow A = 8 \quad \text{OR } \frac{A}{P} = \frac{8}{7}$$

$$\begin{array}{cc} P & A \\ \text{I} & 7 & 8 \end{array}$$

$$\text{II } 7 \quad 8$$

$$(P) \quad 49 \quad 64 (A)$$

$$15$$

$$\text{If } A = 64 \Rightarrow P = 49$$

$$\text{but } A = 4096 \Rightarrow P = \frac{49}{64} \times 4096 = 3136 \text{ Ans}$$

$P = 108000$, $R = 16\frac{2}{3}\%$, $T = 2 \text{ yrs}$, $CI = ?$

$R = 16\frac{2}{3}\% = \frac{1}{6} \rightarrow I$ let $P = (6)^2 \{ (D)^n, \text{ Here } D=6 \}$

$CI_1 = \frac{1}{6} \text{ of } 36 = 6$ $= 36$

$n=2$
↓
No of years

I $6 \swarrow \frac{1}{6}$

II $6 \quad 1 \Rightarrow CI_2 = 6 + 6 + 1 = 13$

if $P = 36 \Rightarrow CI_2 = 13$

but $P = 10800 \Rightarrow CI_2 = \frac{13}{36} \times 10800 = 3900 \text{ Ans.}$

$P = 1024000$, $R = 12.5\%$, $T = 3 \text{ yrs}$, $CI = ?$

$R = 12.5\% = \frac{1}{8} \rightarrow I$ let $P = (8)^3 = 512 \{ \because T=3 \text{ yrs} \}$

$CI_1 = \frac{1}{8} \times 512 = 64$

I $64 \swarrow \frac{1}{8}$

II $64 \quad 8 \swarrow \frac{1}{8}$

III $64 \quad 8 \quad 8 \quad 1$

$CI_3 = 64 + 64 + 64 + 8 + 8 + 8 + 1$
 $= 217$

$\Rightarrow$ If $P = 512 \Rightarrow CI = 217$

but $P = 1024000 \Rightarrow CI = \frac{217}{512} \times 1024000$

$= 434000 \text{ Ans.}$

Explanation $\rightarrow$ Always assume $P = (D)^n$

where $D = \text{denominator of fraction}$, $n = \text{no of yrs}$

e.g If $R = 30\% = \frac{3}{10}$, $T = 3 \text{ yrs}$

Assume $P = (10)^3 = 1000$

Technique 7 (Linear METHOD)

$$20000 \xrightarrow[\text{P}]{\frac{10\% \uparrow}{2000}} 22000 \xrightarrow[\text{A}_2]{\frac{10\% \uparrow}{2200}} 24200 \xrightarrow[\text{A}_3]{\frac{10\% \uparrow}{2420}} 26620$$

$$CI_3 = A_3 - P = 26620 - 20000 = 6620 \text{ Ans}$$

Technique 8 (Net Percentage Change)

$$a + b + \frac{ab}{100}$$

$$\text{Here } a = b = 10\% \Rightarrow 10 + 10 + \frac{10 \times 10}{100} = 21\%$$

$$\text{again } a = 21, b = 10 \Rightarrow 21 + 10 + \frac{21 \times 10}{100} = 33.1\%$$

33.1% is equivalent Rate of 3 yrs

$$\Rightarrow CI_3 = 33.1\% \text{ of } 20000 = 6620 \text{ Ans.}$$

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$P = 7500$ $R = 4\%$ $T = 2 \text{ YRS}$ $CI = ?$, $A = ?$

Solution:- I (Ratio)

$$R\% = 4\% = \frac{1 - I}{25 - P} \Rightarrow A = 26$$

$$\begin{array}{r} P \\ \text{I} \quad 25 \rightarrow 26 \end{array}$$

$$\text{II} \quad 45 \rightarrow 26$$

$$625 \rightarrow 676$$

$$\begin{array}{r|l} | \times 12 & | \times 12 \\ 7500 & 8112 \end{array}$$

$$CI = A - P = 612 \text{ Ans.}$$

II (TRICKY)

$$R = 4\% = \frac{1}{25}$$

$$\text{let } P = (25)^2 = (10)^2 = 625$$

$$\text{I} \quad 25$$

$$\text{II} \quad 25 + 1$$

$$\Rightarrow \text{If } P = 625 \Rightarrow A = 676$$

$$\begin{array}{r|l} | \times 12 & | \times 12 \\ 7500 & 8112 \end{array}$$

$$\Rightarrow CI = 8112 - 7500 = 612 \text{ Ans.}$$

$P = 20000$, $R = 5\%$, $T = 3 \text{ YRS}$, $A = ?$

Solution:- I (Ratio)

$$R\% = 5\% = \frac{1 - I}{20 - P} \Rightarrow A = 21$$

$$\begin{array}{r} P \\ \text{I} \quad 20 \rightarrow 21 \end{array}$$

$$\text{II} \quad 20 \rightarrow 21$$

$$\text{III} \quad 20 \rightarrow 21$$

$$8000 \rightarrow 9261$$

$$\begin{array}{r|l} | \times 1.5 & | \times 1.5 \\ 20000 & 23152.50 \end{array}$$

II (TRICKY)

$$\text{let } P = 5\% = \frac{1}{20} = 6000$$

$$\text{I} \quad 300$$

$$\text{II} \quad 300 + 15$$

$$\text{III} \quad 300 + 15 + 15 + 3$$

$$\text{If } P \rightarrow 6000 \Rightarrow A = 6948$$

$$\begin{array}{r} \text{If } P \rightarrow 20000 \Rightarrow A = \frac{6948}{6000} \times 20000 \\ = 23152.50 \text{ Ans.} \end{array}$$

$P = 1920$ $R = 12\frac{1}{2}\%$ $T = 2 \text{ YRS}$ $CI = ?$

$$\text{let } P = \frac{N}{D} = (10)^2 \quad R\% = 12.5\% = \frac{1}{8}$$

$$\Rightarrow \text{let } P = (8)^2 = 64$$

$$\text{I} \quad 8$$

$$\text{II} \quad 8 + 1$$

$$\Rightarrow \text{If } CI = 17 \Rightarrow P = 64$$

$$| \times 30$$

$$510 \text{ Ans.}$$

$$| \times 30$$

$$1920$$

$R = 10\%$, 3rd year CI = 72.60, $P = ?$

Solution:- $R = 10\% = \frac{1}{10} \Rightarrow P = (10)^3 = 1000$

I 100

II $100 + 10$

III $100 + 10 + 10 + 1 = 121$

IF $CI_3 = 121 \Rightarrow P = 1000$

$\times \frac{1}{10}$

$\times \frac{1}{10}$

72.60

600 Ans.

$R = 5\%$, CI for 3rd year = 220.50, $P = ?$

Solution:- $R = 5\% = \frac{1}{20} \Rightarrow P = (20)^3 = 8000$

I 400

II $400 + 20$

III $400 + 20 + 20 + 1 = 441$

IF $CI_3 = 441 \Rightarrow P = 8000$

$\times \frac{1}{2}$

$\times \frac{1}{2}$

220.50

4000 Ans.

$R = 10\%$, CI for fourth year = 665.50, $P = ?$

Solution:- $R = 10\% = \frac{1}{10} \Rightarrow P = (10)^4 = 10000$

I 1000

II $1000 + 100$

III $1000 + 100 + 100 + 10$

IV $1000 + 100 + 100 + 10 + 100 + 10 + 10 + 1 = 1331$

IF $CI_4 = 1331 \Rightarrow P =$

$\Rightarrow$

$\times \frac{1}{10}$

$\times \frac{1}{10}$

665.50

5000

CI for 2nd year = 440, $R_1 = 10\%$, $R_2 = 20\%$, $P = ?$

Solution:- let $P = 2500$ [LCM of $\frac{1}{10}$ & $\frac{1}{5}$]

I 250

II $250 + 50 = 300$

IF $CI_2 = 300 \Rightarrow P = 250$

$\times \frac{1}{5}$

$\times \frac{1}{5}$

440

2500

CI FOR 3rd year = 330, $R_1 = 10\%$, $R_2 = 20\%$, $R_3 = 5\%$
 $P = ?$

Solution:- Let $P = 1000$

IF $CI_3 = 330 \Rightarrow P = 1000$

I = 100

II $100 + 10$

III $(50 + 5 + 10 + 1) = 66$

$\Rightarrow$

$\begin{array}{c} | \times 5 \\ 330 \end{array} \quad \begin{array}{c} | \times 5 \\ \boxed{5000} \text{ Ans.} \end{array}$

III किसी राशि पर 2 साल का SI 480 रु है और CI 492 रु है तो हलचल और व्याज की दर क्या है?

Solution:- SI FOR 2 year = 480

$\Rightarrow$ SI FOR 1 year = 240

SI FOR 1st year = CI FOR 1st year = $\frac{480}{2} = 240$

CI FOR 2nd year = $492 - 240 = 252$

Diff of CI = $252 - 240 = 12$

$\therefore \frac{240 \times R \times 1}{100} = 12 \Rightarrow R = 5\%$ $P = 4800$

किसी राशि पर 3 साल का साधारण व्याज 1200 रु है और वी साल का CI 820. Find the Rate.

Solution :- SI FOR 3 year = 1200

$\Rightarrow$ CI FOR 1st year = SI FOR 1st year = $\frac{1200}{3} = 400$

I 400

II $\boxed{400 + 40} \Rightarrow R = \frac{20}{400} \times 100 = 5\% \text{ Ans.}$

$CI_2 = 820 - 400 = 420$

एक व्यक्ति 390300 ₹ की राशि को अपने दो पुत्रों में इस प्रकार निवेश करता है कि जब वे दोनों 13 वर्ष के हों तो उनको बराबर धन मिले। यदि उनकी आयु क्रमशः 13 वर्ष और 15 वर्ष है। यदि वार्षिक दर 4% वार्षिक चक्रवृद्धि हो तो प्रत्येक बेटे का हिस्सा क्या होगा?

Solution:- I (Basic)

$$P_1 \left[1 + \frac{4}{100}\right]^5 = P_2 \left[1 + \frac{4}{100}\right]^3$$

$$P_1 \left[\frac{26}{25}\right]^2 = P_2$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{625}{676} \Rightarrow P_1 = \frac{390300}{1301} \times 625$$

$$= 187500$$

$$P_2 = \frac{390300}{1301} \times 676 = 202800$$

II (Tricky)

$$\text{Rate} = 4\% = \frac{1}{25}$$

1st Son 2nd Son

$$(26)^2 : (25)^2$$

$$676 \quad 625 = 1301$$

$$\begin{array}{ccc} | \times 300 & | \times 300 & | \times 300 \\ 202800 & 187500 & 390300 \end{array}$$

$P = ?$, $R = 12.5\%$, $T = 3 \text{ yrs}$
 $CI = 4340$ (TOTAL)

Solution:- $R = 12.5\% = \frac{1}{8}$

$$\text{Let } P = (6)^3 = (8)^3 = 512$$

$$\text{I } 64$$

$$\text{II } 64 \quad 8$$

$$\text{III } 64 \quad 8 \quad 8 \quad 1$$

$$\therefore CI \rightarrow 64 + 64 + 64 + 8 + 8 + 8 + 1$$

$$= 217 \Rightarrow P = 512$$

$$\begin{array}{ccc} 20 \times & & | \times 20 \\ 4340 & & \boxed{10140} \end{array}$$

$R = 14\frac{2}{3}\%$, $T = 2 \text{ year}$

CEAR IInd year = 720

$$P = ?$$

Solution:- $R = 14\frac{2}{3}\% = \frac{1}{7}$

$$\text{Let } P = (7)^2 = (7)^2 = 49$$

$$\text{I} = 7$$

$$\text{II } \boxed{7 + 1} = 8$$

$$\therefore CI \rightarrow 8 \Rightarrow P = 49$$

$$\begin{array}{ccc} | \times 90 & & | \times 90 \\ 720 & & 4410 \end{array}$$

Difference between CI and SI

$$T = 2 \text{ years}$$

$$D = \frac{P R^2}{100^2}$$

$$T = 3 \text{ years}$$

$$D = \frac{P R^2}{100^2} \times \frac{(300 + R)}{100}$$

where $D = \text{Diff between CI \& SI} = \text{CI} - \text{SI}$ # $D = 48$, $T = 2 \text{ years}$, $R = 20\%$, $P = ?$

Solution:- I

$$R = 20\% = \frac{1}{5}$$

$$\text{Let } P = (D)^2 = 25$$

$$\text{I } \boxed{5}$$

$$\text{II } \boxed{5} + (1) \rightarrow D$$

$$\text{If } D = 1 \Rightarrow P = 25$$

$$48 \times \frac{1}{48} = 1200 \text{ Ans.}$$

$P = 2500$, $T = 2 \text{ yrs}$
 $\text{CI} - \text{SI} = ?$, $R = 6\%$ Solution:- $R\% = \frac{6}{100} = \frac{3}{50}$

$$\text{Let } P = (D)^2 = (50)^2 = 2500$$

$$\text{I } 150$$

$$\text{II } 150 + (9) \rightarrow D$$

$$\text{II } D = \frac{P R^2}{100^2} = \frac{2500 \times (6)^2}{100^2}$$

$$= 9$$

$$\text{IV } P : D$$

$$(50)^2 : (9)^2$$

$$2500 : (9) \rightarrow D$$

II

$$R = 20\% = \frac{1}{5}$$

$$P : D$$

$$(5)^2 : (1)^2$$

$$25 : 1$$

$$| \times 48 \quad | \times 48$$

$$1200 \quad 48$$

III (FORMULA)

$$D = \frac{P R^2}{100}$$

$$48 = \frac{P \times (20)^2}{100}$$

$$\Rightarrow P = 1200 \text{ Ans.}$$

$P = 1250$, $T = 2 \text{ years}$
 $\text{CI} - \text{SI} = 2$, $R = ?$

Solution:-

(I) Let Rate of $\% = R$

$$\text{I } \frac{1250 \times R}{100}$$

$$\text{II } \frac{1250 \times R}{100} + \boxed{\frac{1250 \times R \times R}{100 \times 100}} \rightarrow D$$

$$\frac{R^2}{8} = 2 \Rightarrow R = 4\% \text{ Ans.}$$

$$\text{II (FORMULA)} D = \frac{P R^2}{100^2}$$

$$2 = \frac{1250 \times R^2}{100^2} \Rightarrow R^2 = 16$$

$$\Rightarrow R = 4\%$$

$P = 16000$, $R = 5\%$, $T = 3 \text{ years}$, $CI - SI = D = ?$

Solution: (F) Let $P = 16000$ II (TRICKY)

I 400

II $400 + 20$

III $400 + 20 + 20 + 1$ — D

IF $P = 8000 \Rightarrow D = 61$

$\downarrow \times 2$

16000

$\downarrow \times 2$

122

$$5\% = \frac{1}{20}$$

P D

$$D = \frac{PR^2}{100^2} \times \frac{300+R}{100}$$

$$= \frac{16000 \times 5^2}{100^2} \times \frac{305}{100} = 122 \text{ Ans}$$

$P = 8000$, $T = 3 \text{ years}$

$D = CI - SI = 61$, $R = ?$

Solution: (FRICKY)

$$\frac{D}{P} = \frac{3R+1}{R^3} \quad \frac{61}{8000} = \frac{3R+1}{R^3}$$

$$\text{equating } R^3 = 8000 \Rightarrow R = 20$$

$$\text{Rate} = \frac{1}{R} \times 100 = \frac{1}{20} \times 100 = 5\%$$

$P = 2000$, $T = 4 \text{ years}$

$R_1 = 5\%$, $R_2 = 4\%$

$D = CI - SI = ?$

Solution: — $P = 2000$

I 100

II $80 + (4) \rightarrow D$

$P = 5000$, $T = 2 \text{ years}$

$D = CI - SI = 25$, $R_1 = 10\%$

$R_2 = ?$

Solution: — 5000

I 500

$$\text{II } \frac{5000 \times R_2}{100} + \left[\frac{500 \times R_2}{100} \right] - D$$

$$\frac{500 \times R_2}{100} = 25 \Rightarrow R_2 = 5\%$$

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Type 2 (When Rate is Not given)

TRICK Explanation

$$T = 2 \text{ yrs}$$

$$A = P \left[1 + \frac{R}{100} \right]^2$$

$$\frac{A}{P} = \left[1 + \frac{R}{100} \right]^2$$

$$\sqrt{\frac{A}{P}} = \left[1 + \frac{R}{100} \right]$$

$$T = 3 \text{ yrs}$$

$$A = P \left[1 + \frac{R}{100} \right]^3$$

$$\frac{A}{P} = \left[1 + \frac{R}{100} \right]^3$$

$$\sqrt[3]{\frac{A}{P}} = \left[1 + \frac{R}{100} \right]$$

↑ 1 year change ↑
(SI = CI)

$$\left[1 + \frac{R}{100} \right] \begin{cases} \rightarrow T = 2 \text{ yrs} \Rightarrow \sqrt{\frac{A}{P}} \text{ (वर्गमूल कर देंगे)} \\ \rightarrow T = 3 \text{ yrs} \Rightarrow \sqrt[3]{\frac{A}{P}} \text{ (घनमूल कर देंगे)} \\ \rightarrow T = 4 \text{ yrs} \Rightarrow \sqrt[4]{\frac{A}{P}} \\ \rightarrow T = n \text{ yrs} \Rightarrow \sqrt[n]{\frac{A}{P}} \end{cases}$$

e.g if $1 + \frac{R}{100} = \frac{4}{3} \Rightarrow \frac{R}{100} = \frac{1}{3}$ OR $R = 33.33\%$

OR $1 + \frac{R}{100} = \frac{4}{3} \rightarrow A_1$ $I = 1$ $R\% = \frac{1}{3} \times 100$
 $= 33.33\%$

A sum of money becomes 9 times in 2 years at a certain rate of C.I. Find the rate?

Solution:- I (BASIC)

$$\text{Let } P=1 \Rightarrow A=9$$

$$9 = 1 \left[1 + \frac{R}{100} \right]^2$$

$$\text{OR } 1 + \frac{R}{100} = 3$$

$$\frac{R}{100} = 2 \Rightarrow R = 200\% \text{ Ans.}$$

II (TRICKY)

$T=2\text{ yrs}$ take square root

$$\left[1 + \frac{R}{100} \right] = \sqrt{\frac{A}{P}} = \frac{3}{1}$$

$$\Rightarrow \frac{R}{100} = 2 \Rightarrow R = 200\%$$

$$\text{OR } \sqrt{9} : \sqrt{1} = \frac{A}{P} = \frac{3}{1}$$

$$R\% = \frac{2}{1} \times 100 = 200\%$$

कोई राशि C.I. की दर से 2 साल में 9/4 गुनी हो जाती है। क्याज की दर ज्ञात करें?

Solution $T=2\text{ yrs}$, Let $P=1$

$$1 + \frac{R}{100} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$

$$\Rightarrow \frac{R}{100} = \frac{1}{2} \Rightarrow \boxed{R = 50\%}$$

OR

$$\frac{A}{P} = \frac{9}{4}$$

$T=2\text{ yrs}$ take sq root

$$\sqrt{9} : \sqrt{4} = \frac{A}{P} = \frac{3}{2}$$

य 1 साल का Ratio

आ जाता है। अब 50 का use करो.

$$R\% = \frac{1}{2} \times 100 = 50\%$$

₹ 8000 becomes 9261 in 3 years at certain rate of Compound Interest. Find the Rate?

Solution:- I

$T=3\text{ yrs}$ take cube root

$$1 + \frac{R}{100} = \sqrt[3]{\frac{9261}{8000}} = \frac{21}{20}$$

$$\frac{R}{100} = \frac{1}{20} \Rightarrow \boxed{R = 5\%}$$

II

$$\frac{A}{P} = \frac{9261}{8000}$$

$T=3\text{ yrs}$ take cube root

$$\sqrt[3]{9261} : \sqrt[3]{8000}$$

$$21 : 20$$

$$\left[\frac{A}{P} \right] = 1$$

य एक साल का

PA आ जाता है। अब 5 का use करो.

$$R\% = \frac{1}{20} \times 100 = 5\%$$

A sum of money becomes 25 times in 5 yrs and 36 times in 7 years. Find the Rate of interest?

Solution:— I (BASIC)

let $P = ₹$

$$\therefore 25 = \left[1 + \frac{R}{100}\right]^5 \quad \text{--- (1)}$$

$$\text{also } 36 = \left[1 + \frac{R}{100}\right]^7 \quad \text{--- (2)}$$

divide (2) by (1), we get

$$\left[1 + \frac{R}{100}\right]^2 = \frac{36}{25} \Rightarrow 1 + \frac{R}{100} = \frac{6}{5}$$

$$\Rightarrow R = 20\% \text{ ANS}$$

II (TRICKY)

$$\begin{array}{l} \text{4 yrs} \left[\begin{array}{l} A_7 (7 \text{ yrs}) \rightarrow 36 \\ A_5 (5 \text{ yrs}) \rightarrow 25 \end{array} \right. \end{array}$$

$$\Rightarrow 1 \text{ yr change} = \sqrt[2]{\frac{36}{25}} = \frac{6}{5} \frac{A}{P}$$

$$A : P \text{ (For 1 yr)}$$

$$\begin{array}{c} 6 : 5 \\ \hline 1 \end{array} \Rightarrow R.H. = \frac{1}{5} \times 100$$

$$= 20\%$$

कोई राशी CR की दर से 2 साल में 8 गुणी हो जाती है। और 5 साल में 27 गुणी हो जाती है तो व्याज की दर ज्ञात करें?

Solution:—

$$\begin{array}{l} \text{3 yrs} \left[\begin{array}{l} A_5 \xrightarrow{\quad} 27 \\ A_2 \xrightarrow{\quad} 8 \end{array} \right. \end{array}$$

$$\text{change} \Rightarrow \text{In 1 year} \rightarrow \text{Ratio} : \sqrt[3]{8} = \begin{array}{c} A \\ 3 \\ \hline P \\ 2 \end{array}$$

$$R.H. = \frac{1}{2} \times 100 = 50\%$$

कोई राशी CR की दर से 4 साल में 16000 और 7 साल में 18522 हो जाती है। व्याज की दर ज्ञात करें?

Solution:—

$$\begin{array}{l} \text{3 yrs} \left[\begin{array}{l} A_7 \xrightarrow{\quad} 18522 \\ A_4 \xrightarrow{\quad} 16000 \end{array} \right. \end{array}$$

$$\text{change} \Rightarrow 1 \text{ yr change} \begin{array}{c} A : P \\ \sqrt[3]{18522} : \sqrt[4]{16000} = 21 : 20 \end{array}$$

$$R.H. = \frac{1}{20} \times 100 = 5\%$$

If a sum of money becomes N_1 Times in T_1 yrs and N_2 times in T_2 years then

$$(N_1)^{T_2} = (N_2)^{T_1} \text{ OR } N_2 = (N_1)^{T_2/T_1}$$

A sum of money get doubled in 4 years. IN how much time it will become 8 times?

Solution:- I (FORMULA)

$$N_1 = 2, T_1 \rightarrow 4$$

$$N_2 \rightarrow 8, T_2 = ?$$

$$N_2 = (N_1)^{T_2/T_1} \Rightarrow 8 = (2)^{T_2/4}$$

$$\text{OR } 2^3 = (2)^{T_2/4} \Rightarrow T_2/4 = 3$$

$$\Rightarrow T_2 = 12 \text{ yrs.}$$

II (TRICKY)

$$2^1 \text{ गुनी} \rightarrow 4 \text{ }] + 4$$

$$2^2 \text{ times} \rightarrow 8 \text{ }] + 4$$

$$2^3 \text{ times} \rightarrow 12 \text{ }] + 4$$

$$\hookrightarrow 8 \text{ times} \rightarrow 12 \text{ yrs}$$

कोई राशि 4 वर्षों में व्याज की दर से 3 साल में 3 गुनी हो जाती है। कितने समय में यह राशि 81 गुनी हो जाएगी?

Solution:- I (FORMULA)

$$N_1 \rightarrow 3, T_1 \rightarrow 3$$

$$N_2 \rightarrow 81, T_2 \rightarrow ?$$

$$N_2 = (N_1)^{T_2/T_1}$$

$$\Rightarrow 81 = (3)^{T_2/3} \Rightarrow 3^4 = 3^{T_2/3}$$

$$\Rightarrow T_2/3 = 4 \Rightarrow 12 \text{ yrs}$$

II (TRICKY)

$$3^1 \text{ times} \rightarrow 3$$

$$3^2 \text{ times} \rightarrow 6 \text{ }] + 3$$

$$3^3 \text{ times} \rightarrow 9 \text{ }] + 3$$

$$3^4 = 81 \text{ times} \rightarrow 12 \text{ }] + 3$$

$$12 \text{ years Ans.}$$

A sum of money becomes 81 times in 4 years. in how much time it will become 27 times?

Solution:- I (FORMULA)

$$N_1 \rightarrow 81, T_1 \rightarrow 4$$

$$N_2 \rightarrow 27, T_2 \rightarrow ?$$

$$27 = (81)^{T_2/4}$$

$$\text{OR } \Rightarrow 27 = (3^4)^{T_2/4}$$

$$3^3 = 3^{T_2} \Rightarrow T_2 = 3 \text{ yrs.}$$

II (TRICKY)

$$A : P$$

$$T=4 \rightarrow 81 : 1$$

$$T=1 \rightarrow \sqrt[4]{81} : \sqrt[4]{1}$$

$$3 : 1$$

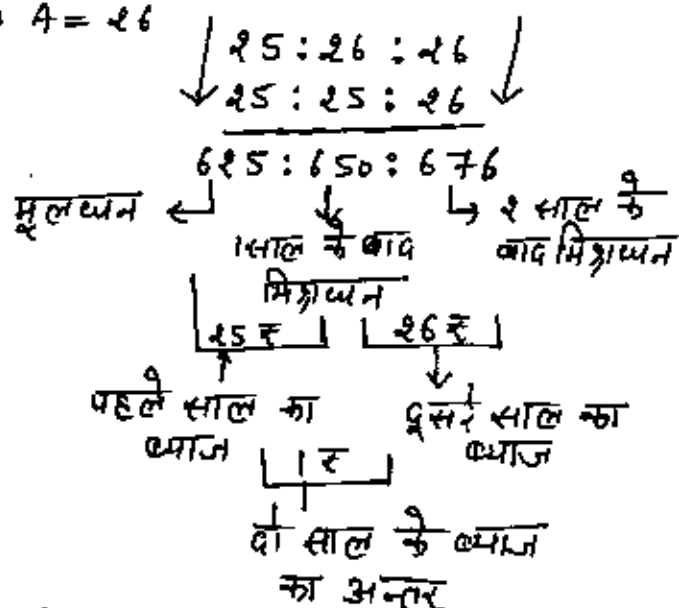
$$\therefore 3^1 \text{ --- } 1 \text{ साल}$$

$$3^2 \text{ --- } 2 \text{ साल}$$

$$27 = 3^3 \text{ --- } 3 \text{ साल} \rightarrow \text{Ans 3}$$

$$\# P = 625 \text{ ₹} \quad R = 4\% \quad T = 2 \text{ yrs}, \text{ CI} = ?$$

$$4\% = \frac{100 - I}{100 - P} \Rightarrow A = 26$$



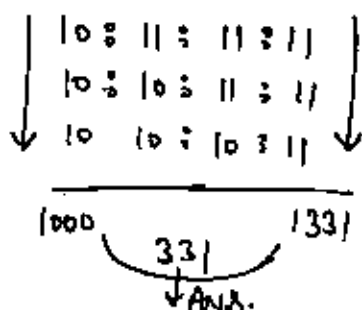
$$I = A - P$$

$$= 676 - 625 = \textcircled{51}$$

$$\# 1000 = P, R = 10\%, T = 3 \text{ yrs}$$

CI

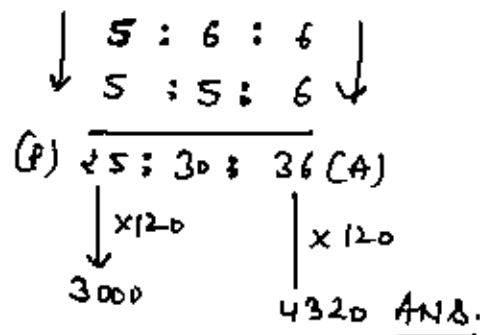
$$10\% = \frac{1}{10}$$



$$\# P = 3000, R = 20\%$$

$$T = 2 \text{ yrs}, A = ?$$

$$20\% = \frac{1}{5}$$



A sum of money becomes ₹ 4800 after 4 years and ₹ 6000 after 8 years. Find the Principal?

Solution:- I (TRICKY)

$$\begin{cases} A_8 \rightarrow 6000 \\ A_4 \rightarrow 4800 \end{cases}$$

$$4 \text{ yr change } \frac{A_8}{A_4} = \frac{6000}{4800} = \frac{5}{4}$$

[Every 4 yrs it will change by $\frac{5}{4}$]

$$\therefore \frac{A_4}{P} = \frac{5}{4} \Rightarrow \frac{4800}{P} = \frac{5}{4} \Rightarrow P = 3840$$

$$\text{OR } P \times \frac{5}{4} = 4800 \text{ OR } P \times \frac{5}{4} \times \frac{5}{4} = 6000$$

$$P = 3840 \text{ Ans.}$$

कोई धनराशि 8 साल के बाद 1600 ₹ हो जाती है और 12 साल के बाद 2000 ₹ हो जाती है तो 16 साल बाद मिलनी हो जाएगी ?

Solution:- I

$$\begin{cases} A_{12} \rightarrow 2000 \\ A_8 \rightarrow 1600 \end{cases}$$

$$4 \text{ साल में } = \frac{2000}{1600} = \frac{5}{4} \text{ गुनी}$$

[हर 4 साल बाद $\frac{5}{4}$ गुनी हो जाएगी]

$$A_{12} \times \frac{5}{4} = A_{16}$$

$$2000 \times \frac{5}{4} = 2500 \text{ ₹ Ans.}$$

II (SUPER TRICKY)

$$\begin{array}{ccccc} & 4 & & 4 & \\ P & \xrightarrow{\quad} & A_4 & \xrightarrow{\quad} & A_8 \\ & & 4800 & : & 6000 \\ & & 4 & : & 5 \\ & & 4 & : & 5 \\ \hline & & 16 & : & 20 : 25 \\ & & \downarrow \times 240 & & \downarrow \times 240 \\ & & 3840 & & 4800 \end{array}$$

$$\begin{array}{ccccc} & 4 \text{ Yr} & & 4 \text{ Yr} & \\ A_8 & \xrightarrow{\quad} & A_{12} & \xrightarrow{\quad} & A_{16} \\ & & 1600 & : & 2000 \\ & & 4 & : & 5 \\ & & 4 & : & 5 \\ \hline & & 16 & : & 20 : 25 \\ & & \downarrow \times 100 & & \downarrow \times 100 \\ & & 2000 & & 2500 \text{ Ans.} \end{array}$$

CI For 2 years and 3 years for a certain sum is 156 and 154 respectively. Find the rate of compound interest? a) $14\frac{2}{3}\%$, b) $16\frac{2}{3}\%$, c) 10%, d) 9%.

Solution:- $CI_3 = P \left[\left(1 + \frac{R}{100} \right)^3 - 1 \right]$
 $CI_2 = P \left[\left(1 + \frac{R}{100} \right)^2 - 1 \right]$ $\Rightarrow \frac{154}{156} = \frac{127}{78} \quad \text{--- (1)}$

Let $1 + \frac{R}{100} = x \Rightarrow \frac{x^3 - 1}{x^2 - 1} = \frac{127}{78} \Rightarrow \frac{(x-1)(x^2+1+x)}{(x-1)(x+1)} = \frac{127}{78}$

$\Rightarrow 78x^2 + 78 + 78x = 127x + 127$

$\Rightarrow 78x^2 - 127x + 78x + 78 - 127 = 0 \Rightarrow 78x^2 - 49x - 49 = 0$

$78x^2 - 91x + 42x - 49 = 0$

OR $13x[6x-7] + 7[6x-7] = 0 \Rightarrow x = \frac{7}{6} \therefore 1 + \frac{R}{100} = \frac{7}{6}$

OR Check by option in (1)

$\Rightarrow R = 16\frac{2}{3}\% \text{ Ans.}$

option b will satisfy.

Find the compound interest on Rs 15625 for 9 months at 16% p.a. compounded quarterly?

Solution:- $R = \frac{16}{4} = 4\%$ and $T = 9 \times 4 = 36 \text{ months} = 3 \text{ yrs}$

$4\% = \frac{1}{25}$

P	A
15	26
25	26
25	26

15625 — 17576

CI = AP = 1951

OR PASCAL 3 3 1

$3A + 3B + C$

$3 \times \frac{15625 \times 4}{100} + 3 \times \frac{625 \times 4}{100} + 1$

$= 625 \times 3 + 25 \times 3 + 1$

$= 1951 \text{ Ans.}$



INSTALLMENT (1624) IN COMPOUND INTEREST

$$P = \frac{X}{\left[1 + \frac{R}{100}\right]^1} + \frac{X}{\left[1 + \frac{R}{100}\right]^2} + \dots + \frac{X}{\left[1 + \frac{R}{100}\right]^T}$$

where $P = \text{Amount / Principal}$, $X = \text{Installment}$

$T \rightarrow \text{Time (Time) OR NO. of Installments}$

A man borrows Rs 21000 at 10% compound interest. How much he has to pay equally at the end of year to settle his loan in two years.

Solution:- I (FORMULA)

$$21000 = \frac{X}{\left[1 + \frac{10}{100}\right]} + \frac{X}{\left[1 + \frac{10}{100}\right]^2}$$

$$\text{OR } 21000 = \frac{10X}{11} + \frac{100X}{121}$$

$$\text{OR } 21000 = \frac{110X + 100X}{121}$$

$$\text{OR } X = 12100 \text{ Ans.}$$

II (TRICKY)

$$R = 10\% = \frac{1}{10} \rightarrow I \Rightarrow \text{Installment} = 11$$

(make each installment equal)

1st year $\frac{10 \times 11}{100} = 11$

2nd year $\frac{(10)^2}{(11)^2} = 121$

110 121

100 121

$$21000 = 100 \times \frac{210}{100} + \frac{121 \times 100}{100} = 12100$$

A man buys a scooter on making a cash down payment of ₹ 16224 and promise to pay two more yearly installment of equivalent amount in next two years. If the Rate of Interest is 4% p.a. Find the cash value of scooter?

Solution:- I (BASIC)

$$P = \frac{16224}{\left[1 + \frac{4}{100}\right]^1} + \frac{16224}{\left[1 + \frac{4}{100}\right]^2}$$

$$= 16224 \left[\frac{25}{26} + \frac{625}{676} \right]$$

$$= 30600$$

$$\text{Cash value} = 30600 + 16224$$

$$= 46824 \text{ Ans.}$$

II (TRICKY) $R = 4\% = \frac{1}{25}$

Principal Installment

$$\text{I } 25 \times 26 \quad 26 \times 26$$

$$\text{II } + 625 \quad 676$$

$$\hline 1275 \quad 676$$

$$1 \times 24 \quad 1 \times 24$$

$$30600 \quad 16224$$

$$\text{CASH VALUE} = 30600 + 16224 = 46824$$

A man borrows Rs 4200 and undertakes to pay back with CI @ 10% p.a in two equal yearly installments at the end of 1st and 2nd year. What is the amount of each installment?

Solution :- $R = 10\% = \frac{1}{10}$ [but installment must be equal]

	P	I
1st Yr	10×11	11×11
2nd Yr	$+ 100$	121
	210	121

$ \times 20$	$ \times 20$
4200	2420 Ans.

OR by FORMULA

$$4200 = \frac{x}{\left[1 + \frac{10}{100}\right]} + \frac{x}{\left[1 + \frac{10}{100}\right]^2}$$

$$\Rightarrow 4200 = x \left[\frac{10}{11} + \frac{100}{121} \right]$$

$$\Rightarrow 4200 = \frac{210x}{121} \Rightarrow x = 2420 \text{ Ans.}$$

A man borrows Rs 1820 and undertakes to pay back with compound interest @ 20% p.a in 3 equal yearly installments at the end of 1st, 2nd and 3rd year. What is the amount of each installment?

Solution :- $R = 20\% = \frac{1}{5} \Rightarrow \text{Installment} = 5+1 = 6$

1st	5×36	$6 \times 36 = 216$	} Installment must be equal
2nd	25×6	$36 \times 6 = 216$	
3rd	125	$216 = 216$	

455	216
$ \times 4$	$ \times 4$
1820	864 Ans.

A man borrows a certain sum of money and pays it back in two equal installments, if CI is reckoned at 5% per annum and he pays back annually Rs 882. What sum did he borrow?

Solution :- $R = 5\% = \frac{1}{20}$

	P	I
1st	20×21	21×21
2nd	$+ 400$	441
	820	441
	$ \times 2$	$ \times 2$
Ans		
	1640	882

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